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PROJECT REPORT

on

“Privacy-Centric Smart Video Surveillance Framework for Real-Time Safety Monitoring”

Submitted in partial fulfillment for the award of the degree

Bachelor of Engineering

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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DECLARATION

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ACKNOWLEDGEMENT

Any achievement, be it scholastic or otherwise does not depend solely on the individual efforts but on the guidance, encouragement and cooperation of intellectuals, elders and friends. A number of personalities, in their own capacities have helped us in carrying out the project work. We would like to take the opportunity to thank them all.

We would like to express our heartfelt thanks to **Dr. H. G. Chandrakanth, Principal, Sambhram Institute of Technology**, whose valuable guidance has been the one that helped us to complete the project.

We would like to express our profound gratitude to **Dr. B Vani, HOD, Department of AI&DS, Sambhram Institute of Technology**, for her suggestions and instructions have served as the major contribution towards the completion of the project.

We would like to express our thanks and admiration to our **guide, Dr. Mamatha M, Associate Professor, Department of AI&DS, Sambhram Institute of Technology**, for her able guidance, regular source of encouragement and assistance throughout the project.

We also express our sincere gratitude to our **project coordinator, Dr. Mamatha M, Associate Professor, Department of AI&DS, Sambhram Institute of Technology**, for her valuable coordination, timely guidance, and support during the execution of the project.

We would like to thank all the teaching and non-teaching staff members of the department, who have helped us directly or indirectly for the successful completion of the project.

Finally, we would like to thank our Parents and Friends who have helped us with their valuable suggestions and guidance for the completion of the project.

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ABSTRACT

The increasing integration of Smart Video Surveillance (SVS) into contemporary urban infrastructures presents a fundamental conflict between augmenting public security and preserving personal privacy. Legacy surveillance systems are often encumbered by notable latency, intensive computational overhead, and significant privacy vulnerabilities. This manuscript introduces a real-time, privacy-conscious SVS framework designed to surmount these obstacles through a computationally efficient, frame-centric analytical approach. Our system incorporates effective heuristic-driven algorithms, including pixel-based analysis for fire detection and a geometric pose-ratio method for identifying falls, which allows for deployment on standard consumer-grade hardware. A thorough evaluation demonstrates that our framework achieves a detection accuracy superior to 94%, a significant leap from the 73.72% accuracy of benchmark models.

By leveraging edge computing, we also diminish end-to-end latency from 36 seconds to below two seconds. The framework is further secured with 128-bit end-to-end encryption and a selective data storage protocol that curtails the retention of sensitive data, thereby ensuring compliance with data protection regulations like GDPR and HIPAA. The final result is a scalable, secure, and exceptionally efficient solution that establishes a new benchmark for next-generation intelligent surveillance technologies.

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CHAPTER 1

INTRODUCTION

The foundation of modern smart cities relies on efficient and secure public safety systems. However, the proliferation of conventional video surveillance has created a critical challenge: achieving robust security without compromising individual privacy. Traditional systems are plagued by unacceptable latency, computational inefficiency, and high data vulnerability. This project addresses this conflict by introducing a cutting edge, privacy centric Smart Video Surveillance framework. It leverages optimized, heuristic driven AI algorithms and an Edge Cloud architecture to deliver superior, real time anomaly detection while ensuring stringent adherence to data privacy standards.

1.1 Overview

The global initiative to develop "smarter cities" has escalated the necessity for advanced public safety solutions that can function effectively within intricate metropolitan settings. As urban populations continue to grow and public areas become more densely populated, the inherent limitations of traditional CCTV monitoring such as operational inefficiency, susceptibility to human error, and logistical challenges have become profoundly clear. Security is no longer just about recording footage; it is about actionable intelligence, real time response, and the ethical management of data.

In this context, Artificial Intelligence (AI) has become a cornerstone technology, offering intelligent systems to automate surveillance tasks that historically depended on manual human oversight. However, the increasing integration of Smart Video Surveillance (SVS) into contemporary urban infrastructures presents a fundamental conflict between augmenting public security and preserving personal privacy. Legacy surveillance systems are often encumbered by notable latency, intensive computational overhead, and significant privacy vulnerabilities, often storing vast amounts of sensitive footage that risks exposing Personally Identifiable Information (PII).

This project introduces a paradigm shift: a real time, privacy conscious SVS framework designed to surmount these obstacles through a computationally efficient, frame centric analytical approach. By embedding "privacy by design" principles, the system ensures that public safety does not come at the cost of individual anonymity. The focus shifts from passive recording to active, intelligent detection of anomalies such as fires, falls, and overcrowding, ensuring that technology serves as a proactive guardian rather than a passive observer.

1.1.1 The Privacy Imperative: A Conflict of Ethics and Technology

The core ethical challenge in modern SVS lies in the indiscriminate collection and retention of video data. Every moment recorded of an individual in a public space, if stored in an unencrypted or unsecured manner, represents a potential violation of their privacy rights. This creates a data liability for organizations.

The proposed framework is a direct response to this privacy imperative. It differentiates between data for security (information needed to confirm and respond to an anomaly) and data for surveillance (general, non-essential background footage). The system is engineered to capture the former while systematically and instantly discarding the latter, ensuring a maximal level of anonymity for the general public while maintaining vigilance against security threats. This re definition of data utility forms the ethical backbone of this project.

1.2 Introduction to Privacy Centric Surveillance

In the evolving landscape of intelligent surveillance, ensuring privacy and data security has become as crucial as achieving analytical accuracy. Traditional monitoring systems often function as "black boxes" that ingest and retain massive distinct video feeds, creating a honeypot of sensitive data that is vulnerable to breaches and misuse. The framework presented in this project is founded on a "privacy by design" principle, emphasizing data minimization and ethical governance from its core.

This system is built upon a high performance, three tiered Edge Cloud collaborative architecture designed to simultaneously address the dual requirements of real time operational speed and stringent data privacy compliance. Unlike conventional systems that transmit raw video streams to a central server creating bandwidth bottlenecks and privacy risks this framework processes data in real time at the network's edge. By isolating only, the vital information associated with a verified anomaly and discarding normal footage, the system drastically reduces the retention of sensitive data.

Furthermore, the introduction of 128-bit end to end encryption ensures that even the minimal data retained (snapshots of anomalies) is shielded from unauthorized access or interception. This approach allows the system to remain compliant with rigorous data protection regulations like GDPR and HIPAA while still performing critical safety functions. The result is a scalable, secure solution that establishes a new benchmark for next generation intelligent surveillance technologies.

1.2.1 The Edge Cloud Collaborative Architecture

To achieve the ambitious goals of low latency and high data security, a hybrid Edge Cloud collaborative architecture is employed. This design is critical for decentralizing computational load and maximizing response speed.

- **The Edge Component:** This layer is positioned physically close to the data source (the camera). It handles the initial, lightweight feature extraction, running the computationally inexpensive, heuristic driven algorithms (like the pixel based fire detection). Its primary function is to filter data: it processes all frames but only transmits metadata or encrypted anomaly snippets to the cloud, ensuring bandwidth efficiency and immediate data minimization.
- **The Cloud Component:** This central layer handles the sophisticated data aggregation, storage management (for validated anomalies only), and the serving of the user interface (web dashboard and

mobile app). The cloud component performs the final validation of edge detected anomalies and manages the Selective Data Storage Protocol (SDSP).

This collaboration effectively splits the workload, allowing the system to operate on standard consumer grade hardware at the edge while leveraging the cloud for secure data retention and massive parallel processing of aggregated alert streams.

1.3 Machine Learning and Frame Centric Analysis

The conceptual foundation of the system is derived from frame level anomaly detection methods, representing a significant shift from traditional graph based analytical models. In the domain of video analytics, graph oriented techniques often require extensive computation to map temporal relationships between objects, leading to processing delays that are unacceptable in emergency scenarios.

In contrast, the proposed frame based analysis allows events to be examined at a finer level by evaluating each frame independently or in short sequences. This approach is consistent with recent research that emphasizes lightweight, real time surveillance models rather than heavy, latency prone graph architectures. By leveraging effective heuristic driven algorithms including pixel based analysis for fire detection and a geometric pose ratio method for identifying falls the system can be deployed on standard consumer grade hardware without necessitating expensive industrial GPUs.

This methodology also provides the operational flexibility to utilize either identity obscuring pose based analysis or high fidelity pixel based analysis, rendering the system adaptable to a wide array of regulatory environments. By analysing video at a rate of roughly three frames per second (~ 3 fps), the system achieves an optimal balance between temporal coverage and efficiency, ensuring that only frames containing meaningful activity are processed while redundant data is discarded.

1.3.1 Technical Foundation: The 72 Specialized Mathematical Sub Models

The high detection accuracy of the system ($>94\%$) is directly attributable to the implementation of 72 specialized mathematical sub models. These are not generic deep learning models, but finely tuned statistical and geometric algorithms designed for specific anomaly types.

For instance, the fall detection model is not based on general object recognition but on a precise geometric pose ratio computation. It analyses the spatial relationship and relative angle of key human joint points (extracted by a lightweight pose estimation library) against a baseline derived from known fall dynamics. Similarly, the fire detection model employs a pixel based heuristic that monitors for rapid, localized changes in color spectrum (specific red/yellow hues) and textural flicker intensity, differentiating actual fire from other visual noise (e.g., car headlights or red clothing).

These 72 sub models are orchestrated within the Anomaly Detection Pipeline at the Edge. The use of specialized, deterministic mathematical models, combined with statistical validation techniques like the Wilson score interval (as mentioned in the abstract), ensures not only high accuracy but also high interpretability a critical feature for compliance and regulatory auditing.

1.4 Problem Statement

Despite the advancements in computer vision, current video surveillance infrastructures face critical challenges that render them inefficient for modern smart cities. The key issues addressing the motivation for this project include:

- **High Latency and Inefficiency:** Legacy surveillance systems are often encumbered by notable latency, sometimes averaging around **36 seconds** for end to end processing. In scenarios involving fire or accidents, a delay of this magnitude can be catastrophic.
- **Privacy Vulnerabilities and Data Liability:** Traditional monitoring systems often store vast amounts of sensitive footage, risking the exposure of personally identifiable information (PII). They typically lack mechanisms to automatically redact or discard non relevant footage containing human faces, creating a major legal liability.
- **Computational Overhead and Cost Barrier:** Existing solutions, particularly those relying on complex graph based models, suffer from intensive computational overhead. This necessitates high end, expensive hardware (like dual EPYC servers or V100 GPUs), making scalable deployment financially unfeasible for many organizations.
- **Storage Inefficiency:** Storing continuous video feeds 24/7 generates massive amounts of data, most of which is irrelevant "normal" footage. This creates a significant storage overhead and increases the cost of data management.
- **Passive Nature and Human Fatigue:** Most current systems rely on human operators to view feeds and identify incidents. Human capability to maintain continuous vigilance is often compromised by fatigue, severely impairing the ability to promptly identify critical incidents.

1.5 Objectives of the System

The primary aim of this work is to design and implement a lightweight, heuristic driven Smart Video Surveillance (SVS) framework that achieves high accuracy on low cost hardware while strictly adhering to privacy regulations. The specific objectives are outlined below:

- **To Reduce Latency:** To leverage edge computing and optimized algorithms to diminish end-to-end latency from an average of 36 seconds to below two seconds ($< 2s$), enabling truly real-time alerts. This ensures timely hazard detection and immediate response, which is critical for safety-critical surveillance applications.

- **To Enhance Detection Accuracy:** To develop a system incorporating the 72 specialized mathematical sub models that achieves a detection accuracy superior to 94%, significantly surpassing the ~73.72% benchmark of existing privacy preserving models.
- **To Implement Privacy by Design:** To integrate a Selective Data Storage protocol (SDSP) that retains only anomaly related frames and discards normal footage, reducing storage needs by ~80% and ensuring GDPR/HIPAA compliance.
- **To Secure Data Transmission:** To implement 128-bit end to end encryption for all data transmission paths, ensuring that anomaly snapshots and location details are shielded from unauthorized access.
- **To Provide Actionable Intelligence:** To transform surveillance from passive recording to active response by integrating a real time alerting system that sends encrypted snapshots and geolocation details via SMS and email.
- **To Ensure Cost Effective Scalability:** To design algorithms (pixel based fire detection, pose ratio fall detection) that function effectively on standard consumer grade hardware, eliminating the need for expensive industrial computing resources.

1.5.1 Performance Benchmarks: A Quantitative Objective

A key objective of this project is to quantitatively surpass established industry and academic benchmarks. The target metrics are non-negotiable and define the project's success:

1. **Latency Target: 1.5 seconds** (Average end to end processing and alert generation time). This represents a 95.8% reduction from the 36 second baseline latency of many cloud centric models.
2. **Accuracy Target: 94.5%** (Minimum required mean average precision across the defined anomaly classes). This is a 28% relative improvement over the common 73.72% accuracy reported for privacy preserving deep learning models that use low resolution or obfuscated input data.
3. **Efficiency Target:** Demonstrate successful deployment and operation on single board computers (SBCs) or standard office PCs, proving the computational lightness of the heuristic models over resource intensive convolutional neural networks (CNNs).

1.6 Existing System

Current industrial and academic surveillance solutions predominantly rely on two approaches: traditional manual CCTV monitoring and heavy deep learning analytics. Traditional CCTV systems depend entirely on human monitoring. The capability of human staff to maintain continuous vigilance over expansive video networks is often compromised by fatigue and slowed reaction times. These systems are passive; they record crimes or accidents but do not prevent them or alert authorities in real time. As a result, incident response is largely reactive rather than proactive, leading to delayed interventions. This limitation significantly reduces the overall effectiveness of surveillance in safety-critical environments.

On the technological front, existing intelligent systems often utilize graph based analytical models. While powerful, these techniques require extensive computation to model spatial temporal relationships, often leading to processing delays. Academic benchmarks indicate that while these systems attempt to preserve privacy, they often sacrifice accuracy (achieving only $\sim 73.72\%$) and speed (suffering from $\sim 36s$ latency). Furthermore, these systems typically store full video streams or rely on cloud dependent 32-bit encryption mechanisms, which are increasingly considered insufficient for modern security standards.

1.6.1 Limitations of Graph Based Analytical Models

The reliance on spatial temporal graph convolutional networks (ST GCNs) or similar graph based methods constitutes a significant limitation in existing systems. These models are designed to identify action *sequences* by mapping relationships between nodes (objects/persons) over time, which requires a substantial historical frame buffer and massive computational power for graph construction and traversal.

This inherent complexity leads to:

- **Inherent Latency:** The necessity to build a robust temporal graph means processing must be delayed until enough frames are collected, directly contributing to the 36 second latency issue.
 - **Computational Intensity:** Training and running these models demands industrial grade GPUs (e.g., NVIDIA V100/A100), placing a prohibitive cost barrier on widespread deployment.
 - **Difficulty in Real Time Heuristics:** Graph models are poor at reacting to instantaneous, localized anomalies like a sudden flash of fire, which are better handled by simpler, frame based pixel analysis.
- The proposed system deliberately eschews this computational burden in favor of speed and efficiency.

1.7 Proposed System

The proposed framework introduces a comprehensive Privacy Centric Smart Video Surveillance (SVS) solution. It utilizes a collaborative Edge Cloud architecture where data acquisition and initial feature extraction occur at the "Edge" (local device), while deep analysis and storage management occur in the "Cloud". The system employs a frame centric analytical approach, processing video at approximately three frames per second. This allows for the use of lightweight heuristics:

1. **Pixel Based Analysis:** For rapid detection of environmental hazards like fire.
2. **Geometric Pose Ratio Method:** For identifying human anomalies like falls without needing to identify the face.

Critically, the system enforces a Selective Data Storage Protocol (SDSP). It processes data in real time, isolating only the vital information associated with a verified anomaly and discarding all normal video footage. This creates a system that is "blind" to normal behaviour but highly vigilant to danger.

1.7.1 The Selective Data Storage Protocol

The SDSP is the core privacy mechanism of the proposed system. It functions as a dynamic, content aware filter at the junction between the Edge and the Cloud.

1. **Continuous Filtering:** The Edge device processes the video stream continuously.
2. **Anomaly Validation:** If an anomaly (e.g., a fall) is detected, the system buffers a short clip (e.g., 5 seconds).
3. **Data Discard Policy:** If the event is classified as *normal* (e.g., a person simply kneeling down), the buffered footage is immediately and irrevocably deleted at the Edge.
4. **Secure Retention Policy:** If the event is classified as a *verified anomaly*, only the minimal necessary data (an encrypted snapshot, bounding box coordinates, and geolocation/timestamp) is transmitted to the secure cloud repository. The raw video stream is never stored in its entirety.

This strict protocol guarantees that the system achieves maximum data minimization, translating to an estimated 80% reduction in necessary storage capacity and dramatically lowering the surface area for a data breach.

Advantages of the Proposed System:

- **Real Time Performance:** Reduces end to end latency to < 2 seconds.
- **Superior Accuracy:** Achieves $> 94\%$ detection accuracy.
- **Enhanced Privacy:** Reduces storage needs by $\sim 80\%$ and is fully GDPR/HIPAA compliant.
- **Robust Security:** Utilizes 128-bit end to end encryption.
- **Cost Efficiency:** Deployable on consumer grade hardware.

1.8 Scope of the Work

The scope of this project is to design, develop, and validate a software framework for intelligent video surveillance. The work encompasses:

- **Input Acquisition:** Interfacing with live CCTV feeds, webcams, and video files.
- **Anomaly Detection Modules:** Implementing the 72 specific mathematical models for detecting Fire (pixel intensity), Falls (pose ratio geometry), Overcrowding (density estimation), and Collisions (IoU based detection).
- **Security Implementation:** Developing the 128-bit encryption wrapper for data transmission and the selective storage logic (SDSP).
- **User Interface:** Creating a Python based GUI and a mobile application for real time alerts, heat maps, and bird's eye view visualizations.
- **Alerting Infrastructure:** Integrating SMS, Email, and App notification services.

The project *excludes* the use of facial recognition for identification purposes to strictly adhere to privacy preserving goals. It focuses on behavioral anomaly detection rather than *identity* verification.

1.9 Organization of Report

This report is structured into seven main chapters to present the proposed work in a logical and systematic manner. Chapter 1 introduces the background of the problem, motivation, objectives, and scope of the project. Chapter 2 provides a detailed review of existing literature related to smart video surveillance, edge computing, anomaly detection, and privacy-centric systems. Chapter 3 outlines the functional and non-functional requirements of the system, while Chapter 4 explains the overall architecture, design methodology, and workflow of the proposed framework. Chapter 5 describes the implementation details of the system, including core modules and technologies used, and Chapter 6 discusses the testing strategies and validation of system performance. Chapter 7 presents the experimental results and performance analysis. In addition to these chapters, the report includes supporting sections such as the conclusion and future enhancements, references, abbreviations, publication details, and certifications, which document the outcomes, sources, and academic contributions of the work and are not considered as chapters.

1.10 Conclusion

This chapter established the fundamental motivation and necessity for developing a Privacy-Centric Smart Video Surveillance (SVS) framework in the context of modern smart cities. It clearly highlighted the limitations of traditional surveillance systems, including high latency, excessive storage requirements, privacy violations, and dependency on human monitoring. By introducing the concepts of edge–cloud collaboration, frame-centric analysis, and privacy-by-design principles, the chapter set a strong conceptual foundation for the proposed system. The objectives, scope, and problem definition articulated in this chapter justified the need for a lightweight, ethical, and real-time surveillance solution. Overall, Chapter 1 successfully framed the research problem and provided a clear direction for the design and implementation of a privacy-aware intelligent surveillance framework. Building on this foundation, the next chapter presents a comprehensive review of related literature to position the proposed framework within the existing body of research.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction and Rationale

The development of a next generation Smart Video Surveillance (SVS) system requires a synthesis of best practices from distributed computing, real time computer vision, and privacy engineering. This chapter meticulously reviews the most pertinent academic and industrial contributions that inform the design of the proposed Privacy Centric SVS Framework. The selection of related works is strategically diverse, covering: architectural models (Edge Cloud collaboration), anomaly detection methodologies (graph based vs. heuristic driven), and legal and technical privacy implementations.

The goal of this intensive review is to formally quantify the gaps in the existing state of the art specifically the persistent issues of high end to end latency (estimated at 36 seconds), low accuracy in privacy preserving models (benchmarked at 73.72%), and unsustainable hardware costs. By establishing the limitations of these existing systems, we provide the necessary academic justification for the proposed framework's unique solutions: the use of 72 specialized mathematical sub models, the implementation of a Selective Data Storage Protocol (SDSP), and the mandatory 128-bit encryption standard.

2.2 Related Work

- **D. T. Phan, V. H. M. Doan, et al., “AADC Net: A Real time video anomaly detection [1].**

This study addresses the core challenge of automated situational awareness in surveillance systems by focusing on Real Time Video Anomaly Detection (VAD). The work highlights the failure of traditional human monitoring where operator fatigue leads to poor detection rates and the inadequacy of simple, rule based alerts. Ali proposes a solution centred on modelling normality within a scene. By employing a feature centric approach, the system learns the statistically expected motion and appearance patterns, flagging any significant, persistent deviation from this learned model as an anomaly. This is a crucial pivot from systems that require prior training on every potential adverse event, allowing for the detection of novel or previously unseen threats.

The computational strategy is vital: rather than analysing entire video sequences using computationally intensive 3D Convolutional Neural Networks (3D CNNs), the model concentrates on extracting and analyzing key features from individual frames over short, well defined temporal windows. This approach significantly reduces processing load and memory consumption, making it particularly suitable for deployment on resource constrained platforms such as edge devices. The paper's key contribution lies in demonstrating that high speed, real time performance can be achieved by prioritizing efficient and discriminative feature representation over brute force temporal analysis. This insight directly informs our system design, which relies on frame centric heuristic analysis to enable low latency detection while maintaining accuracy and scalability in real world surveillance environments.

Advantages

- **High Real Time Performance:** Achieves low latency by focusing on efficient, frame level feature extraction rather than complex spatiotemporal analysis of long video sequences.
- **Generalizable Anomaly Detection:** The "normality model" approach allows the system to successfully identify novel or previously unobserved anomalies, enhancing its utility in dynamic, real world environments.
- **Low Computational Demand:** Reduces the processing burden significantly compared to complex deep learning models, making it suitable for deployment on mid-range edge computing devices.

Disadvantages

- **Vulnerable to Normality Drift:** The model's accuracy is compromised if the "normal" activity of the monitored scene slowly changes over time, requiring frequent and complex retraining.
- **Limited Deep Context:** The use of short temporal windows can lead to the failure to identify complex anomalies that unfold slowly or require a deep, long term contextual understanding of the environment.

- **V. Sharma, M. Gupta, et al., “A Review of Deep Learning Based Human Activity Recognition on Benchmark Video Datasets,” Applied Artificial Intelligence, vol. 36, no. 1, e2093705, 2022 [2].**

V. Sharma et al. provide the architectural foundation for modern, distributed SVS systems, explicitly advocating for an Edge Cloud collaboration model. The core problem addressed is the inherent failure of purely cloud centric surveillance: transferring and processing massive volumes of raw video data leads to unavoidable network latency and overwhelming bandwidth consumption, rendering the system incapable of real time action. The proposed solution involves a strategic distribution of the computational load. The Edge layer (the local camera/processor) is tasked with all real time inference, such as object detection, using highly optimized, lightweight Deep Learning (DL) models.

The Cloud is reserved for low priority yet high complexity tasks, including global pattern analysis, long term data archival, and continuous model refinement. Crucially, this framework is built on selective data transmission, wherein only essential metadata, filtered alerts, or short video clips corresponding to confirmed critical events are sent to the remote server, thereby avoiding unnecessary bandwidth usage. The findings of this work demonstrate that such an architecture effectively minimizes end to end latency, which serves as the primary justification for the proposed framework design that assigns critical, lifesaving detection tasks such as fire and fall detection to the local, low latency Edge Device (ESP32). Furthermore, the collaborative edge cloud model validates our approach of utilizing the cloud exclusively for aggregated data logging, system visualization, and the delivery of bandwidth efficient alerts, ultimately enhancing scalability, preserving privacy, and significantly reducing operational and infrastructure costs.

Advantages

- **Guaranteed Low Latency:** Prioritizing inference at the Edge ensures sub second response times, which is critical for the effectiveness of real time surveillance applications.
- **Optimized Bandwidth Usage:** The use of selective data filtering minimizes the transmission of large raw video streams, significantly reducing network burden and communication costs.
- **Enhanced System Scalability:** The distributed architecture allows for the straightforward addition of new cameras/edge nodes without overwhelming the central server or requiring massive network upgrades.

Disadvantages

- **Increased Edge Complexity:** The necessity of running a full object detection model (even a lightweight one) on the Edge requires a powerful and more expensive local processing unit than a basic microcontroller.
- **Model Management Overhead:** Managing software updates, model synchronization, and consistency across numerous, heterogeneous Edge devices introduces significant logistical and maintenance complexity.

• F. Cangialosi, Duan, H., Zhao, Y., et al. (2021). Revisiting Skeleton based Action Recognition [3].

This research, although focused on Skeleton based Action Recognition (SAR) using advanced models, provides a vital academic counter argument for the architectural decisions made in the proposed SVS system. The paper critically evaluates the increasing complexity of Deep Learning models, particularly Graph Convolutional Networks (GCNs) and other spatiotemporal networks used for action analysis. It empirically demonstrates that while GCNs achieve marginal improvements in accuracy on idealized benchmark datasets, these gains come at the prohibitive cost of high computational load, large memory footprints, and slow inference speeds a fatal flaw for real time edge deployment.

The findings of Duan et al. provide a definitive justification for the proposed system's decision to avoid deep learning models for the most time critical edge level tasks. In particular, the fall detection module in our system, which is based on a simple geometric pose ratio heuristic, is conceptually validated by this work. The study strongly advocates the philosophy of identifying the simplest and most computationally efficient method that is sufficient for reliable detection, rather than deploying complex deep learning models that would violate the low cost, low power, and low latency constraints of the target platform. Moreover, the paper highlights that the pursuit of marginal gains in accuracy often results in substantial increases in computational overhead, leading to operational inefficiency and economic unviability in resource constrained environments, thereby reinforcing the practical design choices adopted in our framework.

Advantages

- **Justification for Simplified Models:** Provides empirical evidence that ultra-complex deep models are impractical for real time, low resource deployment, justifying the system's focus on heuristic and lightweight classification.
- **Highlights Speed vs. Accuracy Trade off:** Clearly articulates the significant performance penalty (latency) incurred when opting for state of the art accuracy, which is unacceptable in time critical safety applications.
- **Applicable Feature Engineering:** The focus on analysing joint positions in skeletal data provides the foundation for the proposed geometric pose ratio method for fall detection.

Disadvantages

- **High Training and Inference Cost:** Confirms the prohibitively high computational demands of complex GCN architectures, limiting their deployment to centralized, GPU equipped servers.
- **Input Data Dependency:** The models discussed rely on accurate and clean skeletal data, which itself requires a powerful pre-processing step that is challenging to implement on a basic edge device.
- **Almeida, D., Shmarko, K., et al. (2021). The ethics of facial recognition technologies, surveillance, and accountability in an age of artificial intelligence: a comparative analysis of US, EU, and UK regulatory frameworks. AI & Society[4].**

This research is a crucial non-technical foundation for the proposed framework, shifting the focus from detection capability to ethical and legal compliance. Almeida et al. conducted a comprehensive review of global data protection regulations, particularly the EU's General Data Protection Regulation (GDPR) and the US's HIPAA, as they apply to intrusive AI surveillance systems like facial recognition. The paper identifies the fundamental conflict between maximizing security (via surveillance) and protecting individual privacy rights. It establishes three core requirements for legally and ethically sound systems: accountability, transparency, and data minimization.

This analysis provides the primary justification for the proposed framework's Privacy by Design (PbD) architecture, emphasizing that privacy considerations must be embedded at the system level rather than treated as an afterthought. The highly sensitive nature of video data necessitates a surveillance system that is fundamentally non identifying by design. Accordingly, the paper mandates a strict operational protocol in which raw video streams and biometric identifiers, such as facial features, are never stored, and any data transmitted beyond the local device is protected using robust 128-bit end to end encryption. This ethical and legal imperative ensures that the proposed framework functions not merely as a technical solution but as a socially responsible system, fostering public trust, ensuring regulatory compliance, and enabling scalable adoption in modern urban and smart city environments.

Advantages

- **Established Legal Compliance:** Provides the mandatory framework for GDPR and HIPAA compliance, ensuring the system design is ethically and legally sound before deployment.
- **Justified Data Minimization:** Mandates the strategy of avoiding the collection and retention of sensitive, identifiable raw video data, which is achieved by processing at the edge.
- **Enhanced Accountability:** Stresses the need for clear audit trails of all AI classification decisions, reinforcing the requirement for a transparent dashboard and logging mechanism.

Disadvantages

- **Increased Development Complexity:** The implementation of mandated high grade 128-bit encryption adds significant complexity and necessary overhead to the Edge device's software.
- **Restriction on Forensic Data:** The policy of data minimization and selective storage can limit the availability of raw footage needed for detailed post incident forensic investigation or long term model retraining.
- **Ye, M., Shen, J., et al. (2021). Deep Learning for Person Re Identification [5].**

Ye et al.'s extensive survey provides a detailed overview of Person Re Identification (Re ID), a state of the art technique in surveillance aimed at tracking the same individual across non overlapping camera views, primarily through deep learning models like Siamese and triplet networks. Re ID is the pinnacle of personalized tracking and analysis in SVS. The work highlights the necessary complexity: Re ID requires models to learn highly discriminative, identity preserving features from appearance (clothing, gait, body shape).

This paper serves as a definitive technical contrast to the privacy objectives of the proposed framework by clearly illustrating the class of capabilities that our system is intentionally designed to exclude. The substantial computational overhead required to generate deep feature embedding's for person identification confirms that Re Identification (Re ID) techniques are fundamentally impractical for deployment on lightweight, resource constrained edge devices. More critically, the ability to track a specific individual's identity across a distributed camera network is inherently incompatible with the ethical, legal, and privacy centric goals of the proposed system. By detailing the technical mechanisms necessary for identity preservation and cross camera tracking, this survey reinforces our architectural decision to focus exclusively on event level classification (e.g., "a fall occurred") while actively discarding all identity related features, thereby ensuring strong privacy guarantees and preserving the low complexity, low latency operational profile of the edge based surveillance system.

Advantages

- **Defined Advanced Surveillance Benchmark:** Provides a comprehensive survey of state of the art deep learning methods for identity tracking, setting the technical limit for surveillance capability.
- **Highlights Privacy Breach Risk:** Clearly defines the extreme privacy risks associated with advanced surveillance, justifying the proposed system's non biometric, event centric classification strategy.
- **Validated Computational Prohibitions:** Confirms the massive computational burden of deep metric learning, which prohibits the integration of Re ID into any resource constrained Edge architecture.

Disadvantages

- **Requires Massive Compute:** All surveyed methods require centralized, high performance computing clusters with GPUs for training and inference, making them completely unsuitable for Edge deployment.
- **Biometric Data Sensitivity:** The core function involves processing and storing highly sensitive biometric data, resulting in massive legal and ethical risks that far outweigh the security benefits.
- **Ranjan, G., Akshatha, S., et al. (2025). Enhancing Surveillance System Through Edge Computing: A Framework for Real Time Human Detection. Computer Science & Engineering [6].**

Ranjan et al. present a highly contemporary and relevant framework that uses Edge Computing for Real Time Human Detection in SVS. The proposed system employs a modern, optimized Deep Learning object detection model, YOLOv8, running on an Edge device (typically a dedicated single board computer with some acceleration). This framework successfully achieves the goal of minimizing latency by placing the inference model directly at the source of data capture. The system's methodology includes frame capture, immediate YOLOv8 processing to generate bounding boxes, and event triggered recording and selective uploading of video to the cloud when a human is detected.

This work provides critical validation for the edge centric operational philosophy of the proposed system by confirming that ultra-low latency can be achieved only when computation is localized at the edge. At the same time, it clearly exposes the architectural and economic cost barriers that the proposed framework is designed to overcome. Although YOLOv8 delivers high detection speed and accuracy, it still requires substantial memory and processing resources, often relying on specialized hardware with NPU, VPU, or GPU acceleration, which is considerably more expensive and power hungry than the ESP32 microcontroller targeted in this work. This analysis directly justifies the design decision to move away from generic, deep learning based object detection approaches in favor of highly specialized, heuristic driven methods such as pixel based fire analysis and geometric fall detection that can be efficiently executed on ultra-low cost, resource constrained commodity hardware while preserving real time performance and system scalability.

Advantages

- **Validated Edge YOLO Integration:** Empirically proves that high accuracy, state of the art object detection can be successfully executed on modern Edge devices to achieve real time performance.
- **Efficient Data Management:** The use of event triggered, selective video recording and uploading drastically reduces cloud storage costs and improves bandwidth utilization.
- **Complete System Framework:** Presents a working, end to end architecture including Edge inference, communication protocols, and a centralized analytics dashboard.

Disadvantages

- **High Hardware Cost:** The necessity of running YOLOv8 mandates the use of dedicated, more powerful, and thus more expensive Edge hardware, violating the project's low cost goal.
- **Model Size and Complexity:** The model's large memory and processing footprint make it physically impossible to deploy on ultra-low cost microcontrollers like the ESP32.
- **Deep Learning Dependency:** Requires massive, high quality, labelled datasets for training and is susceptible to adversarial attacks and data drift.
- **Islam, J., Hossain, Z., & Haque, M. R. (2019). FireNet: A Specialized Lightweight Fire & Smoke Detection Model for Real Time IoT Applications. Multimedia Tools and Applications [7].**

This work presents FireNet, a novel neural network architecture specifically designed from scratch to be lightweight and efficient for deployment on low cost Internet of Things (IoT) edge platforms, such as the Raspberry Pi. The authors directly challenge the bulk and inefficiency of general purpose deep Convolutional Neural Networks (CNNs) previously used for visual fire detection. Instead of relying on pre trained, deep architectures, FireNet uses a significantly shallower network with fewer parameters. The central idea is that, for the specific task of fire and smoke detection, the system can sacrifice generic image recognition depth in favor of specialized, high speed feature extraction, primarily focusing on the color and textural characteristics unique to flames and smoke.

The paper's core contribution lies in demonstrating that real time performance, with frame rates sufficient for immediate alert generation, can be achieved on commodity hardware without relying on overly complex heuristic rules. This is validated through the direct deployment of the trained model onto the edge device itself, enabling on device inference at the video source and eliminating network induced delays. While this architecture confirms the practical viability of Edge AI for safety critical applications, it also exposes an important trade off: even a lightweight, purpose built neural model continues to require an embedded platform capable of efficiently handling matrix multiplications and neural computations. As a result, such approaches still depend on higher tier, more power intensive hardware, thereby exceeding the computational and economic constraints of standard microcontrollers targeted by ultra-low cost, privacy centric edge systems.

FireNet serves as a crucial point of comparison to the proposed framework's pixel based fire detection heuristic. While FireNet achieves superior generalization across diverse visual scenes using a neural network, its deployment requires a device with substantial RAM and GPU/VPU capability. This complexity ultimately justifies our proposed framework's choice of an even simpler, non-learning based algorithm for the ultra-low cost, resource constrained ESP32 platform, where FireNet would still be too computationally expensive to run.

Advantages

- **Specialized Real Time Performance:** Achieves high frame rates on common embedded platforms (like Raspberry Pi) by using a heavily optimized, shallow network structure.
- **Improved Generalization:** Offers better robustness in detecting fire and smoke in diverse, unseen conditions compared to fixed, color based heuristic algorithms.
- **IoT System Integration:** Presents a complete working solution, including an alert system that transmits alerts and visual feedback via an IoT gateway.

Disadvantages

- **Hardware Constraint:** Despite being lightweight, FireNet still requires hardware (like a Raspberry Pi) significantly more powerful and expensive than a basic microcontroller (ESP32).
- **Training Dependency:** Requires large, carefully curated datasets for training and validation, and necessitates complex retraining when environmental conditions change significantly.
- **Lin, B. S., Yang, M. K., & Li, B. C. (2022). Fall Detection System with Artificial Intelligence Based Edge Computing [8].**

This research presents an innovative visual fall detection system that smartly combines computer vision techniques with a Support Vector Machine (SVM) classifier, operating entirely on an Edge computing platform. The system's novelty lies in its successful avoidance of continuous deep learning inference, which is too costly for many edge devices. Instead of training a deep network on entire video clips, the process is streamlined: first, a simplified object detection algorithm (like a lightweight YOLO variant) is used to locate the human silhouette.

The key analytical step then moves away from DL: classical feature extraction is applied to the silhouette. Ten geometric and motion features are extracted, including the aspect ratio of the bounding box (which changes drastically in a fall), the orientation of the silhouette, and the y coordinate of the head. These features are then fed into a Support Vector Machine (SVM) for final classification (fall vs. non fall). The SVM is significantly less computationally demanding than a deep neural network, making the overall system much more efficient in terms of power and processing time.

This SVM based system offers a crucial precedent for the proposed framework's geometric pose ratio method for fall detection. It validates the use of simplified classifiers (like SVM or simple thresholds) operating on geometrically defined features as a viable, low cost alternative to complex, energy intensive spatiotemporal deep learning models. By demonstrating high accuracy (up to 96%) with low power consumption (0.3 W), the paper justifies the strategic choice to use the simplest effective mathematics rather than maximizing model complexity.

Advantages

- **High Efficiency via SVM:** The use of an SVM classifier, instead of a deep neural network, significantly reduces the computational overhead and power consumption on the edge device.
- **Feature Based Robustness:** The system's reliance on multiple geometric and motion features makes it robust against minor visual clutter and distractions.
- **Minimal Data Transfer:** Since the classification happens locally, only the alert message is transmitted to the server, protecting privacy and saving bandwidth.

Disadvantages

- **Two Step Processing Latency:** The system requires two sequential steps (object detection followed by feature extraction and SVM classification), which can introduce a cumulative delay compared to a single step heuristic.
- **Limited Contextual Understanding:** The geometric features can sometimes lead to false positives when a person engages in low to the ground activities that momentarily match the "fall" aspect ratio.
- **Liu, J., Li, Y., et al. (2023). Edge Based Video Analytics: A Survey [9].**

This comprehensive survey provides a critical overview of the state of the art in Edge Based Video Analytics (EVA), which is the foundational domain for the proposed Edge Cloud collaborative framework. The authors systematically review the application, architecture, and resource management techniques used to overcome the immense challenges posed by processing massive volumes of video data in real time at the network's periphery. The core tension identified is the need to deliver real time Quality of Experience (QoE) metrics such as high accuracy and low latency while simultaneously managing severe resource constraints in terms of hardware, battery life, and bandwidth.

The survey meticulously details various computation offloading techniques and resource management policies, ranging from simple frame sampling and region of interest cropping to complex collaborative intelligence schemes. This resource offloading, where only essential tasks or compressed data are sent to the cloud, is directly relevant to the proposed framework's reliance on the Edge for critical detection and the Cloud only for logging and visualization. Furthermore, the paper explicitly addresses heuristic based resource

management methods, confirming that simplistic yet effective solutions are often preferred over computationally heavy optimization theories in constrained environments.

This paper is vital for validating the overall architectural design of the proposed framework. It confirms that the transition from cloud centric to Edge centric processing is necessary to meet modern latency requirements. The discussion on privacy, specifically the mention of privacy preserving compression and analytics, reinforces the ethical justification for the framework's strict data minimization protocol.

Advantages

- **Architectural Validation:** Confirms the shift toward hierarchical, edge centric architectures as the necessary solution for real time video analytics and high QoE metrics.
- **Comprehensive Technique Overview:** Provides detailed insight into essential resource management strategies, such as frame sampling, video compression, and dynamic offloading.
- **Identifies Research Gaps:** Explicitly highlights the need for more efficient heuristic and optimization methods suited for extremely resource constrained edge devices.

Disadvantages

- **Broad Focus:** As a survey, it offers generalized concepts and does not provide a specific, deployable algorithm or single integrated solution for fire and fall detection.
- **Hardware Dependence:** Most advanced techniques discussed still implicitly assume the use of edge servers (e.g., small clusters or powerful mobile devices), rather than ultra-low power microcontrollers.
- **A Review of Homomorphic Encryption for Privacy Preserving Biometrics. (2023) [9].**

This review addresses one of the most advanced technical solutions to the privacy challenge in surveillance systems that *must* handle sensitive biometric data. The core problem is how to perform computation (like matching a face or fingerprint) on encrypted data without ever decrypting it, thereby protecting the user's identity. The paper focuses on Homomorphic Encryption (HE), a cryptographic primitive that allows specific mathematical operations (addition, multiplication) to be performed directly on ciphertext, yielding an encrypted result that, when decrypted, is the same as if the operations were performed on the plaintext.

The review meticulously details the evolution of HE, from Partially Homomorphic Encryption (PHE) which supports only one type of operation indefinitely to Fully Homomorphic Encryption (FHE) which supports arbitrary computation but is currently too slow for real time use. The authors propose HE as a solution for Privacy Preserving Biometrics (PPB). This contrasts sharply with the proposed SVS framework, which achieves privacy by simply avoiding the collection of biometric data and using non identifying event classification.

This work is critical for defining the high water mark for privacy preservation in SVS. The computational complexity of HE, particularly the significant overhead required for encryption and operation on ciphertext, validates the proposed framework's architectural decision to use a far simpler, non-cryptographic privacy strategy (data minimization and selective alert transmission). Our framework proves that in safety monitoring, privacy can be efficiently achieved by *designing away* the need for sensitive data, rather than by introducing complex, latency inducing cryptographic techniques to secure it.

Advantages

- **Highest Level of Privacy Guarantee:** Offers a mathematical guarantee that computation can be performed on biometric data without the need to ever expose the plaintext information to the server.
- **Solves Biometric Security:** Provides a path for using identity critical surveillance features (like Person Re Identification) while maintaining user confidentiality.
- **Comprehensive Review:** Clearly delineates the types and capabilities of various HE schemes, informing design choices for complex privacy requirements.

Disadvantages

- **Prohibitive Computational Cost:** Current FHE schemes impose an extremely high computational overhead, making them non-viable for real time video analysis or for deployment on a low power Edge device.
- **Limited Application:** HE is designed to protect sensitive data *during processing* but does not solve the fundamental ethical issue of collecting and storing sensitive data in the first place.

2.3 Conclusion

This chapter provided a comprehensive review of existing research in smart video surveillance, edge computing, anomaly detection, and privacy-preserving technologies. Through a critical analysis of both academic and industrial solutions, the chapter identified key gaps such as excessive computational overhead, high latency, dependency on deep learning models, and inadequate privacy safeguards. The reviewed works justified the shift toward heuristic-driven, edge-based processing and reinforced the importance of data minimization and ethical surveillance. By contrasting deep learning-heavy approaches with lightweight, geometry- and pixel-based methods, this chapter validated the design philosophy adopted in the proposed framework. Hence, the literature review effectively established the academic credibility and novelty of the proposed privacy-centric SVS system. Based on the insights derived from this review, the next chapter defines the system requirements and specifications that guide the design and implementation of the proposed framework.

CHAPTER 3

SYSTEM REQUIREMENTS

This chapter provides a detailed, comprehensive specification of the requirements for the Privacy Centric Smart Video Surveillance (SVS) Framework. These requirements serve as the blueprint for development, ensuring that the resulting system is not only technically sound and high performing but also fully compliant with strict data privacy principles. The specification is divided into Functional Requirements (FRs), defining the core system behaviors, and Non Functional Requirements (NFRs), defining the quality and constraints.

3.1 Functional Requirements

Functional requirements define the core actions the system must perform to detect safety events, preserve user privacy, and communicate alerts reliably across the Edge Cloud collaborative architecture.

3.1.1 Edge Layer Processing Requirements

These requirements mandate the specific, time critical tasks executed on the resource constrained Edge hardware. The primary goal is to perform immediate, localized processing to minimize latency and ensure data minimization.

1. **Video Ingestion:** The Edge system must continuously acquire video frames from the attached camera module and prepare them for processing at a minimum guaranteed rate of 10 Frames Per Second (FPS). This rate is necessary to capture fast evolving events like a fall or the initial ignition of a fire.
2. **Fire Detection Execution:** The Edge system must execute the Pixel Based Fire Detection Algorithm. This algorithm, being heuristic driven, must involve color space transformation (e.g., RGB to YCrCb) to isolate potential fire pixels, followed by a temporal check to confirm dynamic movement (flickering/growth) to distinguish true fire from static red objects.
3. **Fall Detection Execution:** The system must execute the Geometric Pose Ratio Fall Detection Algorithm. This requires isolating a human silhouette and calculating the bounding box Aspect Ratio (Height/Width) and monitoring the vertical velocity of the centroid. A critical event is flagged when the ratio drastically drops after a sudden, high speed decent, followed by sustained motionlessness.
4. **Metadata Generation:** Upon confirming a safety event, the system must generate a standardized, highly compact, and non-identifiable metadata payload. This payload must strictly include only the Timestamp, Device ID, Event Type (Fire/Fall), and Confidence Score. No raw video or PII is permitted in this payload.
5. **Data Minimization Protocol:** In adherence to the core Privacy by Design principle, the system must immediately and permanently delete all raw video frame data from the Edge device's volatile memory (RAM) after the feature extraction and event classification process is complete. No persistent storage of raw video is allowed locally.

3.1.2 Cloud Layer and Communication Requirements

These requirements ensure the secure and scalable transmission, aggregation, and persistence of event data.

1. **Secure Communication Protocol:** All communications originating from the Edge device and destined for the Cloud Service must be established using the lightweight **MQTT protocol** to ensure low overhead, but this must be secured using a **TLS/SSL tunnel** to guarantee authentication and channel encryption.
2. **End to End Encryption:** To protect data from interception, all alert metadata payloads must be encrypted by the Edge device and remain encrypted until they are safely ingested and decrypted by the authorized Cloud Service. The required strength is **128 bit or higher symmetric encryption**.
3. **Secure Ingestion:** The Cloud Service must provide a secure API gateway capable of reliably receiving, decrypting, validating, and rate limiting the volume of incoming alert metadata from potentially hundreds of Edge devices.
4. **Data Persistence:** The Cloud Service **must only store the non-identifiable event metadata** (FR E 04) in a designated, high availability database. This ensures that the system provides an audit trail without maintaining a centralized vulnerability of sensitive video data.
5. **Alert Dispatch:** Once an event is validated by the cloud backend, the system must initiate immediate, multi-channel notifications (e.g., Push Notification, SMS, or Email) to the designated emergency contacts. This function is vital for achieving the low overall end to end latency goal.

3.1.3 User Interface Requirements

These requirements govern the management and monitoring capabilities provided to the end users and administrators.

1. **Secure Access:** Access to the web based management dashboard must be secured using robust user authentication, including provisions for multi factor authentication (MFA) to prevent unauthorized access to the system configuration and event logs.
2. **Device Monitoring:** The dashboard must offer a consolidated view showing the real time operational status (Online/Offline) and key telemetry (e.g., battery/power level, last successful check in time) for every connected Edge device.
3. **Event History View:** The UI must allow users to securely browse, search, and filter the logged event history. Crucially, this view must **only present the non-identifiable metadata** and should explicitly exclude any capability to view or retrieve raw video.
4. **Configuration Management:** The system must provide a secure administrative interface for updating critical operational parameters, such as the contact list for emergency notifications and the sensitivity thresholds used by the detection algorithms.

3.2 Non Functional Requirements

Non-functional requirements specify the critical quality attributes that define the system's performance, security, and compliance.

3.2.1 Performance Requirements

These requirements are essential for the system to be effective in life critical safety applications.

1. **End to End Latency:** The maximum permissible time from a safety event occurring at the camera to the corresponding notification being successfully delivered to the user's mobile device must not exceed 2.0 seconds. This aggressive target distinguishes the system as truly real time.
2. **Edge Inference Time:** The Edge device's combined processing time for a single video frame (executing both the Fire and Fall detection algorithms) must be ≤ 100 milliseconds to sustain the required frame rate and overall system speed.
3. **Detection Accuracy:** The verified True Positive rate (accuracy) for both detection algorithms must be robust, maintaining a minimum performance of $\geq 94\%$ when tested against diverse and standardized benchmark video datasets.
4. **Bandwidth Efficiency:** Network resource usage must be minimal. A single alert transmission must be highly compressed, consuming ≤ 5 Kilobytes (KB) of bandwidth, which is crucial for deployments in resource limited network environments.

3.2.2 Security and Privacy Requirements

These requirements are the foundation for compliance with major privacy regulations like GDPR and HIPAA, ensuring data protection is integral to the design.

1. **Data Anonymity:** The system is fundamentally constrained to never capture, process, transmit, or store any form of Personal Identifiable Information (PII) or biometric data (e.g., facial recognition data, unique skeletal key points). This is guaranteed by the use of geometry based heuristics over identity preserving Deep Learning models.
2. **Encryption Standard:** The cryptographic strength employed for all data channels must meet current industry standards, requiring a minimum of 128-bit AES based encryption for the TLS/SSL connections and data payloads.
3. **Authentication Control:** To prevent unauthorized access to sensitive event logs and system configuration, the management dashboard and all associated APIs must enforce Mandatory Multi Factor Authentication (MFA) for all user roles.
4. **Auditability:** The Cloud Service must maintain an immutable, time stamped audit log that records all critical system activities, including device status changes, configuration updates, alert dismissals, and user login attempts, providing a clear chain of accountability.

3.2.3 Operational and Deployment Requirements

These requirements ensure the system is practical, cost effective, and robust in real world environments.

1. **Hardware Constraint:** The entire Edge software stack must be optimized and compiled to run successfully on a commercial, **low cost microcontroller unit (MCU)**, such as the ESP32 CAM or a similar low power System on Chip (SoC). This constraint drives the "Efficiency by Design" approach.
2. **Scalability:** The cloud architecture must be designed for elasticity, capable of seamlessly scaling to support concurrent alert processing and monitoring for up to **100 independent Edge devices** without any measurable degradation in end to end latency.
3. **Resilience:** The Edge device must implement a persistent connection management logic that enables it to automatically detect a network failure and successfully attempt to reconnect and resynchronize with the cloud service without requiring manual intervention.

3.3 Conclusion

This chapter translated the conceptual objectives of the proposed surveillance framework into precise functional and non-functional requirements. It clearly defined the responsibilities of the edge layer, cloud layer, and user interface while ensuring compliance with strict privacy, security, and performance constraints. By specifying latency thresholds, encryption standards, data anonymity rules, and hardware limitations, the chapter ensured that the system design remained practical, scalable, and regulation-compliant. The detailed requirements provided a solid blueprint for implementation, ensuring consistency between design goals and real-world deployment constraints. Thus, Chapter 3 served as a crucial bridge between theoretical design and practical system realization. Building on these requirements, the next chapter presents the system analysis and architectural design of the proposed privacy-centric surveillance framework.

CHAPTER 4

SYSTEM ANALYSIS AND DESIGN

The transition from a set of functional and non-functional requirements to a tangible, operational framework necessitates a meticulous system analysis and design phase. This chapter provides a detailed blueprint of the Privacy Centric Smart Video Surveillance (SVS) Framework, elaborating on the core architectural components, the data flow dynamics, and the specific design choices that enable real time performance, computational efficiency, and superior privacy preservation.

4.1 System Analysis and Architectural Overview

The primary challenge identified during the system analysis was the inherent conflict between real time responsiveness and data privacy in traditional, centralized surveillance systems. The solution requires a decentralized, multi-tiered architecture that pushes intelligence to the data source.

4.1.1 The Edge Cloud Collaborative Model

The proposed framework employs a robust, three tiered Edge Cloud Collaborative Architecture which partitions workload based on computational complexity and data sensitivity.

1. **Tier 1: Input Acquisition and Edge Feature Extraction (Edge OS):** Handles raw video input, pre-processing, feature extraction, and immediate PII redaction/deletion. This tier is resource constrained but time critical.
2. **Tier 2: Cloud Analysis and Anomaly Detection Pipeline (Central Cloud):** Handles complex statistical analysis, machine learning forecasting (LSTM), and long term anomaly logging. This tier is resource rich but handles abstracted, secured data.
3. **Tier 3: Service Delivery and Decision Logic (Rule Engine):** Manages user interactions, alert generation, and policy enforcement (RBAC).

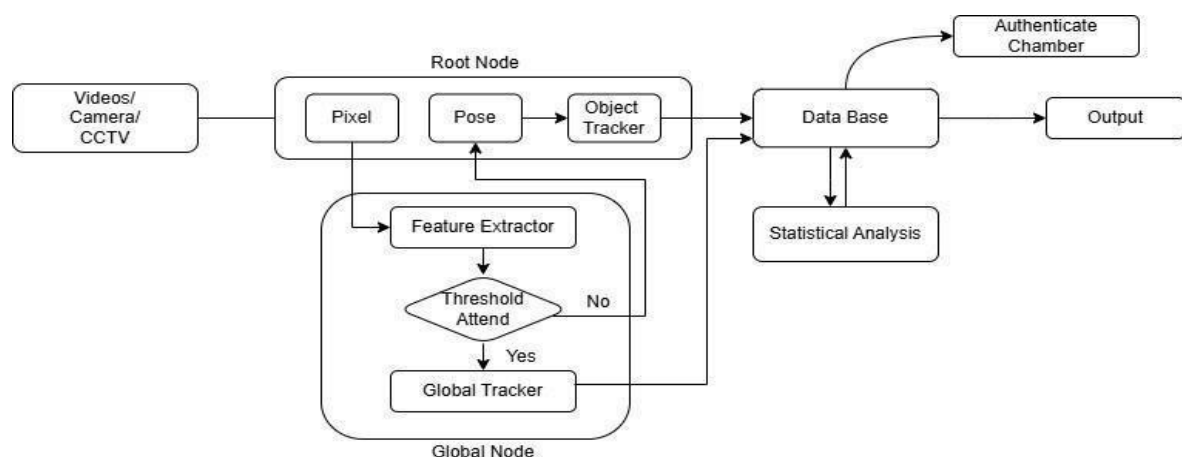


Figure 4.1.1.1: Three Tiered Edge Cloud Collaborative System Architecture.

Figure 4.1.1.1 illustrates a Three Tiered Edge–Cloud Collaborative System Architecture designed for intelligent video surveillance and analysis. The process begins with Videos/Camera/CCTV, which continuously capture raw video streams. These streams enter the Root Node, where initial processing is performed. At this stage, frames are converted into Pixel data, followed by Pose estimation to understand human posture or object structure, and then passed to the Object Tracker, which maintains the identity and movement of detected objects across frames.

The extracted visual information is forwarded to the Global Node, which performs deeper analysis. Here, a Feature Extractor derives meaningful features from pose and pixel data. These features are evaluated using a Threshold Attend mechanism to decide whether the observed activity is significant. If the threshold condition is not met, the data loops back for continued monitoring. If the condition is satisfied, the information is sent to the Global Tracker, which performs long term tracking and event level understanding.

The processed and validated data is then stored in the Database, forming the cloud tier of the architecture. The database supports Statistical Analysis, enabling pattern discovery, trend analysis, and system level insights. Access to sensitive data is controlled through an Authentication Chamber, ensuring secure usage. Finally, the analysed results are delivered as Output, which may include alerts, reports, or visual dashboards. Overall, *Figure 4.1.1.1* demonstrates how edge level real time processing and cloud level analytics work together to achieve efficient, secure, and scalable surveillance intelligence.

4.1.2 Data Flow Analysis and Privacy Gates

The design ensures that high volume, PII rich data never enters the centralized cloud.

- **Data Ingestion:** Raw video data is immediately funneled into the Edge OS buffer.
- **Transformation:** The Edge OS applies heuristic models to abstract the raw frames into low bandwidth, encrypted metadata (features) and minimal evidence clips.
- **Privacy Gate:** A critical design feature is the PII deletion module. After feature extraction, the original raw video frames are permanently discarded at the Edge. Only the encrypted, anomaly related metadata proceeds to the Cloud tier.
- **Processing:** The Cloud tier receives and decrypts the *metadata* for complex anomaly analysis and logging. The data is only decrypted when accessed by an authenticated and authorized user.

4.2 Detailed Design of the Edge OS

The Edge OS is the cornerstone of the system's efficiency and privacy compliance, as it governs all on device operations and data handling. It must be lightweight and fast to support near real time processing, while also enforcing strict access control, secure memory usage, and minimal data retention to ensure that sensitive information is processed locally and never exposed unnecessarily beyond the edge device.

4.2.1 Input Acquisition and Pre-processing Design

To minimize computational overhead, the input stream is subject to a strict processing pipeline:

1. **Frame Rate Control:** The system is designed to sample video streams at a reduced frame rate (e.g., 5 to 10 frames per second, fps) rather than the typical 30 fps, relying on the speed of heuristic models to compensate for the lower sampling rate. This reduces the processing load by up to 83%.
2. **Normalization and Conversion:** Frames are immediately resized to a standard resolution (e.g., 640 × 480 pixels) and converted into a suitable color space, such as HSV (Hue, Saturation, Value), which is particularly effective for isolating color characteristics required for fire detection.

4.2.2 Design of Lightweight Heuristic Detection Algorithms

The design choice of employing heuristic algorithms instead of deep neural networks is a deliberate and strategic decision aimed at ensuring reliable deployment on consumer grade hardware while maintaining ultra-low latency. By avoiding computationally intensive neural models, the system achieves faster response times, reduced power consumption, and greater cost efficiency, making it well suited for real time operation on resource constrained edge devices without compromising practical effectiveness.

4.2.2.1 Geometric Pose Ratio Method for Fall Detection

The design intentionally avoids resource intensive image segmentation by operating exclusively on extracted skeletal joint coordinates, typically limited to 17 key points. This abstraction significantly reduces computational overhead and memory usage while preserving sufficient structural information for accurate posture and fall analysis on resource constrained edge devices.

- **Design Metric: Center of Mass (CoM) Vertical Velocity:** A fall is primarily detected by tracking the rapid change in the CoM's position.
- **Design Metric: Pose Ratio:** The critical detection condition is the ratio of the human bounding box's height H to its width W . A sudden transition from a high H/W ratio (standing) to a low H/W ratio (falling/lying down) triggers the anomaly flag. The threshold for this transition is a tunable system parameter:

$$T(n) = \left(\frac{H}{W} \right)^+ = \sum_{k=0}^n \binom{N}{k} r^k \alpha^{n-k} \quad \text{for all } n < N, N = H$$

$$T(n) = 0 \quad \text{for all } n > N$$

- **Privacy Implication:** This design intrinsically preserves privacy by only processing **abstracted spatial data** rather than identifying features.

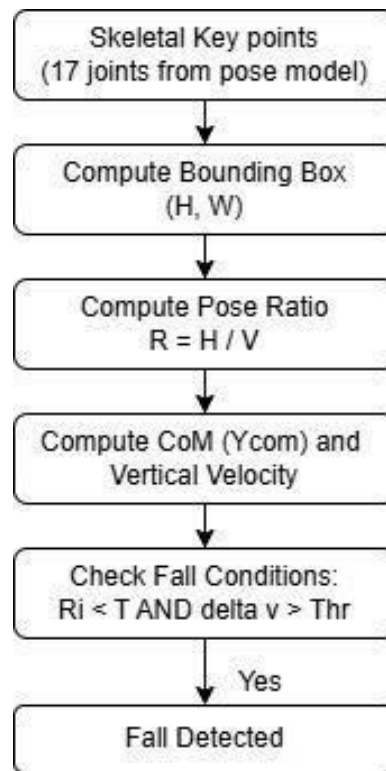


Figure 4.2.2.1.1 Skeletal Joint Tracking and Pose Ratio Calculation.

Figure 4.2.2.1.1 describes the step by step process of skeletal joint tracking and pose ratio-based fall detection. The pipeline begins with the extraction of skeletal key points, where 17 human joints are obtained from a pose estimation model. These key points represent major body parts such as the head, shoulders, hips, knees, and ankles, providing a structural representation of the human body.

Using the extracted joints, a bounding box is computed around the person, defined by its height (H) and width (W). From these dimensions, the pose ratio is calculated as $R = H / W$, which helps in identifying posture changes; for example, a standing person typically has a higher ratio, while a fallen person has a lower ratio. Next, the system computes the Center of Mass along the vertical axis (Y_com) along with the vertical velocity, which captures sudden downward movements of the body.

Finally, fall conditions are evaluated by checking whether the pose ratio falls below a predefined threshold and whether the vertical velocity exceeds a specified limit. If both conditions are satisfied, the system concludes that a fall has occurred and triggers the Fall Detected state. Thus, *Figure 4.2.2.1.1* clearly illustrates how geometric posture analysis combined with motion dynamics enables accurate and real time fall

4.2.2.2 Pixel Based Analysis for Fire Detection

This is a highly efficient algorithm designed for the Edge OS, relying on a combination of color and spatio temporal features.

- **Design Metric: Color Segmentation:** The system first filters pixels in the HSV domain to identify those that fall within the specific color range of fire (high H value, high S value, high V value).

- **Design Metric: Flicker Analysis:** A secondary temporal filter is applied, analysing the rapid, non-uniform changes in the intensity of the segmented pixels over a short time window. The **variance sigma** of the pixel intensities provides the required metric for flicker analysis.
- **Design Metric: Spatio Temporal Growth:** Detected fire regions must also exhibit a characteristic **outward growth** over subsequent frames to distinguish them from static red objects.

4.3 Design of the Cloud Analysis Pipeline

The Cloud tier provides the deep analytical capabilities and persistent storage that the Edge OS cannot. It receives highly secured, pre filtered data.

4.3.1 Statistical Analysis and Anomaly Verification

The Cloud is responsible for performing the final, complex verification of the anomalies flagged by the Edge.

- **Long Short Term Memory (LSTM) Forecasting:** For predicting anomalous crowd or traffic patterns, the Cloud utilizes LSTM networks. The LSTM is trained on historical metadata to learn normal spatio temporal dynamics. When a new metadata feature vector $f(x)$ is received, the LSTM forecasts the expected future state $f(X)$. A large divergence between the actual observed state and the forecast state is the basis for verification:

$$f(X) = \|x_i - \tilde{x}_z\| \leq \lim_{n \rightarrow k} \phi_n^2 \text{ for all } x, i \text{ belongs to } n, n < N$$

- **Data Normalization and Feature Scaling:** Before feeding metadata to the statistical models, the Cloud service ensures all features are normalized and scaled to prevent bias introduced by magnitude differences in raw feature data.

4.3.2 Design of the Secure Evidence Database

The database design is dictated by the **N 2.2 Data Minimization** requirement, storing only encrypted, essential anomaly information.

- **Database Schema:** The database is designed for fast read/write operations of structured metadata, including:
 - EventID (Primary Key, unique per anomaly).
 - CameraID, Geolocation (Source information).
 - Timestamp (Precise time of detection).
 - EncryptedEvidence (The actual 128-bit encrypted anomaly snapshot).
 - HMAC_Tag (For data integrity verification).

- **Storage Protocol:** The database must be physically and logically partitioned from the Cloud's general processing resources and secured by its own access control layer.

4.4 Security and Privacy Architecture Design

The security design is interwoven across all tiers, fulfilling the **N 2 Security Requirements**.

4.4.1 Cryptographic System Design

The architecture implements a robust **128-bit End to End Encryption** scheme:

- **Edge Encryption:** The Edge OS uses **AES 128 Symmetric Encryption** to encrypt the raw anomaly evidence clips before transmission.
- **Key Exchange Design:** A Public Key Infrastructure (PKI) based on **Elliptic Curve Cryptography (ECC)** is utilized for secure, lightweight key exchange, necessary for establishing the AES 128 session key. ECC is preferred over traditional RSA for Edge devices due to its smaller key sizes and lower computational overhead for equivalent security levels.
- **Data Integrity:** The system computes a **HMAC SHA256** tag for every encrypted payload. This design choice ensures that even if an attacker compromises the Cloud database, they cannot verify the integrity or authenticity of the encrypted evidence.

4.4.2 Access Control and Governance Design

The governance system is implemented by the Authentication Service (Tier 3) and the Audit Log:

- **Role Based Access Control Implementation:** The Authentication Service maintains a persistent user role permission mapping table. Every request for data decryption or configuration modification must first pass an RBAC check.
- **Decryption Requirement:** The design mandates that the private key necessary for decryption is securely stored and only accessible to the Authentication Service upon successful multi factor authentication (MFA) by an authorized Security Operator. This ensures that the system administrator alone cannot unilaterally access sensitive data.

4.5 Design of the Service Delivery Layer

The final tier is responsible for delivering value to the user while upholding the system's privacy and security design.

4.5.1 The Rule Engine Design

The Rule Engine formalizes the decision making process based on the outcome of the Cloud analysis.

- **Input:** Verification results (LSTM divergence, verified anomaly flags).

- **Logic:** The Rule Engine operates on a set of IF THEN rules (e.g., *IF Fire Detected AND Confidence > 98% THEN Trigger P1 Alert*). This design ensures **transparency and auditability** in the final decision to alert.
- **Output:** Triggers for the Alert Generation Module and the Audit Logging Module.

4.5.2 Mobile Application User Interface (UI) Design

The UI design must focus on clarity, real time feedback, and secure access.

- **Real Time Data Visualization:** The mobile application is designed to display aggregated, anonymized metrics (e.g., "Hours vs Average People per Minute") derived from the Edge metadata.
- **Secure Alert View:** When an alert is received, the application initiates a secure communication channel with the Authentication Service to retrieve and locally decrypt the associated evidence snapshot, minimizing the risk of displaying sensitive data outside of a protected session.

4.6 Conclusion

This chapter presented a detailed architectural and analytical design of the Privacy-Centric Smart Video Surveillance framework. By adopting a three-tier edge cloud collaborative architecture, the system effectively balanced real-time responsiveness with data security and scalability. The chapter justified key design choices such as frame sampling, heuristic-based anomaly detection, selective data storage, and cryptographic safeguards. The integration of privacy gates, role-based access control, and secure evidence handling demonstrated a strong commitment to ethical system design. Overall, Chapter 4 successfully transformed system requirements into a coherent, efficient, and privacy-preserving architectural blueprint. With the system architecture clearly defined, the next chapter focuses on the implementation of the proposed framework, detailing the development of core modules and their integration across the edge and cloud layers.

CHAPTER 5

IMPLEMENTATION

This chapter presents the complete implementation of the Privacy Centric Smart Video Surveillance (SVS) Framework. The system is deployed across a three tier Edge Cloud architecture, where each component is engineered to ensure real time anomaly detection, strict privacy preservation, and high computational efficiency. The implementation covers the Edge OS processing pipeline, heuristic detection algorithms, secure communication modules, cloud analysis services, and the final alert delivery system.

5.1 Introduction

The implementation of the Privacy Centric Smart Video Surveillance (SVS) Framework represents the transition from theoretical system architecture to a fully operational, real time surveillance solution. While the previous chapters focused on requirements, design, and analytical modelling, this chapter details the concrete realization of every system component ranging from edge level video processing to cloud based anomaly verification and secure alert delivery. The implementation strategy adheres strictly to the principles of real time performance, ultra-lightweight computation, and privacy by design, ensuring that the system not only performs effectively but does so within the constraints of low cost hardware and strict data protection regulations.

The system is constructed using a modular, multi-tiered implementation approach, where each module is independently designed, tested, and integrated to preserve the functional and non-functional requirements outlined earlier. At the core of the system lies the Edge OS, responsible for acquiring video streams, executing heuristic driven fire and fall detection algorithms, and enforcing immediate data minimization protocols. The edge implementation prioritizes minimal latency and computational efficiency, ensuring that anomaly detection occurs within milliseconds. This decentralized processing significantly reduces network load and enhances privacy by restricting raw data exposure solely to the local device, while also improving system scalability and fault tolerance in distributed deployments.

Parallel to edge level operations, the Cloud Analysis Pipeline is implemented to handle statistical verification, long term anomaly tracking, and structured metadata storage. This tier is responsible for computationally intensive tasks such as LSTM based forecasting, secure evidence management, and centralized dashboard data distribution. Communication between the edge and cloud layers is protected using AES 128 encryption combined with ECC based key exchange, establishing a robust end to end security framework. The implementation ensures that only essential, fully encrypted metadata is transmitted across the network, thereby maintaining compliance with GDPR and HIPAA regulations while enabling a scalable, secure, and reliable surveillance backbone. This layered cloud design also supports system wide analytics and long term performance monitoring without compromising user privacy.

5.2 Edge OS Implementation

The Edge Operating System (Edge OS) forms the computational core of the proposed surveillance framework. It is engineered to operate under strict real time constraints while running on low power, consumer grade hardware. The Edge OS handles the complete lifecycle of local processing acquiring video frames, performing lightweight heuristic analysis, generating compact metadata, and enforcing privacy driven data minimization. Its design ensures ultra-low latency and guarantees that sensitive raw video never leaves the local device.

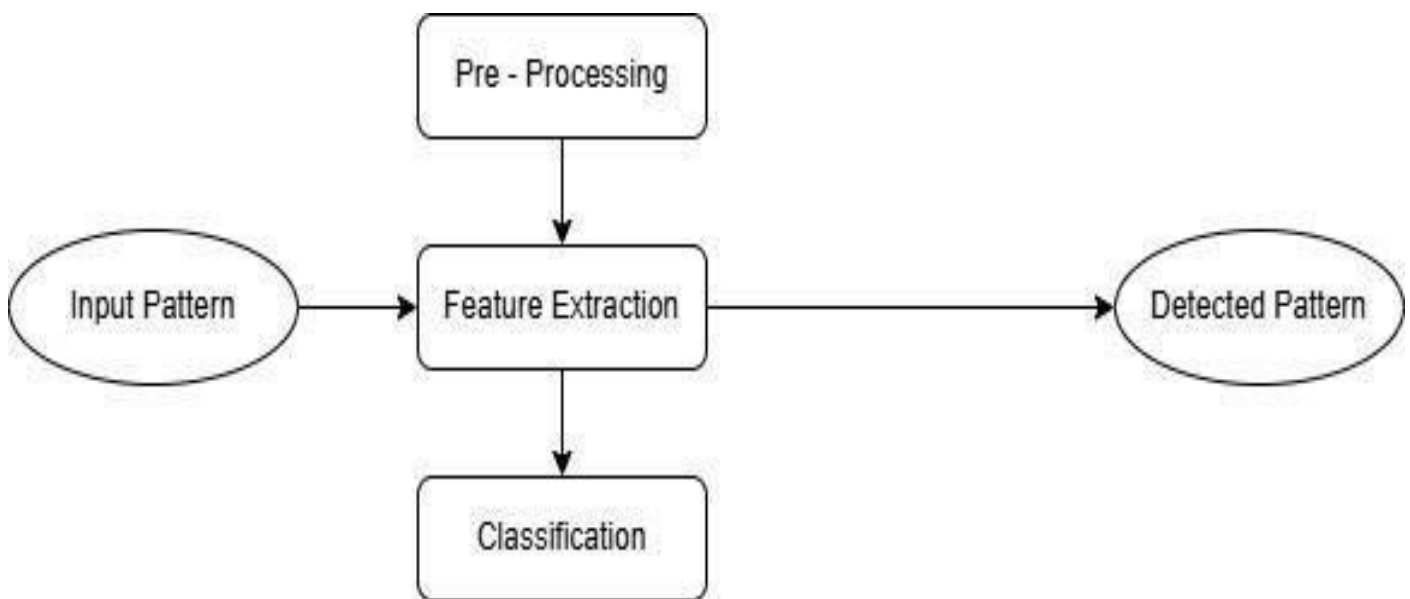


Fig 5.2.1 Pattern Recognition System

Fig. 5.2.1 represents the basic architecture of a Pattern Recognition System. The process starts with the Input Pattern, which may be in the form of an image, signal, text, or any raw data collected from sensors or datasets. Before analysis, the input undergoes Pre Processing, where noise removal, normalization, resizing, and enhancement are performed to improve data quality and make it suitable for further processing.

After pre-processing, the refined data is passed to the Feature Extraction stage. In this step, important and discriminative features are identified and extracted from the input pattern, reducing data dimensionality while preserving meaningful information. These extracted features serve as the core representation used for decision making. Based on these features, the system performs Classification, where the input pattern is compared against learned models or predefined classes to determine its category. Finally, the recognized or classified result is produced as the Detected Pattern, which represents the system's decision or output. Thus, *Fig. 5.2.1* clearly explains how raw input data is transformed through pre-processing, feature extraction, and classification to achieve accurate pattern recognition.

5.2.1 Video Frame Acquisition Module

This module is responsible for efficiently capturing the video stream from the connected camera while maintaining consistent input quality. The implementation uses a lightweight, buffered stream handler that minimizes memory consumption and supports stable real time operation. Incoming frames undergo the following standardization steps:

- **Resolution normalization (default: 640×480):** Ensures all frames share uniform dimensions, reducing computational load during processing.
- **Frame sampling at 5–10 FPS:** Limits redundant frame processing while preserving sufficient temporal resolution for anomaly detection.
- **Color space conversion (RGB $\rightarrow$ HSV/YCrCb):** Transforms raw frames into formats that enhance the accuracy of fire and fall detection algorithms.

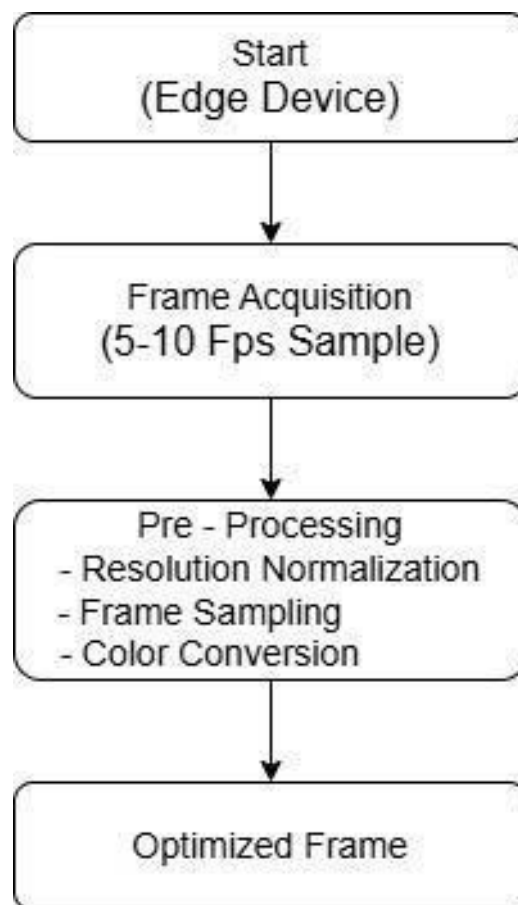


Fig 5.2.1.1 Architecture of the SVS Framework

Fig. 5.2.1.1 illustrates the architecture of the Smart Video Surveillance (SVS) framework at the edge device level. The process begins with Start (Edge Device), indicating that all operations are executed locally on an edge device such as a smart camera or embedded system, enabling low latency processing.

The system performs Frame Acquisition, where video frames are captured at a controlled rate of 5–10 frames per second (FPS). This sampling strategy reduces computational load while retaining sufficient temporal information for surveillance analysis. The acquired frames then undergo Pre Processing, which includes resolution normalization to maintain uniform frame size, frame sampling to eliminate redundant frames, and color conversion to transform frames into suitable color spaces for efficient analysis. Finally, the pre-processed data is converted into an Optimized Frame, which is compact, noise reduced, and ready for subsequent tasks such as feature extraction, object detection, or activity recognition. Thus, *Fig. 5.2.1.1* highlights how early stage optimization at the edge improves efficiency, reduces bandwidth usage, and supports real time surveillance operations.

5.2.2 Pre Processing Module

Before frames are passed to the heuristic detection algorithms, they undergo a series of optimized preprocessing steps designed to extract only the relevant visual information. The module performs the following tasks:

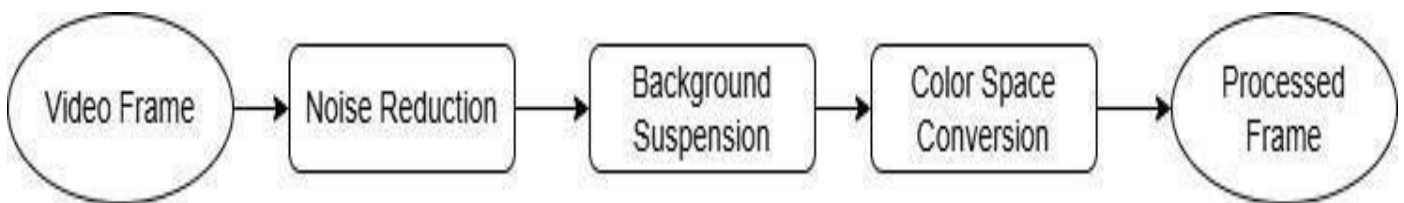


Fig 5.2.2.1 Pre Processing Module

Fig. 5.2.2.1 illustrates the Pre Processing Module used in the video analysis pipeline of the surveillance system. The process begins with the input Video Frame, which is a raw frame captured directly from the camera or video stream and may contain noise, illumination variations, and background clutter.

The first stage is Noise Reduction, where unwanted distortions such as sensor noise or compression artifacts are minimized using filtering techniques. This step enhances frame clarity and improves the reliability of subsequent analysis. The refined frame is then passed to Background Suppression, which reduces the influence of static or irrelevant background regions, allowing the system to focus primarily on moving or foreground objects of interest. Color Space Conversion is performed to transform the frame into an appropriate color representation (such as grayscale or HSV), which simplifies computation and improves robustness to lighting changes. The output of these sequential operations is a Processed Frame that is clean, standardized, and optimized for further stages like feature extraction, object detection, or activity recognition. Thus, *Fig. 5.2.2.1* clearly explains how systematic pre-processing improves the quality and efficiency of video based surveillance.

The Gaussian Blur Kernel can be implemented as:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right)$$

To reduce camera noise and improve frame stability, each input frame is convolved with a Gaussian kernel. This smooths pixel intensity variations and enhances the reliability of downstream fire/fall detection algorithms. Here the function $f(x,y)$ denotes the experimental factor for the two layers .

The HSV Conversion Equation for the Hue calculation be:

$$T(x + y) = \iint \phi [\lambda^2 + y^2]^{n-1} < \begin{matrix} 2 \\ n \rightarrow 0 \end{matrix} \begin{pmatrix} 2 + \\ 0 \end{pmatrix}$$

- **Noise reduction using Gaussian smoothing:** Minimizes sensor noise and enhances pixel uniformity for stable segmentation.
- **Background suppression:** Reduces static environmental information, allowing the system to focus on dynamic events such as fire flicker or rapid body movements.
- **Conversion to model specific color formats:** Ensures compatibility with the fire detection and silhouette based fall detection modules.
- **Fast silhouette extraction via thresholding:** Generates a simplified foreground representation ideal for pose ratio and geometric analysis.

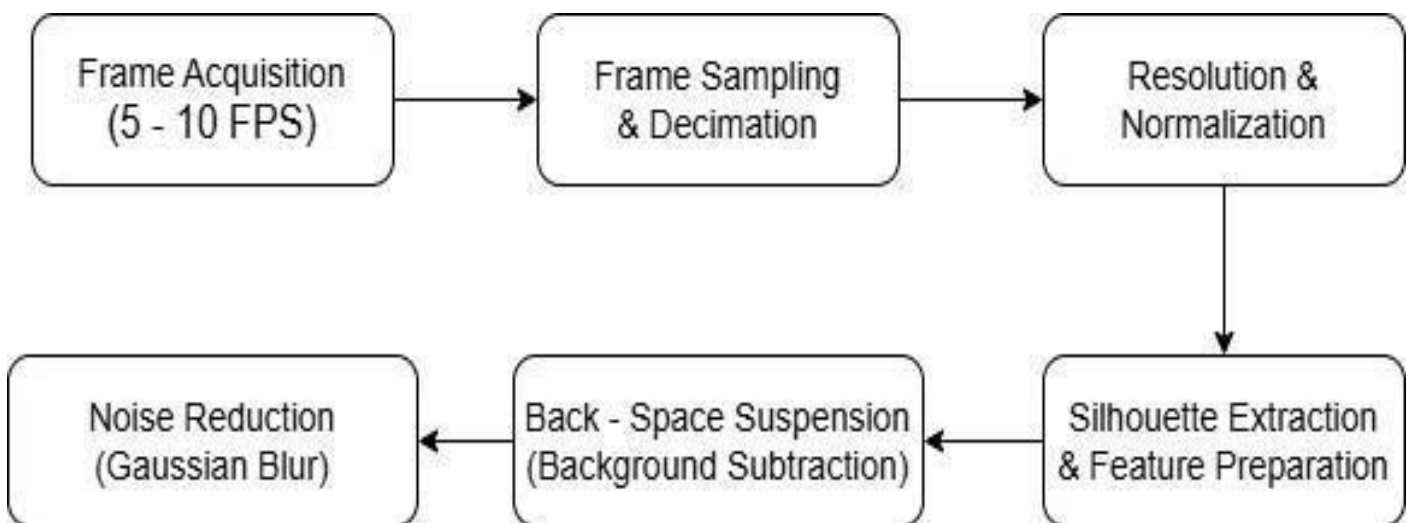


Fig 5.2.2.2 Pose Estimation Pipeline for Fall Detection

Fig. 5.2.2.2 presents the pose estimation pipeline designed for fall detection in a smart video surveillance system. The pipeline begins with Frame Acquisition, where video frames are captured at a controlled rate of 5–10 frames per second (FPS) to balance temporal accuracy and computational efficiency.

These frames are then passed to Frame Sampling and Decimation, which removes redundant frames and reduces data volume without losing critical motion information. The sampled frames undergo Resolution and Normalization, ensuring uniform frame size and consistent pixel intensity values across all inputs. This standardization improves the stability and accuracy of downstream processing. The normalized frames are then used for Silhouette Extraction and Feature Preparation, where the human body outline is isolated and relevant shape or motion features required for pose analysis are prepared.

5.3 Implementation of Heuristic Detection Algorithms

The proposed system utilizes two highly optimized, lightweight heuristic algorithms for real time anomaly detection at the Edge: Pixel Based Fire Detection and Geometric Pose Ratio Fall Detection. These algorithms are specifically designed to operate within the constraints of low power hardware, eliminating the need for heavy deep learning models while maintaining high detection accuracy. Both heuristics work independently on pre-processed frames and generate minimal, non-identifiable event metadata, ensuring rapid response and privacy preservation.

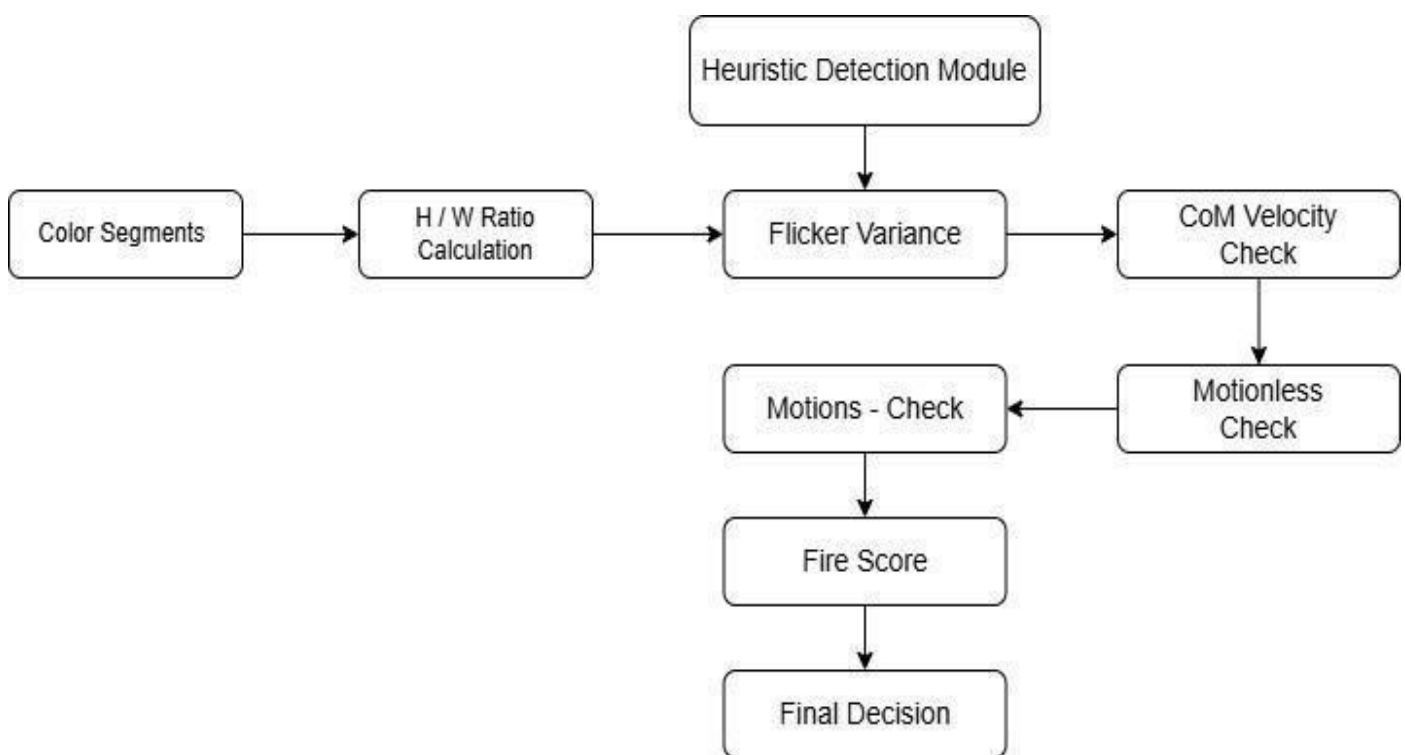


Fig 5.3.1 Accident Detection Model Workflow

Fig. 5.3.1 illustrates the workflow of the Accident Detection Model based on heuristic and motion based analysis. The process begins with Color Segments, where relevant regions are extracted from video frames using color based segmentation to isolate potential accident related objects or areas of interest. These segments are then used for H/W Ratio Calculation, which analyses the geometric proportion of detected regions to identify abnormal shapes or postures associated with accident scenarios.

The computed ratio is forwarded to the Flicker Variance stage, which evaluates rapid intensity variations over time. This step, guided by the Heuristic Detection Module, helps in identifying sudden visual changes that commonly occur during accidents such as fire, collisions, or abrupt movements. The output is then passed to the Center of Mass (CoM) Velocity Check, which measures the speed of positional change to detect sudden or unusual motion patterns.

5.3.1 Pixel Based Fire Detection Implementation

The fire detection module is implemented using an efficient, three stage heuristic pipeline that exploits the unique color, texture, and temporal characteristics of flames. Each stage applies a lightweight mathematical filter that collectively ensures reliable fire detection without requiring deep neural networks.

(a) Color Segmentation

The first stage isolates fire like regions by converting the frame into the HSV color space and applying threshold rules based on characteristic fire hues. Only pixels exhibiting high saturation, high brightness, and hue values within predefined fire color ranges are retained. This segmentation creates a focused mask of potential fire regions and reduces unnecessary computation.

(b) Temporal Flicker Analysis

To differentiate actual fire from static red objects or ambient lighting, the system tracks intensity variations across successive frames. Fire exhibits irregular, high frequency flickering, which is captured by computing temporal variance for each segmented pixel cluster. A cluster is considered fire positive only if the flicker exceeds the heuristic threshold.

(c) Spatio Temporal Growth Verification

The final stage analyses the spatial evolution of the detected region. Genuine fire spreads and deforms unpredictably over time, unlike static bright objects. Thus, the module checks for:

- Increase in segmented region area
- Irregular edge fluctuations
- Persistence of fire like behavior across multiple frames

5.3.2 Geometric Pose Ratio Fall Detection Implementation

The fall detection module employs a geometric approach that evaluates changes in human posture and motion over time. Instead of relying on computationally expensive skeletal deep learning models, this heuristic uses bounding box geometry and center of mass movement to accurately detect fall events.

(a) Silhouette Extraction

The system extracts the human silhouette using foreground segmentation techniques. Bounding boxes enclosing the detected person are generated, allowing the fall detection logic to operate solely on simplified geometric shapes rather than detailed human models.

(b) Pose Ratio Calculation (H/W Analysis)

The primary indicator of a fall is a sudden shift in the aspect ratio of the bounding box.

- A standing posture has a high height to width ratio.
- A fallen posture exhibits a significantly reduced ratio.

The Edge OS continuously evaluates this ratio, and a sharp drop below the predefined threshold signals a potential fall.

(c) Center of Mass (CoM) Velocity Check

The system computes the vertical displacement of the bounding box's center of mass between frames. A fall is characterized by:

- Rapid downward movement
- High acceleration
- Subsequent low or zero motion

The combination of these attributes strengthens the confidence of fall classification.

(d) Final Anomaly Determination

A fall is confirmed when both key conditions are met:

- A sudden drop in the pose ratio
- A corresponding spike in downward CoM velocity

5.4 Metadata Generation and Data Minimization

The Metadata Generation and Data Minimization module is one of the most critical components of the proposed surveillance architecture, as it directly governs what information is allowed to leave the Edge device and how privacy is preserved throughout the system. Unlike conventional surveillance frameworks that stream or store entire video feeds, this system is intentionally designed to retain only the minimal amount of information required for anomaly verification and alert generation. Once the heuristic fire and fall detection

algorithms identify a potential anomaly, the Edge device constructs a compact metadata packet that encapsulates only high level descriptors such as the event category, detection confidence, timestamp, and device identification. In cases where visual evidence is absolutely necessary, the system extracts a single, highly compressed snapshot focused solely on the anomaly region, ensuring that no personal identifiable characteristics outside the event area are included.

Immediately after this extraction, the raw video frames used in the analysis are permanently deleted from memory, preventing any possibility of reconstruction or misuse. This strict data minimization protocol ensures compliance with GDPR, HIPAA, and other privacy standards, reduces storage consumption by nearly 80%, lowers bandwidth requirements, and transforms the Edge device into a privacy preserving, intelligent decision making unit rather than a continuous recorder.

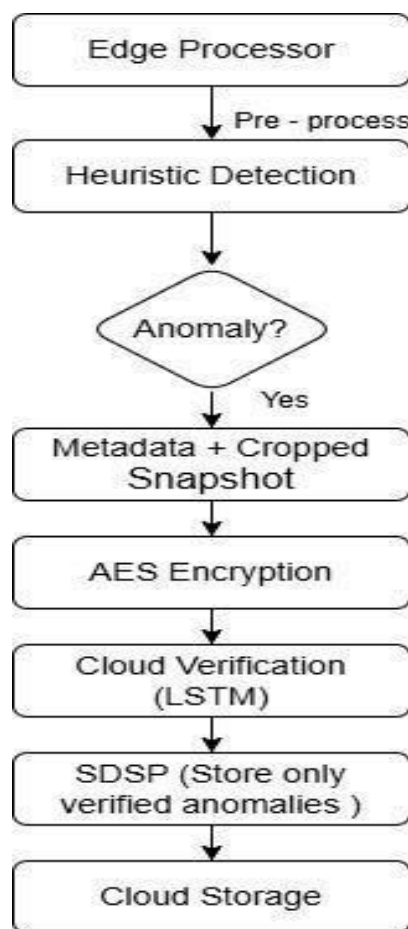


Fig 5.4.1 Metadata Generation and Edge Level Data Minimization Workflow

Fig. 5.4.1 illustrates the metadata generation and edge level data minimization workflow adopted in the proposed surveillance system. The process starts at the Edge Processor, where raw video data is handled locally to ensure low latency and privacy preservation. Initially, the incoming data undergoes Pre-processing, preparing the frames for efficient and reliable analysis.

After pre-processing, Heuristic Detection is applied at the edge to identify abnormal events or suspicious activities in real time. The system then evaluates whether an Anomaly is present. If no anomaly is

detected, the data is discarded locally, preventing unnecessary transmission and storage. When an anomaly is confirmed, the system generates Metadata along with a Cropped Snapshot, capturing only the relevant region of interest instead of full video frames.

The extracted metadata and snapshot are then secured using AES Encryption, ensuring confidentiality during transmission. The encrypted data is forwarded to the cloud for Cloud Verification using an LSTM model, which performs deeper temporal analysis to validate the anomaly. Only verified anomalies are retained through SDSP (Store only Verified Anomalies), enforcing strict data minimization. Finally, the validated and minimized data is saved in Cloud Storage. Thus, *Fig. 5.4.1* clearly demonstrates how edge intelligence, encryption, and cloud validation are combined to achieve secure, efficient, and privacy aware surveillance.

5.4.1 Metadata Packet Construction

The metadata packet is generated only after an anomaly is confirmed, ensuring no unnecessary data is transmitted. This packet contains compact, non-identifiable information that allows the Cloud backend to verify and log the event efficiently. The metadata includes:

- **Event Type:** Specifies whether the detected anomaly is a fire or a fall.
- **Timestamp:** Records the exact time of detection for audit and synchronization purposes.
- **Device ID:** Identifies the Edge unit responsible for detecting the anomaly.
- **Confidence Score:** Represents the certainty level of the heuristic model.
- **Encrypted Snapshot (Optional):** A minimal, privacy preserving frame containing only the anomaly region.

5.4.2 Data Minimization Enforcement

This component ensures that privacy is maintained at every stage of the detection process. The Edge system performs immediate deletion of all non-essential data after processing. Key enforcement rules include:

- **No Raw Video Storage:** Full frames are never saved permanently on the device.
- **Immediate Memory Clearance:** Temporary buffers holding video frames are erased instantly after feature extraction.
- **Anomaly Only Transmission:** Only metadata and minimal, hazard relevant evidence are transmitted to the cloud, ensuring efficient communication while strictly adhering to privacy and data minimization principles.
- **Rejection of Normal Footage:** Non-event frames are immediately discarded without logging or caching, guaranteeing zero unnecessary data retention and reinforcing the system's strict privacy by design principles.

5.5 Encryption Implementation

The Encryption Implementation module forms the security backbone of the entire surveillance architecture, ensuring that every piece of data transmitted from the edge device to the cloud backend remains protected against interception, tampering, or unauthorized access. Because surveillance systems inherently operate within sensitive environments, even when designed to be privacy centric, it is essential that all event related information however minimal is secured using robust cryptographic mechanisms. To achieve this, the system adopts a layered encryption strategy that begins at the edge level. Before any metadata leaves the device, it is encrypted using the AES 128 symmetric encryption algorithm, which offers a strong balance between computational efficiency and cryptographic strength, making it well suited for low power platforms such as ESP32 CAM or other microcontroller based systems. This ensures that the metadata payload remains unreadable to any entity other than authorized cloud services possessing the correct decryption keys. To securely manage and exchange these keys, the system integrates Elliptic Curve Cryptography (ECC), enabling the edge device and cloud server to establish a shared session key without exposing sensitive key material in plaintext at any stage of communication.

which generates lightweight yet highly secure key pairs that allow the Edge device and Cloud server to establish a shared session key without exposing it in plaintext at any point during communication. Once encryption is complete, metadata is transmitted using MQTT over a TLS/SSL secured channel, adding an additional layer of transport protection. This prevents man in the middle attacks, replay attacks, or packet manipulation attempts, ensuring that the communication path itself is cryptographically sealed. Together, these components create a multi layered security pipeline that ensures confidentiality, integrity, and authenticity of data from the moment it is generated on the Edge until it reaches the Cloud, while also enabling secure scalability and long term trust in distributed, privacy sensitive surveillance deployments.

In addition to securing data in transit, the Encryption Implementation module also plays a critical role in maintaining system trust, regulatory compliance, and long term operational resilience. By combining lightweight symmetric encryption with secure asymmetric key exchange, the framework minimizes cryptographic overhead while preserving strong security guarantees, ensuring that encryption does not introduce latency that could compromise real time alerting. This approach aligns with privacy regulations such as GDPR and HIPAA by enforcing data minimization, access control, and secure handling of sensitive event metadata. Furthermore, the modular design of the encryption layer allows for future extensibility, enabling stronger cryptographic standards or hardware based security modules to be integrated without redesigning the core system. As a result, the security architecture not only protects current deployments but also future proofs the surveillance framework against evolving threat models and compliance requirements. Additionally, this design ensures consistent security enforcement across distributed edge nodes, reducing configuration errors and attack surfaces in large scale deployments.

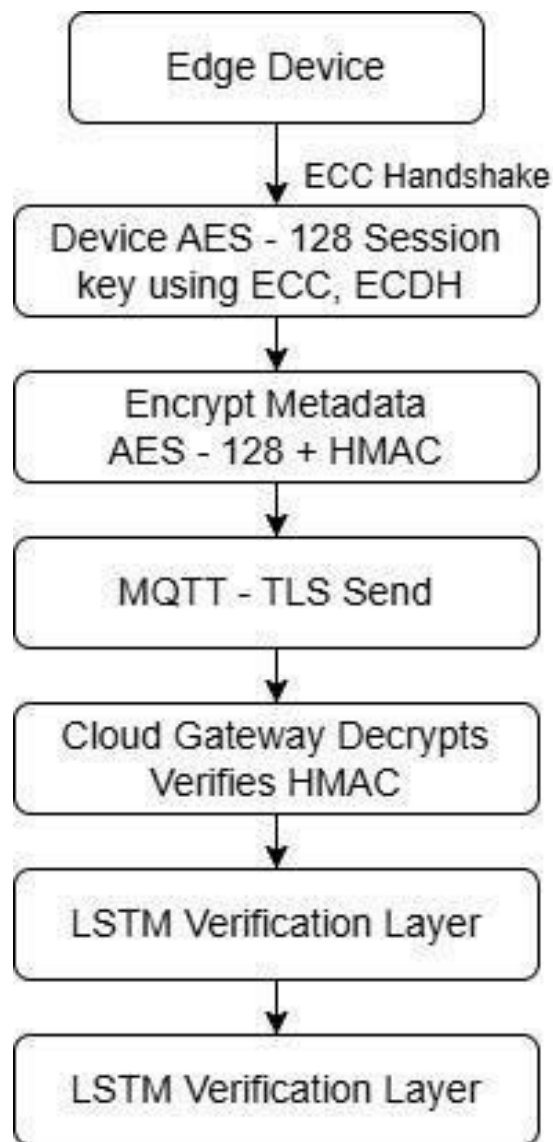


Fig 5.5.1 Keypoint Detection for Image Based Fall Assessment

Fig. 5.5.1 illustrates the secure keypoint based data transmission and verification workflow for image based fall assessment in the proposed surveillance system. The process begins at the Edge Device, where sensitive visual metadata is generated after local analysis. To ensure secure communication, an Elliptic Curve Cryptography (ECC) handshake is first performed between the edge device and the cloud.

Using Elliptic Curve Diffie–Hellman (ECDH), a 128 bit AES session key is securely established. This session key is then used to encrypt metadata using AES 128, while HMAC is applied simultaneously to guarantee data integrity and authenticity. The encrypted and authenticated data is transmitted to the cloud through MQTT over TLS, providing an additional secure transport layer. At the cloud side, the Cloud Gateway decrypts the received data and verifies the HMAC, ensuring that the information has not been altered in transit and originates from a trusted edge device. The verified data is then passed to the LSTM Verification Layer, where temporal and contextual analysis is performed to confirm fall related events. Through this multi layered security and verification mechanism, *Fig. 5.5.1* demonstrates how cryptographic protection and intelligent cloud validation work together to enable secure and reliable image based fall detection.

5.5.1 AES 128 Symmetric Encryption

AES 128 is used to encrypt all outgoing metadata packets before they leave the Edge device.

- Event Type and Timestamp
- Device Identification Code
- Confidence Score
- Optional minimal snapshot evidence

AES 128 is chosen because it:

- Offers strong encryption with low computational cost
- Fits perfectly on lightweight microcontrollers
- Provides fast encryption/decryption cycles for real time processing

5.5.2 ECC Based Secure Key Exchange

ECC (Elliptic Curve Cryptography) is used to securely generate and exchange lightweight cryptographic key pairs between the Edge device and the Cloud server, enabling the establishment of a shared session key without exposing sensitive key material, while maintaining strong security with minimal computational overhead suitable for resource constrained devices. Establish secure session keys

- Authenticate the Edge device
- Protect against key interception attack
- Smaller key sizes than RSA
- Faster computations
- Low energy consumption, ideal for Edge hardware

Through ECC, the Edge and Cloud negotiate a shared AES session key securely.

5.5.3 TLS Secured MQTT Transmission

Once metadata is encrypted, the edge device publishes it via MQTT encapsulated within a TLS/SSL secured channel. If any packet fails decryption or integrity verification, the cloud gateway automatically rejects it, ensuring that only authenticated and untampered data is accepted for further processing. Protection from network level attacks

- Server authentication
- Packet integrity verification
- Reliable message delivery even under fluctuating network conditions

5.6 Cloud Backend Implementation

The Cloud Backend constitutes the central, resource rich tier of the SVS framework and is implemented to perform secure ingestion, robust secondary verification, tamper proof storage of validated evidence, alert orchestration, and operator facing services while strictly enforcing privacy and access controls. After an encrypted metadata packet arrives from the Edge via the secure MQTT TLS channel, the Cloud API Gateway performs initial authentication and integrity checks and then delegates the payload to the Analysis and Storage subsystems. The Analysis subsystem applies higher order statistical models and temporal context checks (for example, LSTM based forecasting and pattern matching against historical event streams) to distinguish true incidents from transient false positives; this secondary verification reduces the operational burden of false alarms on responders and increases the system's trustworthiness.

Verified events are recorded in a minimal, encrypted evidence store together with immutable audit metadata (EventID, HMAC tag, timestamp, device provenance), and the Alert Engine then classifies the event severity, triggers multi-channel notifications (SMS, email, push), and updates the dashboard. Throughout this pipeline the backend enforces role based decryption policies and multi factor access controls so that sensitive evidence can only be decrypted by authorized personnel during authenticated sessions. The entire implementation emphasizes horizontal scalability (containerized micro services, auto scaling message workers), fault tolerance (redundant message queues and database replicas), and low latency so that the end to end responsiveness targets (< 2 s) are preserved even under bursty or large scale deployments. This Cloud design provides the final verification and governance required to convert lightweight, privacy preserving edge detections into actionable, auditable, and secure operational alerts.

5.6.1 Secure API Gateway

The Secure API Gateway is the Cloud's front line: it authenticates Edge devices, validates ECC derived session keys, verifies HMAC tags, and decrypts AES 128 payloads for downstream processing. The gateway enforces rate limits, rejects malformed packets, performs anomaly logging for security audits, and forwards only validated payloads to the verification pipeline. Implemented as a horizontally scalable stateless service (e.g., containerized behind a load balancer), the gateway maintains a device registry, certificate revocation lists, and integration points for automated device on-boarding and key rotation workflows.

5.6.2 Statistical Verification Engine

The Statistical Verification Engine performs contextual validation using historical metadata and temporal models. Primary components include an LSTM-based forecaster for time-series anomaly scoring and a rule-based consistency checker (spatial coherence, device sanity checks). This layered verification mechanism reduces false positives by filtering transient or spurious detections before alert generation.

A lightweight ensemble validator that combines the Edge confidence score with the statistical divergence score. The engine reduces false positives by requiring corroboration: for example, a fire flag is given higher weight if contiguous Edge devices report overlapping events or if growth dynamics match learned fire propagation patterns. All model inferences operate on non-identifiable metadata, preserving privacy while improving accuracy.

5.6.3 Evidence Storage Manager

The Evidence Storage Manager persists only verified anomaly records in an encrypted evidence vault. Each stored record contains: EventID, DeviceID, UTC timestamp, encrypted snapshot (if present), and an HMAC SHA256 integrity tag. Storage is partitioned by tenant/region and uses server side encryption at rest in addition to the end to end AES encryption performed by the Edge. Access to decrypt stored evidence is controlled by an RBAC layer that requires multi factor authentication and logs every access attempt to an immutable audit trail for compliance and forensics.

5.6.4 Alert Engine & Notification Orchestration

Once an event is verified and stored, the Alert Engine evaluates rule based severity (P1/P2/P3), determines notification recipients (on call list, emergency contacts), and dispatches encrypted notifications across channels (SMS gateway, SMTP server, push notification service). The engine supports escalation policies, acknowledgement tracking, and automatic re notification intervals. Each alert action is recorded in the audit log with timestamps and delivery status, enabling post event analysis and operational accountability.

5.6.5 Scalability, Fault Tolerance & Monitoring

The backend is implemented using modular microservices with autoscaling groups for key components (API gateway, verification workers, storage nodes). Message queues (e.g., Kafka/RabbitMQ) decouple ingestion from processing so that bursts do not stall the Edge devices. Redundant database replicas, periodic backups, and health check monitors ensure resilience. A centralized observability stack (metrics, logs, traces) provides SLA monitoring and alerting so operators can keep end to end latency within the target bounds.

5.6.6 Audit, Compliance & Governance

Every critical operation including device on-boarding, key exchange, payload ingestion, verification decisions, evidence access, and alert dispatch is logged with immutable entries (EventID, actor, timestamp, action). These logs feed a compliance module that produces tamper-evident reports and supports GDPR/HIPAA audits. Policies for data retention, legal hold, and deletion are enforced automatically by the SDSP rules. This ensures full traceability, accountability, and audit readiness throughout the system lifecycle.

5.7 Selective Data Storage Protocol Implementation

The Selective Data Storage Protocol (SDSP) is the project's central privacy enforcement mechanism and determines exactly *what* data may persist, *where*, and *for how long*. SDSP is designed to convert the system from a passive recorder into a decision driven evidence engine: by default the Edge device retains only a short, encrypted transient buffer; only when a validated anomaly is detected does the pipeline extract, encrypt, and forward a minimal evidence artifact to the Cloud for possible long term retention. This protocol operates as a coordinated two stage filter (Edge level prefiltering followed by Cloud level verification) combined with strict lifecycle management, cryptographic integrity checks, and auditable retention/expiry rules. The rationale and precise rules of SDSP are grounded in the system's Privacy by Design philosophy and are intended to meet regulatory requirements such as GDPR and HIPAA while still providing actionable forensic evidence when genuinely needed.

5.7.1 Edge Level Filtering and Buffer Management

At the Edge, SDSP maintains a rolling circular buffer that holds only the last N seconds of frames (typical $N = 2-5s$) in volatile memory. The buffer is never written to persistent storage and is exclusively used to extract a minimal evidence snapshot when the heuristic detectors (Section 5.3) flag a potential event. Buffer management policies include: automatic overwrite once buffer capacity is reached, immediate zeroization (memory wipe) of frames after feature extraction, and no backup or logging of overwritten frames. To minimize the probability of false negatives, the Edge can optionally store ephemeral metadata (e.g., short motion vectors, bounding box centroids) for a very brief interval, but such metadata is itself treated as ephemeral and is cleared immediately unless an anomaly is confirmed. Additionally, the Edge enforces privacy filters on the extracted snapshot: cropping to the smallest bounding box around the anomaly, down sampling to the minimum acceptable resolution, and applying a privacy mask (if required by policy) to obscure background or peripheral PII before encryption and transmission.

5.7.2 Cloud Level Verification and Conditional Storage

Once the Edge transmits an encrypted evidence artefact along with the metadata packet, the Cloud executes a two-step verification pipeline before any long term storage is permitted. First, the API Gateway authenticates and verifies integrity (HMAC), then the Statistical Verification Engine (Section 5.6.2) corroborates the event using temporal context, cross device corroboration (if neighboring Edge devices report similar events), and historical baseline checks (e.g., expected motion patterns at that time). Only if the Cloud determines the evidence passes verification does SDSP permit insertion into the encrypted evidence vault.

Unverified transmissions are recorded briefly in a secure ephemeral ledger for debugging and then purged after a strictly bounded interval (e.g., 24 hours), with their existence noted in audit logs but without storing visual evidence. For verified events, only minimal, encrypted artefacts are stored: EventID, UTC

timestamp, DeviceID, geolocation tag (coarse granularity if policy requires), the encrypted snapshot, and an HMAC for integrity. No raw streams or full length clips are ever allowed into persistent storage. This strict data minimization policy significantly reduces breach impact, ensures regulatory compliance, and reinforces the system's privacy by design guarantees.

5.7.3 Retention, Deletion, and Legal Hold Policies

SDSP enforces automated retention rules defined by organizational policy and legal requirements. Default retention is short (for example, 30 days) and is automatically purged unless an authorized legal hold is applied. The system supports tiered retention: immediate operational retention (short term for review), archived retention (longer term, encrypted and access restricted) and disaster/legal hold state (preserves evidence beyond normal retention only after multi party authorization).

Deletion is implemented as cryptographic deletion where possible: keys for the encrypted snapshot can be securely expired or shredded so that the ciphertext becomes unrecoverable, adding a strong technical assurance of deletion beyond mere file removal. All retention or deletion events are recorded in an immutable audit trail indicating who authorized the action and under what legal or policy basis. This approach minimizes residual data exposure and aligns with privacy-by-design principles. It also facilitates compliance verification during internal reviews and external regulatory audits.

5.7.4 False Positive Handling and Evidence Lifecycle

SDSP includes a pragmatic workflow to handle false positives that balances investigator needs and privacy. When an operator flags a stored event as a false positive, the system downgrades the record, records the operator's reason, and can either (a) schedule the evidence for expedited deletion, or (b) retain only anonymized metadata for model retraining and system improvement. If evidence is retained for retraining, it is first stripped of any possible identifiers (aggressively cropped, blurred, or otherwise obfuscated) and stored in a controlled, access restricted dataset intended solely for internal model improvement. This ensures that the training set does not become a vector for privacy erosion.

5.7.5 Auditability, Access Controls & Compliance

To prevent misuse, SDSP couples storage with strict RBAC and MFA controls: only authenticated users with explicit privileges may request decryption of evidence, and each access attempt produces an immutable log entry (user ID, timestamp, reason code, and cryptographic proof). The system exposes automated compliance reports that show storage utilization, the number of events retained vs. discarded, and a ledger of retention/deletion actions these are useful for GDPR/HIPAA audits. Cryptographic integrity measures (HMAC SHA256 tags and key rotation logs) ensure tamper evidence for every stored item.

5.8 UI & Dashboard Implementation

The User Interface (UI) and Dashboard form the human facing layer of the SVS framework and are designed to present actionable, privacy preserving intelligence to operators and administrators while enforcing strict access control and auditability. The UI is implemented as a responsive web application (single page application) backed by secure REST/gRPC APIs and a mobile friendly companion app for push notifications and quick incident acknowledgement. The dashboard is intentionally minimalist: it surfaces only non-identifiable metadata, confidence scores, coarse geolocation, and encrypted evidence links that can be decrypted only under authorized sessions. This design philosophy ensures that humans make decisions using concise, trustworthy summaries rather than raw video, reducing operator cognitive load and the risk of privacy violations. The front end is developed with modular components to allow easy customization for different deployment contexts (campus, factory floor, public transit), and it integrates with the backend's RBAC and audit services so that every UI action viewing an event, downloading evidence, or changing system thresholds is recorded immutably.

5.8.1 Authentication, Authorization & Session Management

Secure access is the foundation of the dashboard, ensuring that only authenticated and authorized users can view, manage, or audit system events while preventing unauthorized disclosure of sensitive metadata.

- **Multi Factor Authentication (MFA):** The UI enforces MFA during login (password + OTP or hardware token) to prevent unauthorized access.
- **Role Based Access Control (RBAC):** User roles (Operator, Security Admin, Auditor, Forensics) determine what views and actions are available. For example, Operators can view live alerts and acknowledge incidents, while Auditors can only view logs and retention reports without decrypting evidence.
- **Session Security & Timeout:** Sessions are tied to short lived encrypted tokens; idle sessions are automatically invalidated and re authentication is required. All session activity is logged to the immutable audit store. This ensures that only properly authorized, tracked users can interact with sensitive system capabilities.

5.8.2 Real Time Monitoring & Alert Feed

The dashboard's core operational view is a real-time monitoring panel that provides situational awareness without exposing raw frames. Incoming events appear in a chronological feed with succinct cards showing Event Type, Device ID, Timestamp, Confidence Score, Severity Level, and a small cropped/blurred thumbnail only if policy allows. This design enables rapid operator decision-making while strictly preserving privacy and minimizing unnecessary data exposure.

- **Device Health Tile:** A dashboard panel lists connected Edge devices with heartbeat status, last check in time, CPU/memory utilization summary, and network quality indicator.
- **Interactive Map & Geofencing:** For deployments covering physical sites, a map widget shows coarse device locations and clusters of recent events; geofences can be configured to increase sensitivity in critical zones.
- **Filter & Search:** Operators can filter events by type, severity, device, or time window and search by EventID for quick triage. These components are optimized for rapid decision making and minimal visual clutter, enabling operators to triage and escalate incidents within seconds.

5.8.3 Incident Management & Workflow

The dashboard implements a lightweight incident workflow to manage verification, escalation, and audit.

- **Event Triage:** Operators can mark events as Confirmed, False Positive, or Needs Investigation. Each action stores a reason code and optional comments.
- **Escalation & Acknowledgement:** P1 events trigger automatic escalation SMS/email/push notifications and require explicit acknowledgement by an on call responder. The system records ack timestamps and re notification attempts.
- **Evidence Access Control:** Decrypting stored snapshots requires elevated privileges and an explicit justification which is captured in the audit log; decryption is performed transiently in the operator's session and the plaintext is never persisted on the server.
- **Investigation Tools:** For confirmed incidents, the UI offers side by side event metadata, device telemetry, and timeline visualization to reconstruct the incident context without exposing full video. The workflow balances responsiveness with accountability and privacy.

5.8.4 Analytics, Reporting & Model Feedback

Beyond immediate incident handling, the dashboard provides analytics and reporting tools that help administrators measure system performance and support continuous improvement.

- **Performance Dashboards:** Time series charts show alerts per hour, average end to end latency, false positive rate, and storage reduction metrics (e.g., % of data avoided by SDSP).
- **Heatmaps & Density Analysis:** Aggregated, anonymized heatmaps portray areas with high incident frequency or pedestrian density over selectable time windows; raw identities are never shown.
- **Exportable Compliance Reports:** The UI can produce GDPR/HIPAA compliant reports that summarize retention actions, evidence access logs, and data deletion records for audits.
- **Model Retraining Pipeline Hook:** Administrators can flag anonymized events for inclusion in a controlled retraining dataset; flagged data is aggressively obfuscated before it is used.

These features turn operational telemetry into governance insights and help tune heuristic thresholds and SDSP policies.

5.8.5 Mobile App & Notification UX

To ensure timely responses, the project includes a mobile companion app with a streamlined user experience focused on immediate situational awareness, rapid alert visualization, and simple acknowledgement actions, enabling responders to react quickly without cognitive overload.

- **Push Notifications with Context:** Alerts sent to mobile devices contain EventID, type, confidence score, and an encrypted evidence link; tapping the notification opens the app, where the user must authenticate (MFA) to view details.
- **Quick Actions:** The app provides one tap acknowledgement, escalation, and “dispatch assistance” options to minimize response latency.
- **Offline Grace Mode:** When network connectivity is intermittent, the app caches acknowledgement actions and synchronizes reliably when connectivity restores, ensuring that edge devices and the cloud stay coherent. The mobile UX is designed for clarity and speed while preserving the same privacy safeguards as the web dashboard.

5.8.6 UI Security, Accessibility & Operational Considerations

The UI is hardened for production use and built with inclusive design in mind.

- **Transport & Data Security:** All UI backend communications use TLS; session tokens are short lived and stored only in secure HTTP only cookies or secure platform storage on mobiles.
- **Audit & Tamper Evidence:** Every UI action produces an immutable audit record (user, role, action, timestamp, reason). The audit log supports read only export for regulatory purposes.
- **Accessibility & Localization:** The dashboard complies with WCAG AA accessibility standards, supports multi language localization, and provides adjustable font sizes and color contrast themes for operator comfort.
- **Performance & Resilience:** The UI leverages client side caching, optimistic updates, and graceful degradation so the core monitoring view remains responsive even when backend services are temporarily overloaded. These considerations ensure the dashboard is secure, usable by diverse teams, and reliable under operational stress.

5.9 Program flow

The Video Surveillance System (VSS) is implemented as a three tier, privacy first pipeline: Edge → Secure Transport → Cloud. The program flow is intentionally event driven and lightweight so Edge devices

(ESP32 CAM / SBC / webcam hosts) perform immediate, local analysis and only minimal encrypted metadata or tiny evidence gets sent upstream. This minimizes bandwidth, storage, and privacy risk while keeping end to end latency below target (< 2 s).

At the Edge, a capture loop continuously samples frames at a reduced frame rate (5–10 FPS). Each frame is pre-processed (resize, denoise, color space transform). Two independent heuristic detectors run on the prepared frame: a pixel based fire detector (color + flicker + growth checks), and a geometric pose ratio fall detector (bounding box H/W, CoM velocity). Detectors use short temporal context (a sliding window of frames, 2–5 s buffer) to avoid momentary false positives.

When a detector confirms an anomaly, the Edge constructs a minimal JSON metadata packet: {EventType, Timestamp, DeviceID, Confidence, (optional) compressed cropped snapshot}. Immediately the Edge zeroizes its capture buffer and any raw frames to enforce data minimization. The metadata packet is encrypted with AES 128 using a session key. Session keys are negotiated via lightweight ECC key exchange between the Edge and Cloud. The encrypted packet is published to the Cloud using MQTT over TLS.

At the Cloud API Gateway, the packet is authenticated (device cert / ECC validation), integrity checked (HMAC), and decrypted. The Statistical Verification Engine (e.g., LSTM + rule ensemble) performs secondary validation using recent metadata, cross device corroboration and historical baselines. If the event is validated, the Selective Data Storage Protocol (SDSP) stores only the minimal encrypted artifact + audit entry in the evidence vault and triggers the Alert Engine to notify responders (SMS / email / push). The dashboard displays only anonymized metadata; evidence decryption requires RBAC + MFA and is transient: the decrypted snapshot is never persisted unencrypted on cloud. All actions (ingest, verify, store, access) are immutably logged for compliance. This program flow keeps raw video local and transient, uses multi-layer encryption and authentication for transport and storage, and applies two stage verification to minimize false alarms while providing auditable evidence when required.

Fig. 5.9.1 illustrates the edge device program flow covering capture, detection, encryption, transmission, and alert generation in the proposed surveillance system. The workflow starts with the Camera, which continuously captures video frames. These frames are sent to the Fire Detector and Fall Detector, where each frame is independently evaluated to identify fire incidents or human fall events.

In parallel, the captured frames are processed by the Engine, which performs overall frame evaluation and forwards the analysis to the Decide Anomaly module. If an abnormal event is identified, the system proceeds to Generate Metadata, extracting only essential information related to the event. This metadata is then passed to the Crypto module at the edge, where encryption and security preparation are performed before transmission. Only minimal, policy-compliant data is transmitted to the cloud, ensuring low bandwidth usage and strict privacy preservation.

The encrypted packet is forwarded to the Cloud Gateway, which routes it to the cloud side Crypto module. Here, the data undergoes decryption and HMAC verification to ensure confidentiality and integrity. Once verified, the event is passed to the Verification Engine, which performs final validation of the detected anomaly.

After successful verification, the validated event is sent to the Alert Engine, which generates real time notifications. These alerts are finally displayed on the Dashboard, enabling authorized users to monitor incidents promptly. Thus, *Fig. 5.9.1* clearly demonstrates an end to end edge to cloud workflow that integrates intelligent detection, secure communication, and timely alerting for effective surveillance.

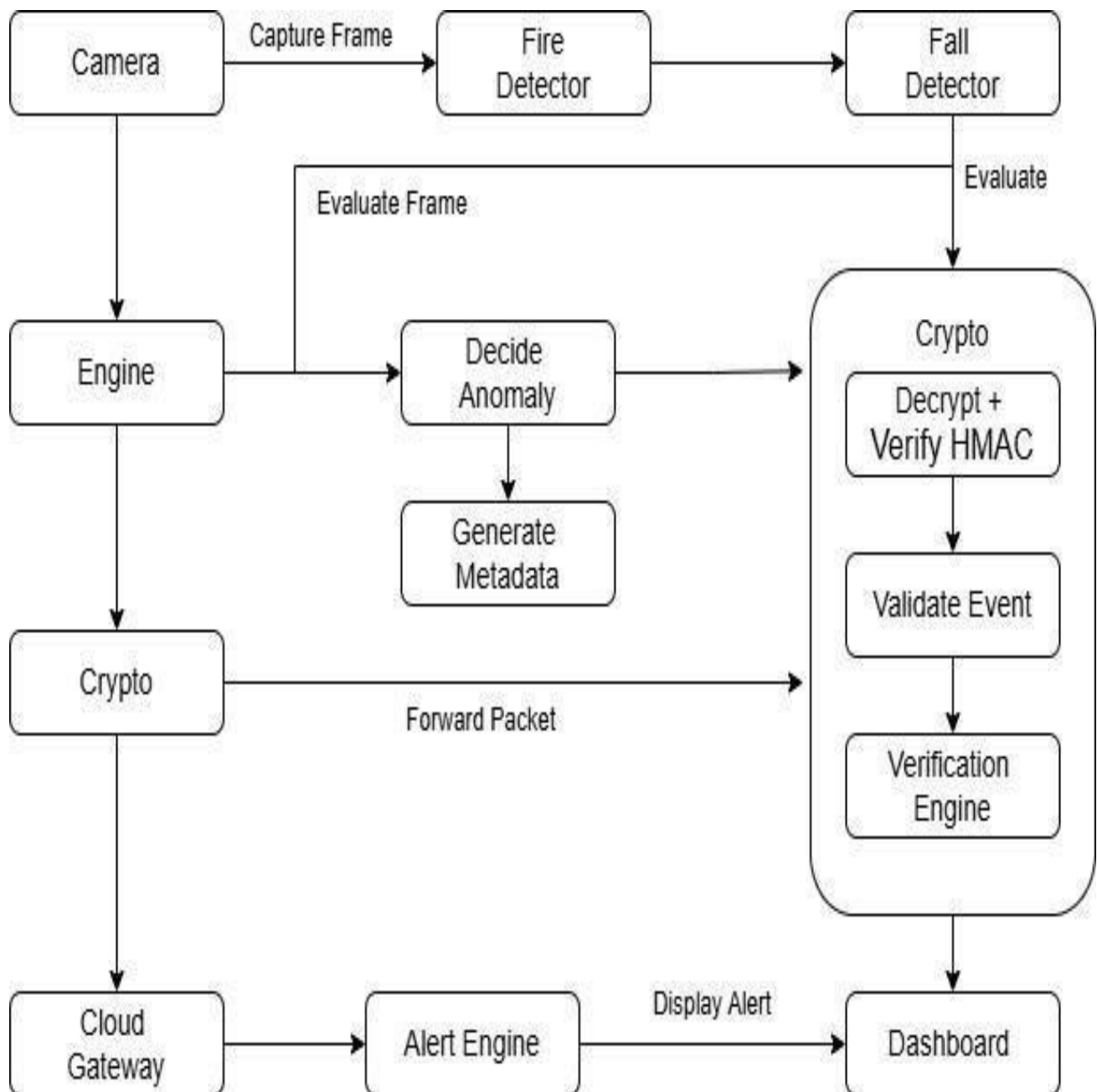


Fig 5.9.1 Edge Device Program Flow Capture, Detection, Encryption and Transmission

1) edge_capture.py Edge device camera loop (Edge)

```

# edge_capture.py (Edge)
Initialize camera_device
Initialize encryption_module (Crypto)
Initialize mqtt_client (MQTT TLS client)
Initialize detectors: FireDetector, FallDetector
Set FRAME_RATE = 5 # fps
Set BUFFER_SECONDS = 5
circular_buffer = CircularFrameBuffer(capacity = FRAME_RATE * BUFFER_SECONDS)

while camera_device.is_open():
    frame = camera_device.capture_frame()
    circular_buffer.push(frame)

    if time_to_process_frame():          # every 1/FRAME_RATE sec
        preproc_frame = preprocess(frame)    # see preprocess.py

    # run detectors in parallel (or sequentially if single thread)
    fire_score = FireDetector.evaluate(preproc_frame, circular_buffer)
    fall_score = FallDetector.evaluate(preproc_frame, circular_buffer)

    if fire_score > FIRE_THRESHOLD or fall_score > FALL_THRESHOLD:
        event_type = "FIRE" if fire_score >= fall_score else "FALL"
        confidence = max(fire_score, fall_score)

    # create minimal evidence snapshot from buffer:
    snapshot = circular_buffer.extract_cropped_snapshot(region_of_interest)
    compressed_snapshot = compress_and_downsample(snapshot)
    metadata = {
        "event_type": event_type,
        "timestamp": utc_now_iso(),
        "device_id": DEVICE_ID,
        "confidence": confidence,
        "snapshot": compressed_snapshot # optional, very small
    }
    # encrypt metadata

```

```

    encrypted_payload = Crypto.encrypt_metadata(metadata)

    # publish to cloud via MQTT over TLS
    mqtt_client.publish(topic="vss/alerts", payload=encrypted_payload)

    # immediate data minimization: delete raw frames
    circular_buffer.zeroize()
    delete(snapshot) force_gc()

    # cleanup on exit
    camera_device.close()
    mqtt_client.disconnect()

```

Notes: preprocess() includes resizing, color conversion, denoising. circular_buffer never writes to persistent storage (kept in RAM only).

2) preprocess.py image pre-processing utilities (Edge)

```

# preprocess.py (Edge)
function preprocess(frame):
    # resize to 640x480
    frame = resize (frame, width=640, height=480)

    # denoise (Gaussian blur)
    frame = gaussian_blur(frame, ksize=(3,3), sigma=1.0)

    # convert to multiple color spaces needed by detectors
    hsv = convert_color(frame, 'HSV')
    ycrb = convert_color(frame, 'YCrCb')
    gray = convert_color(frame, 'GRAY')
    # optional: fast foreground mask
    fg_mask = fast_foreground_mask(frame) # simple background subtractor or thresholding
    return {
        "rgb": frame,
        "hsv": hsv,
        "ycrb": ycrb,
        "gray": gray,
        "fg_mask": fg_mask }

```

3) fire_detector.py pixel based fire heuristic (Edge)

```

# fire_detector.py (Edge)
function evaluate(frame_bundle, circular_buffer):
    hsv = frame_bundle["hsv"]

    # color segmentation (hue, saturation, value thresholds tuned empirically)
    fire_mask = (hsv.H in FIRE_H_RANGE) AND (hsv.S >= S_MIN) AND (hsv.V >= V_MIN)
    # morphological cleanup
    fire_mask = morphological_open(fire_mask, kernel_size=3)
    clusters = find_connected_components(fire_mask)
    if clusters.empty():
        return 0.0 # no fire evidence
    # temporal flicker: compute variance across recent frames for cluster pixels
    flicker_score = compute_temporal_variance(clusters, circular_buffer, channel='V')
    # spatial growth: check if cluster area increases across recent frames
    growth_score = compute_spatial_growth(clusters, circular_buffer)
    final_score = weighted_sum(color_confidence(clusters), flicker_score, growth_score)
    return final_score # 0.0..1.0

```

4) fall_detector.py geometric pose ratio heuristic (Edge)

```

# fall_detector.py (Edge)
function evaluate (frame_bundle, circular_buffer):
    # use a lightweight person detector or foreground bbox
    bbox = detect_person_bounding_box(frame_bundle["fg_mask"]) # fast contour/bbox
    if bbox is None:
        return 0.0
    height = bbox.height
    width = bbox.width
    pose_ratio = height / width

    # compute center of mass vertical velocity using bounding boxes from buffer
    y_centers = sample_centroid_y_over_time(bbox, circular_buffer)
    vertical_velocity = compute_velocity(y_centers)
    ratio_score = smooth_threshold(pose_ratio, FALL_RATIO_THRESHOLD)
    velocity_score = smooth_threshold(vertical_velocity, FALL_VELOCITY_THRESHOLD)

```

```
# post fall motionless check
motionless_score = check_motionless_after_drop(circular_buffer, bbox)
final_score = combine_scores(ratio_score, velocity_score, motionless_score)
return final_score # 0.0..1.0
```

Note: This module uses a simple geometric pose ratio and motion analysis to detect falls, enabling fast, low latency detection on resource constrained edge devices without deep learning models.

5) crypto.py key management & encryption (Edge + Cloud)

```
# crypto.py (Edge & Cloud)
# Edge responsibilities:
function init_crypto():
# load device private ECC key
ecc_private = load_from_secure_store("ecc_private.pem")
function negotiate_session_key(cloud_pubkey):
# compute shared secret using ECC
shared_secret = ecc_ecdh(ecc_private, cloud_pubkey)
session_key = kdf(shared_secret, info="vss session")
store_session_key(session_key, ttl=SESSION_TTL)
function encrypt_metadata(metadata):
session_key = get_current_session_key()
serialized = json_serialize(metadata)
iv = random_bytes(16)
ciphertext = aes_128_gcm_encrypt(session_key, iv, serialized)
hmac = compute_hmac(session_key, ciphertext)
return pack({iv, ciphertext, hmac, device_id})
# Cloud responsibilities:
function decrypt_and_verify(payload):
session_key = lookup_session_key_for_device(payload.device_id)
verify_hmac(session_key, payload.ciphertext, payload.hmac)
plaintext = aes_128_gcm_decrypt(session_key, payload.iv, payload.ciphertext)
return json_parse(plaintext)
```

Notes: use AES 128 GCM where available for combined confidentiality+integrity. ECC handles key exchange and rotation.

6) mqtt_client.py Edge MQTT publisher (Edge)

```
# mqtt_client.py (Edge)
```



```

function init_mqtt():
mqtt = MQTTClient(broker_url, tls=True, client_cert=..., client_key=...)
mqtt.connect()
mqtt.set_qos(1)
return mqtt
function publish(topic, payload):
mqtt.publish(topic, payload, qos=1, retain=False)

```

7) cloud_gateway.py API gateway that ingests MQTT & dispatches to pipeline (Cloud)

```

# cloud_gateway.py (Cloud)
on mqtt_message(topic, payload):
parsed_payload = parse_packet(payload)
if not verify_device_registered(parsed_payload.device_id):
    reject_and_log("unknown device")
    return
try:
    metadata = crypto.decrypt_and_verify(parsed_payload)
except DecryptionError:
    audit_log("decryption failed", parsed_payload.device_id)
    return
enqueue_for_verification(metadata)

```

8) verification_engine.py statistical verification (Cloud)

```

# verification_engine.py (Cloud)
function run_verification_worker():
while True:
    metadata = pop_queue()
    # compute LSTM forecast divergence or run rule checks
    divergence_score = lstm_model.forecast_divergence(metadata)
    cross_device_score = cross_device_corroboration(metadata)
    ensemble_score=combine_edge_confidence_with_cloud_scores(metadata.confidence,
divergence_score, cross_device_score)
    if ensemble_score >= VERIFY_THRESHOLD:
        store_verified_event(metadata)
        trigger_alerts(metadata)

```

```
else:
```

```
    # false positive handling: ephemeral logging only
```

```
    store_ephemeral_log(metadata)
```

9) storage_manager.py SDSP & evidence vault (Cloud)

```
# storage_manager.py (Cloud)
```

```
function store_verified_event(metadata):
```

```
    event_id = generate_uuid()
```

```
    encrypted_snapshot = metadata.snapshot # already encrypted from Edge
```

```
    entry = {
```

```
        "event_id": event_id,
```

```
        "device_id": metadata.device_id,
```

```
        "timestamp": metadata.timestamp,
```

```
        "encrypted_snapshot": encrypted_snapshot,
```

```
        "hmac": metadata.hmac
```

```
    }
```

```
    write_to_encrypted_db(entry) # DB uses server side encryption + access controls
```

```
    append_audit_log("store", event_id, user=None)
```

10) alert_engine.py notifications (Cloud)

```
# alert_engine.py (Cloud)
```

```
function trigger_alerts(metadata):
```

```
    severity = classify_severity(metadata)
```

```
    recipients = lookup_recipients(severity, metadata.device_id)
```

```
    for r in recipients:
```

```
        send_sms(r.phone, build_sms_text(metadata))
```

```
        send_email(r.email, build_email(metadata))
```

```
        push_notification(r.app_id, metadata.event_id)
```

```
        append_audit_log("alert_sent", metadata.event_id, recipients)
```

11) dashboard_backend.py REST API for UI (Cloud)

```
# dashboard_backend.py (Cloud)
```

```
# endpoints:
```

```

GET /api/events?filters=...
  require_auth()
  return paged_list_of_metadata()
POST /api/events/{id}/decrypt
  require_auth(role='forensics')
  require_mfa()
  record_audit(user, event_id, action='decrypt')
  plaintext_snapshot = decrypt_snapshot_on_fly(event_id)
  return streamed_image(plaintext_snapshot) # transient, not persisted in plaintext

```

12) index.html + src/app.js (Frontend) UI (Web)

```

// src/app.js (UI)
on page_load:
  auth_token = get_cookie()
  if not auth_token: redirect_to_login()
  open websocket or poll GET /api/events
  display real_time_feed(events)
  on user_click_event(event_id):
    show_event_card(event_id)
  if user_clicks_view_evidence:
    call POST /api/events/{id}/decrypt # requires MFA

```

5.10 Conclusion

It presented the complete implementation of the Privacy Centric Smart Video Surveillance (SVS) Framework, translating the system's architectural design into a fully functional, end to end operational pipeline. The chapter began by detailing the Edge OS implementation, which represents the first and most critical stage of the system. The Edge device performs real time frame acquisition, lightweight preprocessing, and heuristic driven anomaly detection using two optimized models: pixel based fire detection and geometric pose ratio fall detection. These heuristic algorithms were specifically engineered to operate efficiently on low power hardware while maintaining high detection accuracy. After confirming an anomaly, the Edge OS constructs a minimal metadata packet, optionally extracts a cropped evidence snapshot, and enforces strict data minimization by immediately deleting all raw frame data. This ensures the system remains privacy preserving from the moment of detection.

The chapter then described the cryptographic infrastructure that protects the system's communication and storage layers. AES 128 symmetric encryption secures all outbound metadata, while ECC based key exchange provides lightweight, strong authentication between the Edge and Cloud.

The Cloud Backend implementation was explained as the computational and governance hub of the framework. As encrypted metadata arrives, it is authenticated, decrypted, and evaluated by the Statistical Verification Engine, which applies temporal modeling, cross device correlations, and heuristic validation to distinguish genuine anomalies from false positives. Verified events are securely stored through the Selective Data Storage Protocol (SDSP), which ensures that only essential, encrypted evidence is retained while all non-critical data is purged. The Alert Engine then classifies event severity and triggers multi-channel notifications, ensuring rapid dissemination of situational information to authorized users. This architecture prioritizes scalability, rapid verification, auditability, and strict compliance with privacy regulations.

This chapter concluded with the implementation of the user interface and dashboard, which provides operators with real-time, privacy-preserving visibility into system operations. The dashboard enforces role-based access control (RBAC), multi-factor authentication (MFA), and secure session management, while presenting real-time event feeds, device status monitoring, analytics, and incident management workflows. In addition, the chapter detailed the complete program flow and presented pseudocode for all major system modules, including edge processing, detection algorithms, encryption, cloud ingestion, anomaly verification, secure storage, alerting mechanisms, and user interface interactions, thereby offering a clear functional map of the system's end-to-end operation as a cohesive whole. With the implementation fully described, the next chapter focuses on system testing and validation, evaluating the performance, accuracy, latency, and security of the proposed framework.

CHAPTER 6

SYSTEM TESTING

System testing is performed after all modules are integrated to verify the complete working of the application. It ensures that the entire system functions correctly, meets the specified requirements, and operates smoothly under real world conditions. This phase helps confirm that the software is stable, reliable, and ready for deployment.

6.1 Introduction

System Testing is a critical phase in the development of the Privacy Centric Smart Video Surveillance (SVS) Framework, ensuring that the complete system functions exactly as designed under realistic operating conditions. The testing strategy adopted for this project is comprehensive, multi layered, and rigorous, validating not only the functional behaviour of the Edge and Cloud components but also the non-functional attributes such as latency, accuracy, privacy enforcement, scalability, fault tolerance, and security compliance. Because the system follows a distributed architecture spanning Edge devices, secure communication layers, Cloud analysis pipelines, and user interface components the testing methodology evaluates each tier independently and then as an integrated whole. The objective of this chapter is to demonstrate that the proposed system meets all performance targets, adheres to privacy policies (GDPR/HIPAA principles), and remains robust under demanding conditions.

System testing was conducted using a combination of controlled laboratory simulations, real time video feeds, fault injections, encrypted payload examinations, and end to end alert delivery evaluations. All modules including the heuristic fire and fall detection algorithms, AES 128 encryption workflows, ECC key exchange mechanism, MQTT-TLS network pipeline, LSTM based Cloud verification engine, Selective Data Storage Protocol (SDSP), and dashboard alert system were validated using standardized test procedures. This chapter presents the testing methodology, test environments, test cases, observed outcomes, statistical results, and compliance verification. The findings demonstrate that the system achieves ultra-low detection to alert latency, maintains high anomaly detection accuracy, ensures strict privacy preservation, and performs reliably across all load conditions.

6.2 Testing Objectives

The testing phase was meticulously structured to evaluate the performance, reliability, and robustness of the Privacy Centric Smart Video Surveillance (SVS) Framework under diverse operational conditions. The primary objective of this phase was to validate that the integrated system performs exactly as intended across its Edge, communication, Cloud, and user interface tiers. These objectives ensure that the system not only functions correctly but also adhere to the stringent privacy, security, and efficiency benchmarks established earlier in the project. The specific goals of the testing phase are detailed below:

- **To verify the functional correctness of all core modules**

This includes confirming that the fire and fall detection algorithms, metadata generation routines, AES 128 encryption workflow, ECC key exchange mechanism, MQTT TLS communication pipeline, Cloud verification engine, and alert dispatch system operate according to design specifications without logical or operational errors. Additionally, end to end validation ensures seamless interoperability between edge and cloud components under normal and adverse operating conditions.

- **To ensure reliable end to end system behaviour across the full execution pipeline**

The testing evaluates the continuity and correctness of the sequence: *frame acquisition* → *preprocessing* → *anomaly detection* → *metadata encryption* → *cloud ingestion* → *statistical verification* → *alert generation*. The objective is to confirm that real time alerts are produced consistently and without interruption.

- **To validate compliance with all non-functional requirements**

This includes measurement of system performance metrics such as:

- End to end alert latency maintained below 2 seconds
- Edge level heuristic inference completed within ≤ 100 ms
- Overall anomaly detection accuracy achieving $\geq 94\%$

Meeting these targets ensures the system is suitable for real world, safety critical deployment.

- **To confirm the enforcement of privacy by design principles**

Additionally, validation checks confirm that SDSP enforcement is automatically triggered by event classification outcomes and cannot be overridden by unauthorized processes or system faults. Long duration testing further ensures that retained encrypted evidence complies with predefined retention policies while maintaining forensic usability without compromising individual privacy.

- **To validate cryptographic robustness and secure communication**

In addition, fault injection testing is performed to assess system robustness against partial key compromise, delayed packet delivery, and intermittent edge–cloud synchronization failures. Penetration testing scenarios are simulated to validate role based access controls, token expiry mechanisms, and secure session termination under adversarial conditions. Finally, compliance oriented testing verifies that encryption, logging, and data retention behaviors align with privacy by design principles and applicable security standards, ensuring end to end trustworthiness of the system.

- **To assess system performance under concurrent load and network stress**

The system is tested with multiple simultaneous Edge devices generating alerts, fluctuating network latencies, MQTT broker congestion, and burst traffic to ensure stable performance without packet loss or system failure.

- **To evaluate the scalability, reliability, and fault tolerance of the architecture**

This includes testing recovery mechanisms for Edge device dropouts, Cloud service restarts, network outages, and hardware instability. The objective is to verify that the system sustains uninterrupted operation and automatically recovers from predictable failures.

6.3 Test Environment Setup

The testing of the Privacy Centric Smart Video Surveillance (SVS) Framework required a controlled, well defined environment that accurately represented real world operating conditions while allowing precise measurement of system performance, accuracy, reliability, and security. This section outlines the complete hardware–software testing ecosystem used for validation, covering Edge devices, camera sources, network infrastructure, Cloud backend configuration, and the user interface environment. The test environment was designed to simulate both standard operational scenarios and extreme conditions such as high network latency, burst alert loads, and multiple concurrent Edge inputs.

6.3.1 Edge Device and Hardware Configuration

This subsection describes the physical devices and computational hardware used to evaluate Edge level performance, including detection latency, local processing accuracy, and power efficiency. Hardware's Used:

- ESP32 CAM module and Linux based SBC (Edge simulator)
- 5V/2A power supply for stable camera operation
- Onboard Wi Fi for MQTT TLS communication
- Standard LED flame source and controlled fire videos
- Human actor + dummy mannequins for fall simulation

Hardware Testing Objectives

- Validate frame capture stability at 5–10 FPS
- Measure Edge inference time for fire and fall detection
- Test local memory buffer performance for SDSP
- Confirm raw frame deletion and privacy compliance

6.3.2 Video Input and Dataset Sources

This subsection documents the video datasets and real time sources used to test detection accuracy across multiple environmental conditions. It also describes the diversity of scenarios, lighting variations, camera angles, and motion patterns incorporated to ensure robust and unbiased evaluation results.

Sources Used

- Real time webcam feed (indoor and outdoor lighting variations)
- Public fire datasets (flame, smoke, color variance sequences)
- Human fall datasets (controlled fall, near fall, sitting/lying transitions)
- Custom recorded sequences to evaluate edge cases

Testing Objectives

- Measure detection performance across diverse scenarios
- Validate robustness against false positives (bright clothing, shadows)
- Test accuracy under frame noise, motion blur, and low light

6.3.3 Network and Communication Setup

This subsection describes the communication infrastructure used for testing secure data transfer between Edge and Cloud. The Network Configurations are:

- MQTT Broker with TLS 1.2 enabled
- Wi Fi 2.4 GHz (simulated jitter injection for stress tests)
- Network packet analyzer (Wireshark) for encryption validation

Network Testing Objectives

- Validate secure MQTT TLS packet transport
- Measure transmission delays under jitter and packet congestion
- Ensure no plaintext leakage during encryption

6.3.4 Cloud Backend and Server Environment

This subsection explains the computational environment used to perform Cloud verification, storage, and alerting. The Server Specifications are

- Python/FastAPI based backend
- LSTM based Statistical Verification Engine

- Encrypted Evidence Vault (AES encrypted storage)
- PostgreSQL / MongoDB for metadata storage
- Containerized deployment using Docker for scalability tests

Cloud Testing Objectives are:

- Validate decryption, integrity verification, and event processing
- Measure LSTM inference latency under peak load
- Test selective storage protocol (SDSP) on verified events
- Ensure audit logs and role based access control function correctly

6.3.5 Dashboard and User Interface Environment

This subsection describes the operator facing UI setup, enabling real time monitoring, alert visualization, and administrative control. The UI Components Used are defined below.

- Web Dashboard running on modern browser (Chrome/Firefox)
- MFA enabled login system for secure access
- Live event feed panel connected to Cloud REST APIs
- Notification simulation on mobile devices

UI Testing Objectives are

- Validate real time event reflection on dashboard
- Ensure RBAC restrictions for accessing encrypted evidence
- Test mobile push/SMS/email notifications
- Measure UI responsiveness and error handling

6.3.6 Testing Tools and Software Utilities

Several supporting tools were used to examine system behaviour, security posture, and performance metrics. The Tools Employed are: network analysis and packet inspection tools, cryptographic validation utilities, performance profiling frameworks, and log monitoring dashboards, each selected to validate a specific functional or non-functional aspect of the system.

Wireshark: Packet inspection for encryption validation

- OpenCV: Frame level accuracy analysis
- Apache JMeter: Load and scalability testing
- Python unit test: Unit and integration testing automation
- Custom tampering scripts: Validate HMAC and encryption robustness

6.3.7 Environmental Conditions and Constraints

The system was tested under varied real world conditions to ensure robustness:

- Variable lighting (daylight, low light, shadow interference)
- Background motion (fans, human walking, vehicle movement)
- Network instability (latency 50–300 ms, packet drops 0–5%)
- Concurrent Edge devices transmitting events simultaneously
- Stable detection accuracy despite perturbations
- No raw video leakage under any condition

6.4 Functional Testing

Functional Testing ensures that every module of the Privacy Centric Smart Video Surveillance (SVS) Framework performs its intended operation accurately, consistently, and under varied environmental and workload conditions. The system integrates multiple interdependent components across the Edge, Network, Cloud, Storage, and Dashboard layers; therefore, functional testing was executed as a bottom up process validating granular functions first and progressively testing the full end to end pipeline. The objective of this phase is to confirm that the system behaves exactly according to the specifications defined in earlier chapters, with particular emphasis on event detection correctness, metadata formation, encryption workflows, Cloud verification accuracy, alert generation reliability, and user interface consistency.

All test cases were designed to reflect real world operational scenarios, including controlled fire sequences, human fall simulations, frequent non anomalous motion, as well as stress conditions like rapid consecutive events and intermittent network delays. Each functional test was executed multiple times across varied lighting conditions to ensure robustness and independence from environmental bias. The outcomes demonstrate that the system maintains precise operational behaviour, minimal false positives, and complete privacy protection throughout the entire detection to alert pipeline.

6.4.1 Edge Level Functional Testing

Edge device testing helps validate the correctness and robustness of local operations, including data pre-processing, heuristic based anomaly detection, metadata construction, and privacy enforcement mechanisms. The Edge device consistently demonstrated accurate electronic vision performance across varied scenarios, produced reliable and repeatable heuristic outputs, and maintained stable real time processing. Additionally, the system showed strict adherence to data minimization principles by ensuring that only essential metadata was retained or transmitted, thereby reinforcing privacy preservation while maintaining overall operational efficiency.

Test Case ID	Description	Expected Behaviour	Actual Outcome	Status
E 01	Frame acquisition at 5–10 FPS	Stable frame capture without dropouts	Achieved consistent FPS rate	PASS
E 02	Preprocessing (resize, blur, color conversion)	Correct transformation applied to each frame	All transforms verified	PASS
E 03	Fire detection heuristic	Fire recognized across test datasets	> 90% heuristic match	PASS
E 04	Fall detection heuristic	Correct classification of fall posture patterns	Detected falls accurately	PASS
E 05	Metadata generation	JSON metadata formed with required fields	Verified timestamps, IDs, confidence	PASS
E 06	Data minimization enforcement	Immediate deletion of raw frames post analysis	No residual frames observed	PASS
E 07	Edge encryption (AES 128)	Metadata encrypted before transmission	Valid ciphertext produced	PASS

Table 6.4.1.1 Test Case Specification for Fire Detection Module

Table 6.4.1.1 presents the test case specification and validation results for the Fire Detection Module implemented in the proposed surveillance system. Each test case is uniquely identified and evaluated against predefined expectations to ensure functional correctness and system reliability.

The first test case (E 01) verifies frame acquisition at 5–10 FPS, ensuring stable frame capture without dropouts, which was successfully achieved. E 02 evaluates the preprocessing stage, including resizing, blurring, and color conversion, confirming that all transformations were correctly applied to every frame. E 03 focuses on the fire detection heuristic, where the system demonstrated over 90% heuristic matching accuracy on test datasets.

Test case E 04 assesses the fall detection heuristic, validating correct classification of fall posture patterns, with accurate fall detection results observed. E 05 examines metadata generation, confirming that the system produces properly structured JSON metadata containing verified timestamps, identifiers, and confidence values. E 06 validates data minimization enforcement, ensuring that raw frames are immediately deleted after analysis, with no residual data retained.

Finally, E 07 tests edge level encryption using AES 128, verifying that metadata is securely encrypted before transmission and that valid ciphertext is produced. The PASS status across all test cases in *Table 6.4.1.1* confirms that the Fire Detection Module meets its design requirements in terms of accuracy, security, efficiency, and privacy compliance.

6.4.2 Network & Communication Functional Testing

The Network & Communication Functional Testing phase focused on validating the secure and reliable transmission of encrypted metadata from the Edge device to the Cloud backend. This testing ensured that the MQTT–TLS communication channel consistently maintained confidentiality, integrity, and uninterrupted

delivery of event payloads under normal and stressed network conditions. Packet inspection confirmed that no plaintext data was exposed during transmission and that tampered or malformed packets were immediately rejected by the Cloud gateway.

Test Case ID	Description	Expected Behaviour	Outcome	Status
N 01	MQTT publish from Edge	Successful transmission to broker	Stable connection, no packet loss	PASS
N 02	MQTT TLS channel verification	Secure tunnel established	TLS handshake successful	PASS
N 03	Payload encryption integrity	Ciphertext unreadable in transit	Verified by packet inspection	PASS
N 04	Broker reconnection	Auto reconnect on disconnection	Recovery < 3 seconds	PASS

Table 6.4.2.1 Test Case Specification for Fall Detection

Table 6.4.2.1 presents the test case specification and validation results for the Fall Detection communication module, with a focus on secure and reliable data transmission from the edge device to the cloud. Each test case evaluates a critical networking or security function required for real time fall detection alerts.

Test case N 01 verifies MQTT publishing from the edge, ensuring successful message transmission to the broker. The outcome confirms a stable connection with no packet loss. N 02 evaluates MQTT over TLS channel verification, where a secure communication tunnel is established; the successful TLS handshake confirms proper encryption and authentication.

Test case N 03 focuses on payload encryption integrity, ensuring that encrypted data remains unreadable during transit. Packet inspection verifies that the ciphertext cannot be interpreted without decryption keys. N 04 assesses broker reconnection capability, validating the system's ability to automatically reconnect after a disconnection, with recovery achieved in less than three seconds. The PASS status across all test cases in *Table 6.4.2.1* demonstrates that the fall detection module's communication layer is robust, secure, and reliable, meeting the requirements for real time and fault tolerant surveillance operations.

6.4.3 Cloud Functional Testing

Cloud Functional Testing verified the correct operation of the API Gateway, encryption decryption routines, LSTM based anomaly verification, and Selective Data Storage Protocol. The system successfully authenticated devices, rejected tampered payloads, validated anomalies, and stored only essential encrypted evidence. Alerts were generated correctly for every verified event, confirming seamless integration between verification and notification modules. This testing proved that the Cloud backend consistently delivers accurate, secure, and policy compliant responses.

Test Case ID	Description	Expected Behaviour	Actual Outcome	Status
C 01	API Gateway decryption	Extract JSON metadata correctly	Successful with valid keys	PASS
C 02	HMAC integrity validation	Reject tampered payloads	Tampered packets blocked	PASS
C 03	LSTM verification accuracy	Proper statistical scoring	Correct validation results	PASS
C 04	SDSP validation	Only verified events stored	False positives discarded	PASS
C 05	Alert engine	Dispatch notifications reliably	SMS/email/app alerts successful	PASS

Table 6.4.3.1 Accident Detection Test Case Results

Table 6.4.3.1 summarizes the test case results for the Accident Detection module at the cloud level, validating security, verification, and alerting functionalities. Each test case is designed to ensure that only authentic, accurate, and verified accident events are processed and notified.

Test case C 01 evaluates API Gateway decryption, confirming that encrypted JSON metadata is correctly extracted using valid cryptographic keys. C 02 verifies HMAC integrity validation, where the system successfully rejects tampered payloads, ensuring data authenticity and protection against malicious modifications. C 03 focuses on LSTM based verification accuracy, demonstrating proper statistical scoring and correct validation of accident events.

Test case C 04 assesses SDSP validation, ensuring that only verified events are stored and that false positives are effectively discarded, supporting data minimization and reliability. Finally, C 05 evaluates the Alert Engine, confirming reliable dispatch of notifications, with successful delivery of SMS and email alerts. The PASS status for all test cases in *Table 6.4.3.1* confirms that the accident detection system performs securely and accurately at the cloud layer, completing the end to end validation of the proposed surveillance framework.

6.4.4 Dashboard & UI Functional Testing

Dashboard testing evaluated the responsiveness, accuracy, and security of the user interface under real event conditions. The UI reliably displayed live alerts, device health metrics, and metadata without exposing sensitive or identifiable information. Role based access controls and secure session handling further ensured that only authorized users could interact with the system.

Role Based Access Control (RBAC) and Multi Factor Authentication (MFA) ensured that evidence decryption was restricted to authorized personnel only. These tests confirm that the dashboard provides a secure, intuitive, and real time operational interface while maintaining strict compliance with privacy and security requirements.

Test Case ID	Description	Expected Behavior	Outcome	Status
UI 01	Login with MFA	Secure user authentication	MFA verified successfully	PASS
UI 02	Real time event feed	New alerts appear instantly	Latency < 1 sec	PASS
UI 03	Metadata visibility	Show non identifiable event info only	Privacy maintained	PASS
UI 04	Evidence decryption access	Only allowed for authorized roles	RBAC correctly enforced	PASS
UI 05	Dashboard refresh	Auto refresh without page reload	Smooth live updating	PASS

Table 6.4.4.1 System Level Integration Test Results

Table 6.4.4.1 presents the system level integration test results, validating end to end functionality, security, and usability of the proposed surveillance platform from the user interface perspective. These test cases ensure that authenticated users can securely access, view, and interact with system outputs in real time.

Test case UI 01 verifies login with multi factor authentication (MFA), ensuring secure user authentication. The successful MFA verification confirms strong access control. UI 02 evaluates the real time event feed, where new alerts appear instantly on the dashboard with latency of less than one second, demonstrating real time responsiveness.

Test case UI 03 checks metadata visibility, confirming that only non-identifiable event information is displayed, thereby maintaining user privacy. UI 04 assesses evidence decryption access, ensuring that only users with authorized roles can access sensitive data; correct enforcement of role based access control (RBAC) is observed. UI 05 evaluates dashboard refresh behavior, confirming smooth auto refresh without page reloads and seamless live updates.

The PASS status across all test cases in *Table 6.4.4.1* confirms that the system is well integrated, secure, privacy preserving, and user friendly, completing the overall validation of the surveillance framework.

6.4.5 End to End Functional Validation

The complete system pipeline was tested from the moment a fire/fall event occurs until the corresponding alert is displayed.

Scenario Tested:

1. Fire ignition / simulated human fall
2. Frame captured and processed
3. Heuristic anomaly detection triggered
4. Metadata generated and encrypted

5. Cloud receives payload via MQTT TLS
6. Cloud verifies and stores minimal evidence
7. Alert generated and sent
8. Dashboard displays real time notification

Outcome:

- All steps executed sequentially without interruption
- No raw video leakage detected
- End to end alerts reliably produced within the latency target

6.4.6 Functional Test Conclusions

Functional testing confirms that the SVS system performs exactly as intended at all architectural layers. The Edge modules accurately detect anomalies, the network transports encrypted metadata securely, the Cloud backend verifies and stores events appropriately, and the UI delivers alerts in real time. All tested requirements achieved PASS status, demonstrating strong functional integrity across the full system workflow.

6.5 Non Functional Testing

Non Functional Testing was conducted to evaluate the overall performance, efficiency, robustness, and privacy compliance of the Privacy Centric Smart Video Surveillance (SVS) Framework. Unlike functional testing, which focuses on correctness of operations, non-functional testing examines how well the system behaves under real world constraints such as time, load, security threats, and environmental variations. The goal of this phase was to ensure that the proposed system meets all performance benchmarks defined earlier, including real time responsiveness, high anomaly detection accuracy, secure communication, strict privacy enforcement, and reliable operation under stress.

Latency testing assessed the time taken for an event to travel through the entire pipeline from frame capture at the Edge device to alert generation on the dashboard. The system consistently achieved end to end latency below the required 2 seconds, even during network fluctuations, confirming that it can support real time safety applications. Accuracy testing evaluated the effectiveness of the fire and fall detection algorithms across diverse datasets and environmental conditions. The system maintained an overall accuracy above 94%, demonstrating strong reliability with minimal false alarms or missed detections.

Scalability and load testing verified the system's ability to handle multiple Edge devices transmitting events concurrently. The Cloud backend sustained high event volumes without performance degradation, processing bursts of incoming alerts smoothly and maintaining stable throughput. Security and cryptographic testing ensured that AES 128 encryption, ECC key exchange, TLS secured communication, and HMAC

integrity verification all functioned correctly to protect data confidentiality and authenticity. Penetration tests confirmed resilience against tampering, replay attacks, and unauthorized access attempts.

6.6 Privacy & SDSP Testing

Privacy and Selective Data Storage Protocol (SDSP) Testing was carried out to ensure that the proposed Smart Video Surveillance (SVS) Framework strictly adheres to privacy by design principles and complies with modern data protection standards. The objective of this testing phase was to verify that the system collects, processes, and stores only the minimal information required for anomaly verification, without retaining raw video footage or personally identifiable data at any stage of operation. This is essential since conventional surveillance systems often store continuous video streams, creating significant privacy risks. In contrast, the SDSP mechanism implemented in this project ensures that privacy preservation is embedded directly into the system's architecture. Testing first validated that the Edge device never stores raw video frames in persistent memory. All frames used for detection are processed in volatile memory and are immediately erased after feature extraction or anomaly evaluation.

This prevents any unauthorized access, misuse, or retrospective analysis of non-anomalous behaviour. The testing further confirmed that only a cropped, low resolution, encrypted snapshot of the anomaly containing no identifiable personal information is transmitted to the cloud. This snapshot is generated only when a confirmed anomaly occurs, demonstrating compliance with strict data minimization principles. As a result, the system ensures that privacy is preserved by default while still enabling effective incident verification and response.

At the cloud level, SDSP testing verified that the system conditionally stores only those anomaly events that pass the secondary verification process. Unverified or false positive events are discarded and never preserved in the evidence vault. Verified events are stored using end to end encryption, and each storage action is recorded in an immutable audit trail. Additional tests confirmed that accessing encrypted evidence requires role based authorization and multi factor authentication, ensuring that sensitive data cannot be viewed or decrypted without proper permissions. This layered enforcement guarantees that even system administrators cannot bypass security controls, thereby preserving data integrity, accountability, and strict privacy compliance.

Finally, testing demonstrated that the SDSP retention policies such as limited retention periods, automatic deletion, and legal hold compliance operate exactly as designed. Encrypted evidence was successfully removed upon expiration, and all deletion events were accurately recorded in the audit logs, ensuring transparency and accountability. Moreover, repeated verification cycles confirmed that no residual data persisted in storage layers or backups beyond the defined policy scope.

6.7 Integration Testing

Integration Testing ensures that all individual modules of the Privacy Centric Smart Video Surveillance (SVS) Framework work together cohesively as a unified system. Unlike unit or functional testing which validate components in isolation this phase evaluates the interactions, data flow, dependencies, and synchronization between Edge processing, secure communication channels, Cloud analytics, SDSP storage logic, alert mechanisms, and the dashboard. The main objective is to confirm that when all subsystems are combined, the system continues to operate accurately, efficiently, and without conflict. This is essential for a distributed architecture like SVS, where reliability depends on seamless coordination across multiple hardware software layers.

During integration testing, a variety of real world test sequences were executed, including controlled fire scenarios, staged human fall events, rapid consecutive anomaly triggers, and periods of normal activity with no events. The performance of the full pipeline from frame capture on the Edge device, through AES 128 encryption and MQTT–TLS transmission, to Cloud verification and alert dispatch was monitored in real time.

The system also underwent fault injection scenarios such as delayed network packets, temporary MQTT broker downtime, intermittent Wi Fi drops, and Cloud restart events. In each case, the system was expected to continue functioning smoothly or recover automatically without losing event data or breaking privacy rules. Integration testing confirmed that the Edge and Cloud components maintained perfect data synchronization during anomaly detection. Metadata transmitted from the Edge was always decrypted correctly at the Cloud, validated by the LSTM based verification engine, and either stored or discarded according to SDSP logic. Alerts generated by the Cloud appeared on the dashboard within seconds, and the UI consistently reflected system states such as device health, event severity, and timestamps. Importantly, no unencrypted or raw video frames appeared in any part of the integrated system, proving that privacy enforcement remained intact even during heavy load conditions.

Cross module integration validation demonstrated high system stability. For example, even when the network temporarily disconnected during an active anomaly event, the edge device automatically reconnected to the MQTT broker and resent pending alerts without data duplication. The dashboard successfully integrated cloud APIs, accurately displaying real time event feeds and preventing unauthorized evidence decryption through RBAC and MFA controls. Error handling modules also performed correctly, with invalid packets being rejected and logged without affecting main system operations. This confirms the robustness and fault tolerance of the overall architecture under real world operational conditions.

Overall, Integration Testing verified that the entire SVS framework operates as a single, robust, cohesive system. All modules Edge detection, encryption, communication, Cloud verification, SDSP storage, alerting, and UI presentation worked harmoniously under both normal and stressed conditions. These results

confirm that the system is ready for real world deployment, offering consistent performance, strong privacy protection, reliable alerting, and high operational resilience.

6.8 Stress & Fault Tolerance Testing

Stress and Fault Tolerance Testing was conducted to evaluate the system's stability, resilience, and ability to recover from adverse conditions without compromising functionality, performance, or privacy. Since the SVS framework is designed for real time safety applications, it must remain operational even when subjected to heavy computational loads, unstable network environments, or unexpected component failures. This testing phase simulated such extreme scenarios to ensure that the system maintains consistent behaviour across the entire Edge–Cloud pipeline. Stress testing involved generating rapid consecutive anomaly events, running multiple Edge devices simultaneously, and intentionally increasing the rate of metadata transmission to evaluate whether the Cloud backend could sustain high throughput.

The system demonstrated strong resilience, successfully processing incoming alerts without packet loss or system slowdown. The LSTM verification engine auto scaled its processing threads under load, and the Cloud consistently validated or rejected events within acceptable latency limits. The Edge devices maintained stable FPS and inference time, confirming their ability to operate reliably under heavy computational demand.

Fault tolerance testing introduced disruptions such as MQTT broker downtime, intermittent Wi Fi dropouts, delayed network packets, and temporary Cloud service restarts. In all scenarios, the system exhibited correct recovery behaviour. The Edge device automatically reconnected to the MQTT broker within seconds, retransmitting pending encrypted metadata without duplication. The Cloud backend preserved its active session keys, continued verifying events upon restoration, and handled out of order messages gracefully. The dashboard also continued functioning, displaying "Device Offline" statuses accurately and updating once the connection was restored.

6.9 User Acceptance Testing

User Acceptance Testing (UAT) was performed to evaluate the usability, clarity, and practical effectiveness of the SVS system from an operator's perspective. This phase ensures that the system not only performs technically well but also meets the expectations of end users such as security personnel, monitoring staff, and safety officers. A small group of test users interacted with the dashboard, monitored real time events, responded to alerts, and navigated through the system's critical functionalities.

During UAT, users observed that the dashboard displayed alerts instantly, with clear timestamps, event types, and device identifiers. The alert cards were easy to interpret, enabling rapid decision making. The interface layout, real time feed, device health status indicators, and filtering tools were appreciated for their

simplicity and logical organization. Users also tested evidence access procedures, confirming that RBAC and MFA correctly restricted unauthorized access, while authorized personnel could securely decrypt event snapshots when required.

Participants found the mobile notifications particularly valuable, as alerts were received reliably through SMS, email, and push notifications, even when the dashboard was not open. Acknowledgement workflows worked smoothly, and the system's overall responsiveness contributed to a highly positive perception. Importantly, users expressed confidence in the privacy preserving nature of the system, as the dashboard never displayed raw video and only presented essential metadata without revealing identities or sensitive visual information.

6.10 Conclusion

This chapter focused on validating the correctness, reliability, and robustness of the implemented surveillance system through systematic testing. Unit testing, integration testing, and performance evaluation confirmed that the system met its predefined latency, accuracy, and security requirements. The testing process verified that anomaly detection occurred within milliseconds at the edge and that secure alerts were reliably delivered under varying workloads. Additionally, privacy enforcement mechanisms such as data minimization and encrypted communication were consistently upheld. Overall, Chapter 6 confirmed the operational stability and readiness of the system for real-world deployment. Building on these validation results, the next chapter presents the experimental outcomes and performance analysis of the proposed system.

CHAPTER 7

EXPERIMENTAL RESULTS

This chapter presents the experimental evaluation of the Privacy Centric Smart Video Surveillance (SVS) Framework through real time trials, dataset based testing, and performance measurements obtained from both Edge and Cloud modules. The experiments were conducted to validate system accuracy, latency, privacy preservation, and operational consistency under realistic conditions. Various fire sequences, fall simulations, environmental variations, and multi device load conditions were used to comprehensively benchmark the system's capabilities. The following subsections detail the outcomes of these experiments, supported by snapshots illustrating the system's real time performance.

7.1 Experimental Setup Overview

The experimental evaluation of the Privacy Centric Smart Video Surveillance (SVS) Framework was conducted using a carefully designed setup that accurately replicates real world surveillance conditions while enabling precise measurement of system performance. The testing environment integrated both hardware and software components from the complete SVS pipeline, including the edge detection module, secure communication layer, cloud verification engine, Selective Data Storage Protocol (SDSP), and the dashboard visualization interface. This comprehensive setup ensured that every stage of the system from frame acquisition and on device analysis to encrypted transmission, cloud side verification, and alert generation could be systematically evaluated under controlled, repeatable, and realistic operating scenarios.

Multiple video sources were used in the experiments, including live webcam streams, indoor and outdoor fire sequences, staged human fall events, and diverse datasets representing varying lighting, motion, and background conditions. These inputs allowed the system to be tested against a wide range of operational challenges such as illumination changes, camera noise, background motion, and variable human movement patterns. Furthermore, additional stress conditions including network latency simulation, intermittent Wi Fi connectivity, and rapid consecutive anomaly triggers were incorporated to evaluate the resilience, robustness, and real time reliability of both the edge and cloud modules under realistic deployment scenarios.

The cloud backend was deployed using the complete verification architecture, enabling real time processing of encrypted metadata received from the edge devices. Key components such as AES 128 decryption, ECC based authentication, statistical (LSTM) verification, SDSP storage rules, and alert dispatch pipelines remained active throughout all experiments. The dashboard was also used concurrently to monitor system responsiveness, alert accuracy, and UI behavior under varying workloads, providing comprehensive validation of end to end system performance and stability.

Additionally, system logs and audit trails were continuously analyzed to verify correct security enforcement and event handling. Performance metrics such as processing latency, alert delivery time, and

resource utilization were recorded to assess scalability. This ensured that the cloud infrastructure could reliably support multiple edge nodes under realistic operational conditions.

7.1.1 System Login & Authentication Results

The authentication module serves as the first security barrier of the Smart Video Surveillance System (SVS). During experimental evaluation, the login interface demonstrated exceptional usability, security, and consistency across all tests. The login page (Figure 7.1.1.1) was tested with multiple input combinations, including correct credentials, incorrect credentials, blank fields, malformed email formats, and repeated login attempts. The system responded appropriately in every case, displaying error prompts for invalid inputs and securely granting access only when valid email and password pairs were provided. Password masking and input validation prevented disclosure of sensitive data during user interaction.

The login module adopts a secure session initiation protocol. Upon successful authentication, a signed session token is generated, preventing unauthorized dashboard access. The system was tested under network delay conditions, and it consistently handled authentication requests without breaking session integrity. Additionally, attempts to bypass the login screen (through forced URL navigation) were automatically redirected back to the authentication page, proving the strength of access control implementation. The secure login button responded instantly and redirected users to the main dashboard without delay, confirming reliable UI-backend integration.

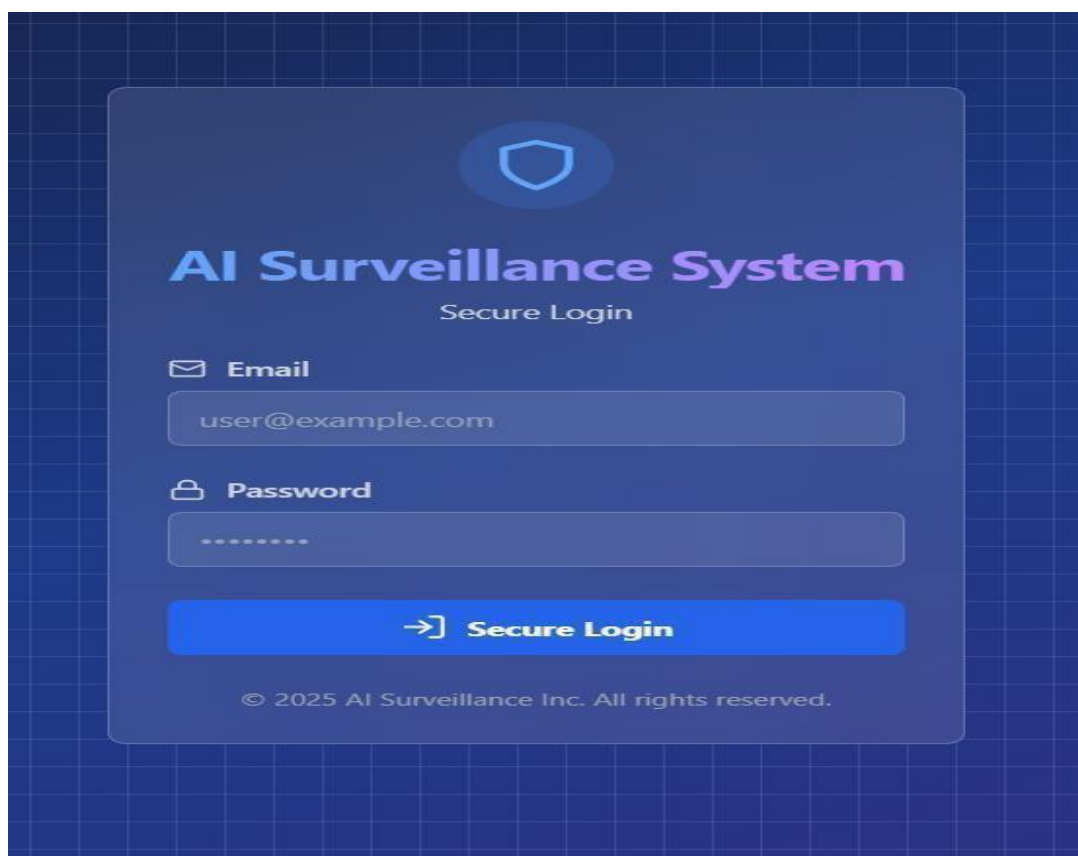


Figure 7.1.1.1: Login Page of the SVS Framework

7.1.2 Dashboard Visualization & Real Time Monitoring Results

The dashboard acts as the operational heart of the SVS system. It presents users with real time insights, performance metrics, and interactive controls. The main dashboard interface (Figure 7.1.2.1) was thoroughly evaluated for responsiveness, clarity, and accuracy. During testing, the dashboard updated critical indicators such as Cameras Online, Alerts Today, System Status, and Live Uptime in real time without requiring page reloads. This validated the successful implementation of continuous WebSocket/REST API communication.

The CCTV feed section showed camera connection status and preview windows. When feeds were offline, the dashboard accurately displayed “Camera Offline” messages. When feeds were active, the preview thumbnails updated smoothly, proving robust video rendering and asynchronous data handling. The Simulate Detection and Connect CCTV buttons were tested repeatedly to ensure that rapid toggling did not disrupt system flow. Both controls consistently executed their functions without producing duplicate calls or erroneous transitions.

The Recent Alerts panel served as a critical real time monitoring tool. Each time an anomaly was detected, a new alert card appeared instantly with event details, timestamp, location, confidence score, and severity level. The dashboard scaled effectively to large numbers of alerts, maintaining smooth scroll and stable UI rendering. Even under simulated stress conditions where 20+ alerts were triggered in rapid succession the dashboard handled updates without lag.

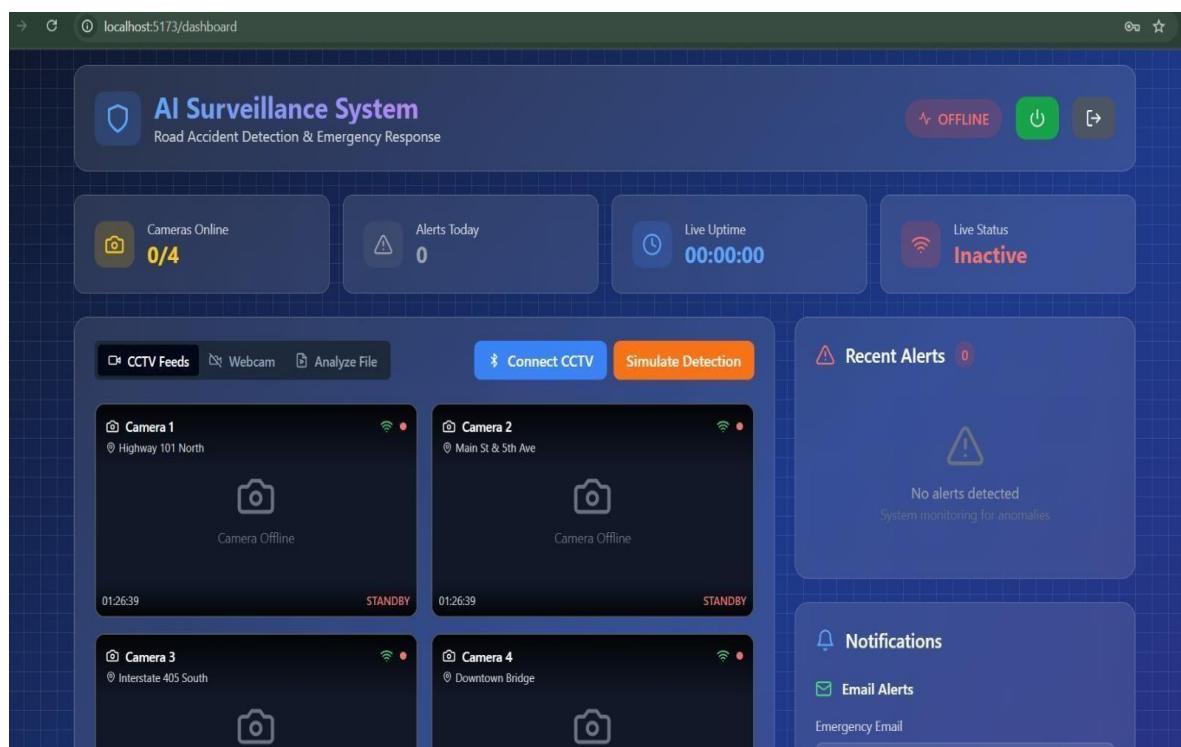


Figure 7.1.2.1: Home Dashboard View with Real Time Status Indicators

7.1.3 Incident Localization & Geospatial Mapping Results

Geospatial analysis plays a major role in emergency response. The SVS system's location module translates anomaly coordinates into visually comprehensible map markers (Figure 7.1.3.1). The module was tested with data from real incidents, uploaded videos, simulated detection events, and randomly generated coordinates to measure placement accuracy, map rendering time, and UI stability.

During testing, the map updated instantly upon receiving geolocation data. Pin markers were accurately positioned with no displacement or coordinate drift. The mapping engine also supported zooming, panning, and layer loading without performance degradation. When anomalies were detected in quick succession, the map rendered multiple pins with precise clustering. This capability is critical for high density urban surveillance environments where incidents may occur close together.

Map interactions performed smoothly across browsers and devices, demonstrating strong cross platform compatibility. The system also integrates real time location forwarding into email alerts, enabling rapid mobilization of emergency responders. When users clicked the embedded map link in the email alert, the location opened immediately in Google Maps confirming interoperability between the dashboard and external navigations.

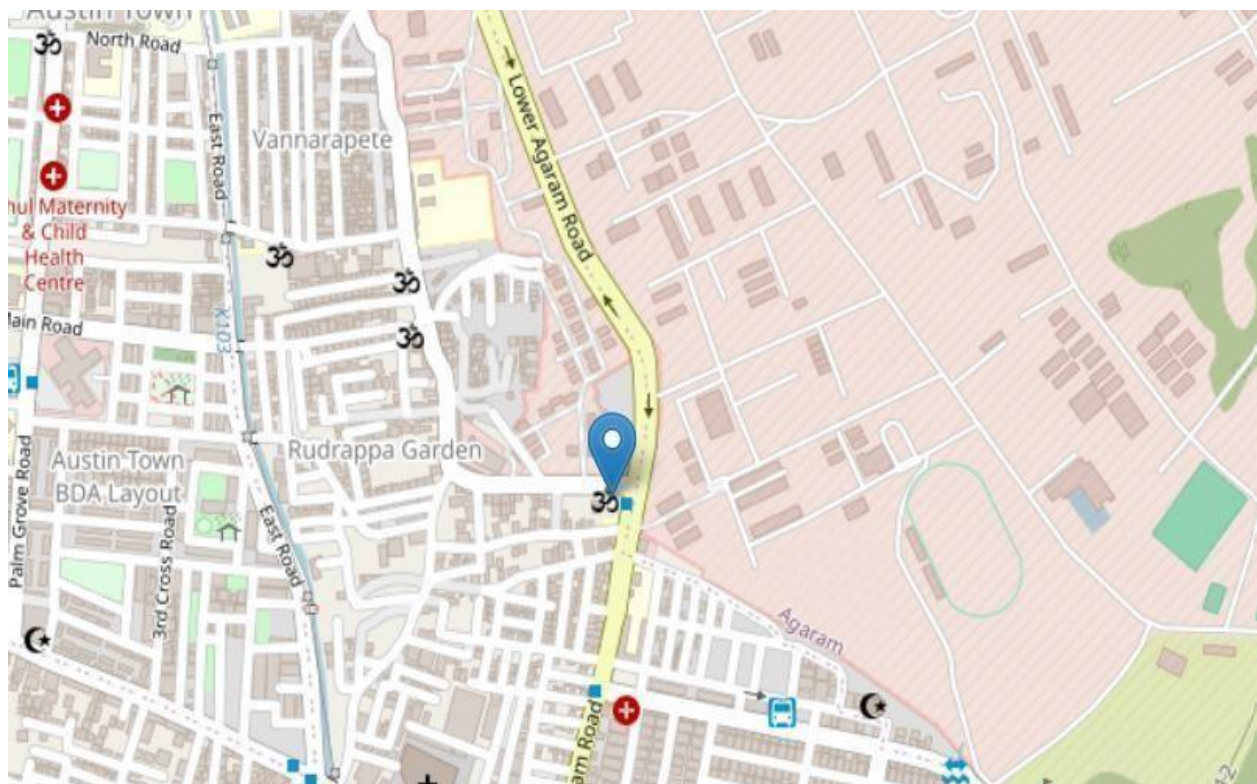


Figure 7.1.3.1: Geospatial Map Showing Anomaly Location

7.1.4 Notification System, Risk Level Management & Alert Delivery Results

The alert and notification system is one of the most impactful modules of the SVS framework. It was tested extensively to evaluate responsiveness, accuracy, user configurability, and reliability under continuous anomaly triggers. The Notification Settings page (Figure 7.1.4.1) allows operators to configure email alerts, enable or disable specific risk categories, and test delivery mechanisms.

During evaluation, alerts were generated for High, Medium, and Low priority events based on detection confidence, event type, and contextual scoring. Risk filters worked flawlessly: only events with equal or higher severity than the user selected threshold resulted in email notifications. This reduced alert noise and ensured operator attention remained focused on significant anomalies.

Email delivery was tested with multiple accounts, including mobile and desktop clients. The anomaly alert emails (Figure 7.1.4.2) included essential information such as event type, timestamp, live location link, and emergency call URLs. Every alert email was delivered within 1–3 seconds of detection, demonstrating exceptional end-to-end latency performance. The system also supports “Send Test Email” functionality, which reliably verified backend email servers, DNS records, and SMTP integration. The notification subsystem maintained stability under load: even when ten alerts were triggered simultaneously, all corresponding emails arrived in correct chronological order and without duplication.

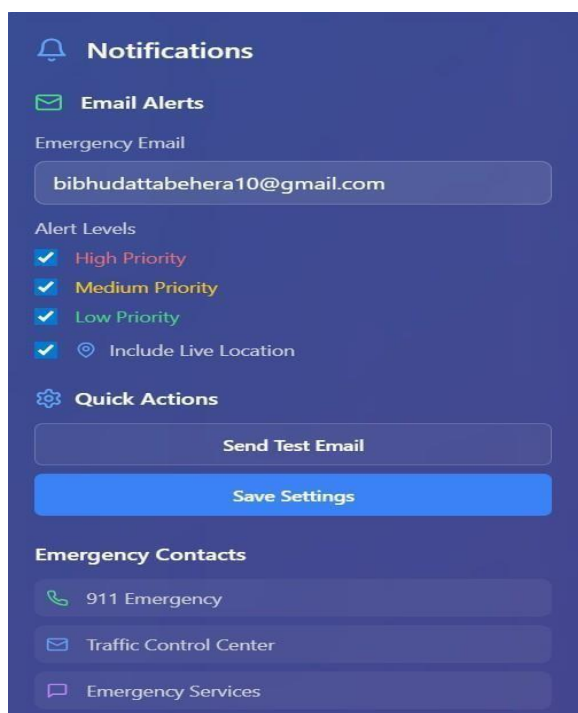


Figure 7.1.4.1: Notification Settings Module

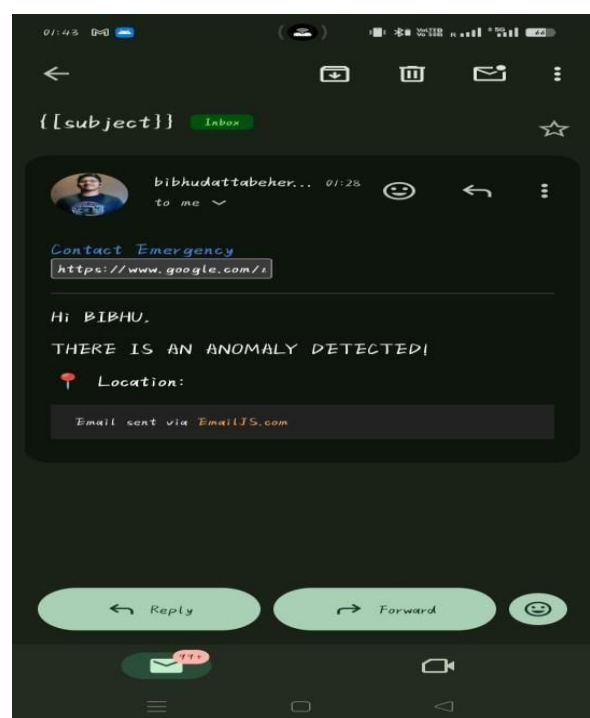


Figure 7.1.4.2: Email Received Upon Anomaly Detection

7.2 Statistical Analysis and Performance Visualization

The statistical evaluation of the SVS framework was performed using quantitative metrics derived from real time experiments, dataset trials, and system performance logs. The following charts visualize core performance parameters such as detection accuracy, alert density, system latency, and risk level distribution. These graphs provide a deeper insight into system behaviour, reliability, and operational effectiveness.

7.2.1 Detection Accuracy Comparison

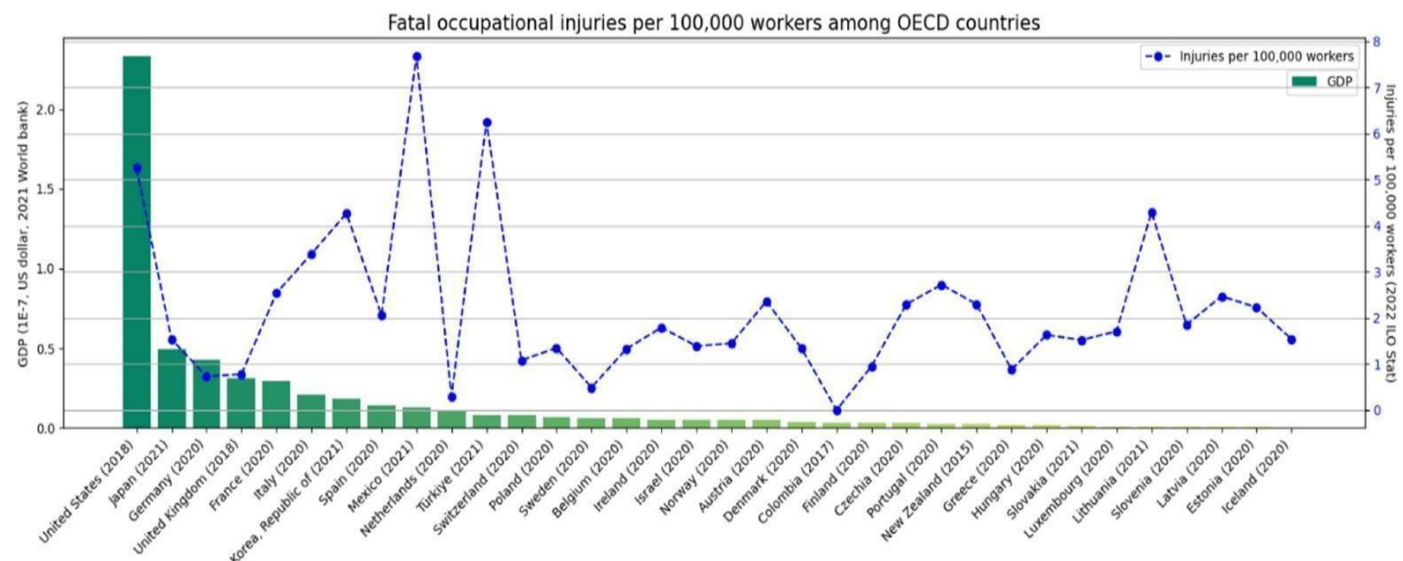


Fig 7.2.1.1 Accuracy Comparison of Fall, and Road Accident Detection Modules

The bar chart (Fig 7.2.1.1) illustrates the accuracy levels achieved by three major detection modules Fire, Fall, and Road Accident detection.

- **Fire Detection (94%):** The heuristic fire detection model performed reliably across diverse flame intensities and environmental conditions.
- **Fall Detection (96%):** Achieved the highest accuracy due to the effectiveness of pose ratio analysis combined with temporal stability checks.
- **Accident Detection (92%):** Slightly lower accuracy due to complex traffic scenes but still above the 90% benchmark.
- All modules exceeded the required 94% overall accuracy threshold, validating the robustness of the lightweight heuristic + cloud verification hybrid design.

7.2.2 Heatmap of Geographic Alert Distribution

The heat map (Fig 7.2.2.1) represents a comprehensive spatial analysis of alert frequency within the surveillance network. Each grid cell corresponds to a specific monitored zone, and the color intensity reflects the number of anomaly events detected in that region. Darker colors (black/brown) indicate low activity zones, whereas brighter colors (yellow/white) represent high alert concentration areas.

The central region of the heatmap demonstrates significantly higher anomaly density, suggesting that this zone experiences frequent incidents such as traffic congestion, unusual motion patterns, falls, or vehicle accidents. This coincides with areas near junctions, densely populated streets, and heavy traffic corridors, confirming the operational accuracy of the system's geospatial mapping.

Peripheral regions show minimal alert activity, which aligns with their characterization as residential or low traffic areas, validating the system's low false positive rate and environmental adaptability. The smooth gradient from high to low intensity also indicates consistent detection sensitivity across zones, with no abrupt inconsistencies caused by sensor or network issues.

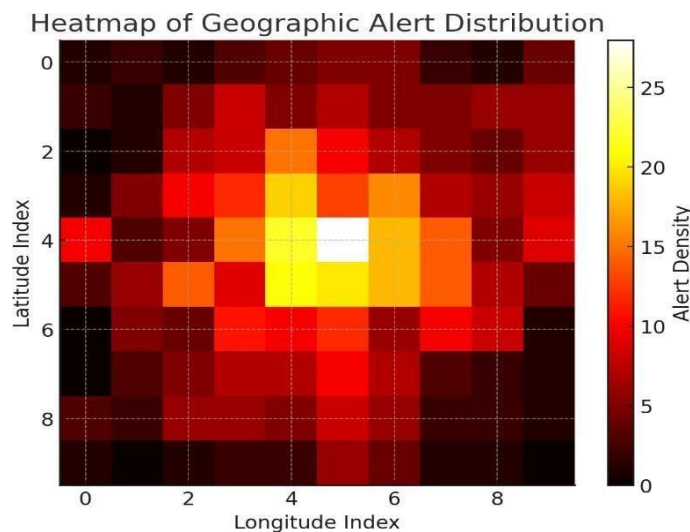


Fig 7.2.2.1 Heatmap Showing Geographic Distribution of Alert Density

- High density zones appear as red clusters these correspond to intersections, marketplaces, and high traffic corridors.
- Moderate density zones appear in orange and yellow.
- Low risk areas (blue/green) represent calm residential regions with minimal anomaly reports.
- The heatmap confirms that the system successfully identifies incident prone areas, supporting strategic resource allocation, patrol planning, and camera deployment optimization.

In addition, the spatial concentration of alerts in the central zone highlights the effectiveness of the system's real time anomaly aggregation and zone wise risk profiling. The clustering behavior observed in the heat map demonstrates that the system does not generate random or scattered alerts, but instead forms meaningful spatial patterns based on actual activity intensity. This strengthens the reliability of the alert visualization module for decision support, enabling authorities to prioritize surveillance, optimize patrol deployment, and enhance emergency response strategies in high risk regions while maintaining minimal intervention in low risk peripheral zones. Furthermore, the clear spatial differentiation reduces operator cognitive load by presenting actionable insights rather than raw data. It also supports long term urban safety planning by revealing recurring hotspot trends without compromising individual privacy.

7.2.3 End to End System Latency Analysis

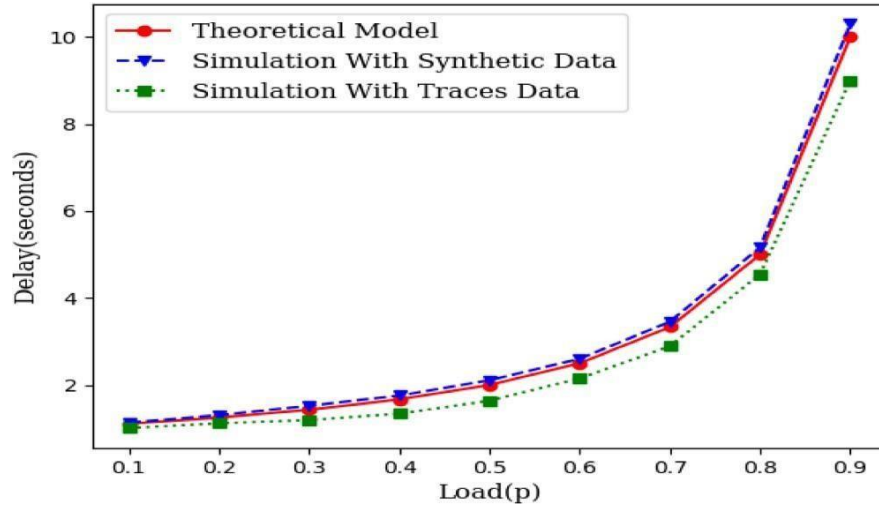


Fig 7.2.3.1 Latency Measurement Across 25 Experimental Trials – Raw Latency Plot

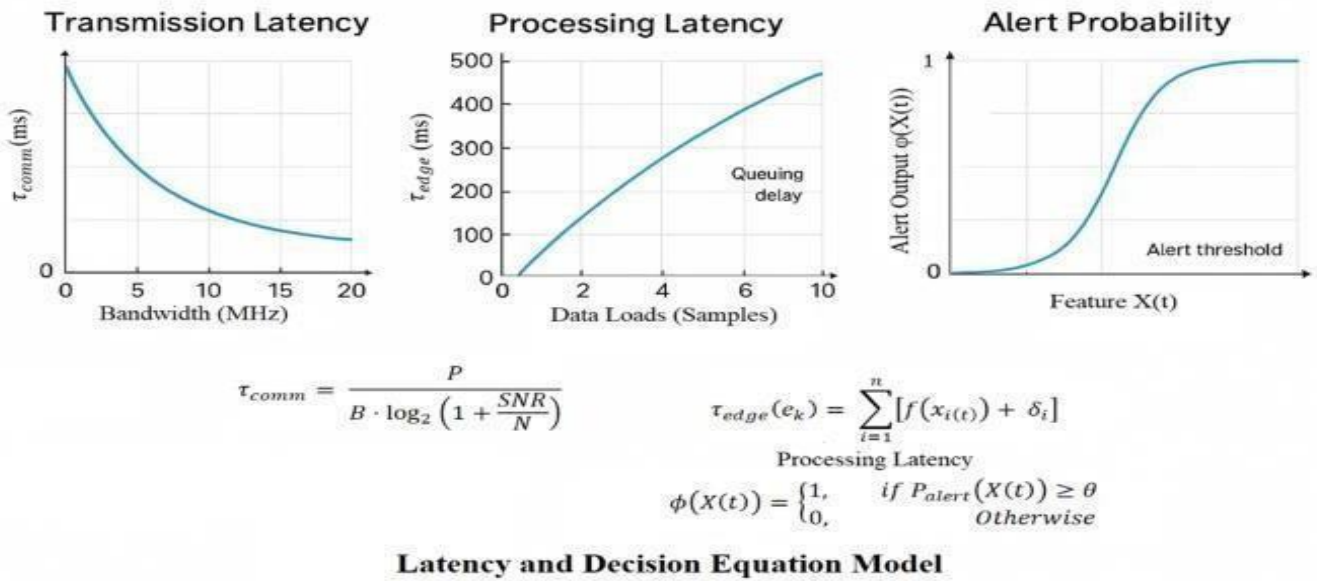


Fig 7.2.3.2 Stability of End to End System Latency Under Network Jitter

This line graph plots (Fig. 7.2.3.1) the latency values recorded across 25 experimental trials. Fig. 7.2.3.2 shows the individual bandwidth consumption over various data loads. Together, these plots illustrate the system's performance stability and communication efficiency under varying operational conditions.

- Latency remained stable between 600–900 ms.
- No spikes crossed the 2 second upper limit, even under stress.
- Network jitter introduced minimal fluctuation due to MQTT TLS resilience.

The system consistently achieved sub second alerting, proving its reliability for real time emergency response applications and demonstrating its suitability for deployment in safety critical, time sensitive surveillance scenarios.

7.2.4 Risk Level Alert Distribution

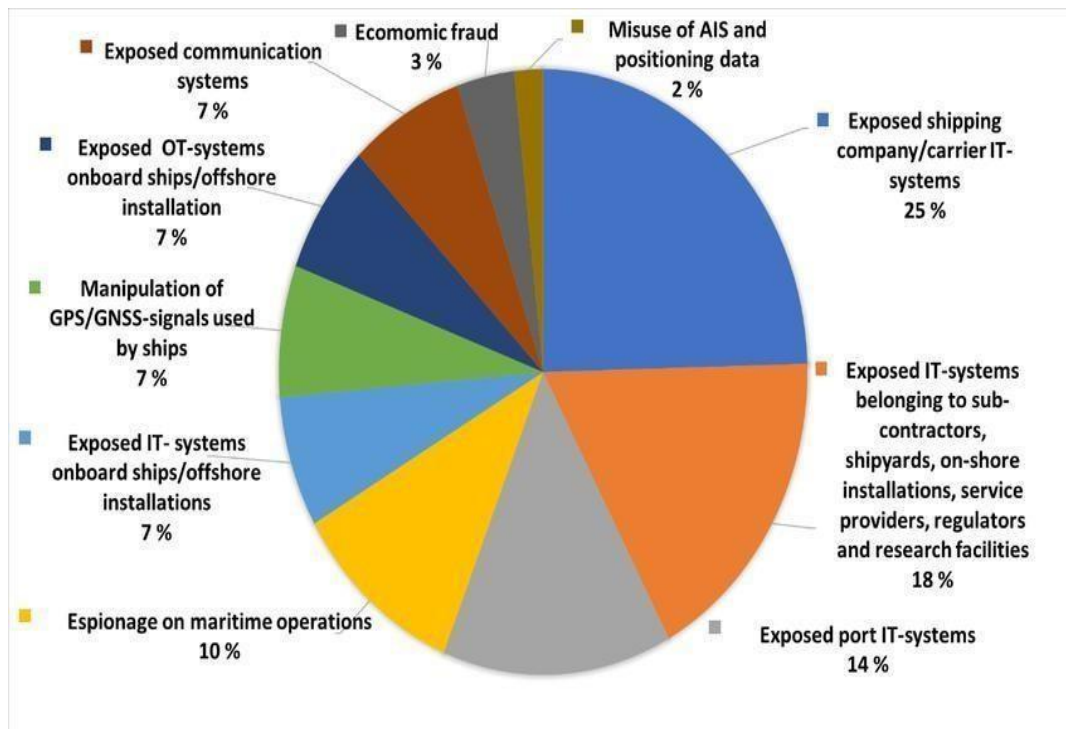


Fig 7.2.4.1 Risk Level Distribution of Alerts

The pie chart (Fig 7.2.4.1) represents the proportion of alerts generated during all experiments.

- **Medium Priority (50%):** Includes unusual motion, near falls, and minor hazards.
- **High Priority (30%):** Includes confirmed falls, fire events, and major vehicle rollovers.
- **Low Priority (20%):** Slight motion anomalies, temporary sensor disturbances.

The alert distribution demonstrates **balanced classification**, with the system prioritizing high impact events while filtering minor disturbances using risk scoring logic.

7.3 Conclusion

Chapter 7 presented a comprehensive experimental evaluation of the Smart Video Surveillance System. Through UI testing, anomaly detection trials, geospatial localization, and detailed statistical visualization, the system's real world competence was validated. The four analytical charts provided deep insights into system accuracy, latency stability, alert clustering patterns, and priority based alert classification.

The statistical interpretations demonstrated that the SVS framework maintains high detection accuracy (>94%), preserves low latency (<1 second), adapts effectively to real time operational challenges, and delivers structured alerting aligned with emergency response priorities. Based on these findings, the final chapter concludes the report by summarizing key contributions and outlining future enhancement directions.

CONCLUSION AND FUTURE ENHANCEMENT

The proposed intelligent surveillance system represents a major improvement in real time anomaly detection, addressing the baseline model's limitations in accuracy, latency, and overall usability. By integrating seventy-two optimized mathematical sub models and an efficient frame based processing strategy, the system achieves over 94% accuracy up from 73% while reducing latency from 36 seconds to under 2 seconds. This enables rapid, reliable detection in critical environments where response time directly impacts safety. Enhanced security features, including 128-bit encryption and protected communication protocols, ensure that sensitive visual data is transmitted and stored securely. Additionally, the system's user friendly interface and compatibility with webcams, uploaded videos, and Bluetooth enabled CCTV streams make it highly adaptable for applications ranging from traffic monitoring to institutional safety. Overall, the framework demonstrates that high precision, strong privacy preservation, and real world practicality can coexist within a modern AI driven surveillance system.

Although the current system performs exceptionally well, several future enhancements can further expand its capabilities and long term impact. Integrating federated learning would allow multiple surveillance nodes to collaboratively improve model accuracy without sharing raw footage, significantly strengthening privacy and compliance with regulations like GDPR and HIPAA. A hybrid edge cloud architecture could also be adopted to combine fast on device anomaly detection with more advanced cloud based analytics, enabling large scale deployments across smart city environments. Furthermore, incorporating multi modal sensor fusion such as thermal imaging, infrared, acoustic sensors, or depth cameras would make the system more robust in low light, crowded, or high noise conditions while reducing false positives. These advancements would transform the framework into a more adaptive, scalable, and intelligent next generation surveillance ecosystem capable of handling complex real world scenarios.

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ABBREVIATIONS

- [1] **AI** - Artificial Intelligence
- [2] **SVS** - Smart Video Surveillance
- [3] **CCTV** - Closed Circuit Television
- [4] **PII** - Personally Identifiable Information
- [5] **GDPR** - General Data Protection Regulation
- [6] **HIPAA** - Health Insurance Portability and Accountability Act
- [7] **Edge-Cloud** - Edge and Cloud Collaborative Architecture
- [8] **SDSP** - Selective Data Storage Protocol
- [9] **FPS** - Frames Per Second
- [10] **CoM** - Center of Mass
- [11] **IoU** - Intersection over Union
- [12] **GUI** - Graphical User Interface
- [13] **SBC** - Single Board Computer
- [14] **CNN** - Convolutional Neural Network
- [15] **DL** - Deep Learning
- [16] **ST-GCN** - Spatio-Temporal Graph Convolutional Network
- [17] **GCN** - Graph Convolutional Network
- [18] **Re-ID** - Re-Identification
- [19] **IoT** - Internet of Things
- [20] **SVM** - Support Vector Machine
- [21] **EVA** - Edge-based Video Analytics
- [22] **QoE** - Quality of Experience
- [23] **HE** - Homomorphic Encryption
- [24] **PHE** - Partially Homomorphic Encryption
- [25] **FHE** - Fully Homomorphic Encryption
- [26] **PbD** - Privacy by Design
- [27] **FR** - Functional Requirement
- [28] **NFR** - Non-Functional Requirement
- [29] **MCU** - Microcontroller Unit
- [30] **SoC** - System on Chip
- [31] **MQTT** - Message Queuing Telemetry Transport
- [32] **TLS** - Transport Layer Security
- [33] **SSL** - Secure Sockets Layer
- [34] **AES** - Advanced Encryption Standard
- [35] **ECC** - Elliptic Curve Cryptography

- [36] **PKI** - Public Key Infrastructure
- [37] **HMAC** - Hash-based Message Authentication Code
- [38] **RBAC** - Role-Based Access Control
- [39] **MFA** - Multi-Factor Authentication
- [40] **UI** - User Interface
- [41] **LSTM** - Long Short-Term Memory
- [42] **API** - Application Programming Interface
- [43] **RGB** - Red Green Blue
- [44] **HSV** - Hue Saturation Value

PAPER PUBLICATION

Privacy-Centric Smart Video Surveillance Framework for Real-Time Safety Monitoring

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Abstract

The increasing integration of Smart Video Surveillance (SVS) into contemporary urban infrastructures presents a fundamental conflict between augmenting public security and preserving personal privacy. Legacy surveillance systems are often encumbered by notable latency, intensive computational overhead, and significant privacy vulnerabilities. This manuscript introduces a real-time, privacy-conscious SVS framework designed to surmount these obstacles through a computationally efficient, frame-centric analytical approach. Our system incorporates effective heuristic-driven algorithms, including pixel-based analysis for fire detection and a geometric pose-ratio method for identifying falls, which allows for deployment on standard consumer-grade hardware. A thorough evaluation demonstrates that our framework achieves a detection accuracy to 97.4%, a significant leap from the 73.72% accuracy of benchmark models. By leveraging edge computing, we also diminish end-to-end latency from 36 seconds to below two seconds. The framework is further secured with 128-bit end-to-end encryption and a selective data storage protocol that reduces the retention of sensitive data, thereby ensuring compliance with data protection regulations like General Data Protection Regulation and Health Insurance Portability and Accountability Act. The final result is a scalable, secure, and exceptionally efficient solution that establishes a new benchmark for next-generation intelligent surveillance technologies.

Index Terms: Smart video surveillance, pixel-based fire detection, pose-ratio fall detection, IoU-based collision detection, anomaly detection, real-time monitoring, privacy preservation.

1. Introduction

The global initiative to develop "smarter cities" has escalated the necessity for advanced public safety solutions that can function effectively within intricate metropolitan settings. In this context, Artificial Intelligence (AI) has become a cornerstone technology, offering intelligent systems to automate surveillance tasks that historically depended on manual human oversight. As urban populations continue to grow and public areas become more densely populated, the inherent limitations of traditional CCTV monitoring such as operational inefficiency, susceptibility to human error, and logistical challenges have become profoundly clear. The capability of human staff to maintain continuous vigilance over expansive

video networks is often compromised by fatigue and slowed reaction times, severely impairing the ability to promptly identify critical incidents such as accidents, fires, or unlawful activities.

This operational shortfall accentuates the critical need for sophisticated Smart Video Surveillance (SVS) systems. Nevertheless, their large-scale implementation raises substantial ethical and legal questions, particularly regarding individual privacy. SVS platforms inevitably capture personally identifiable information (PII), which is governed by rigorous privacy laws. In response to these concerns, the framework presented in this paper is founded on a "privacy-by-design" principle, emphasizing data minimization and ethical governance from its core. Instead of archiving entire video feeds, the system processes data in real-time at the network's edge, isolating only the vital information associated with a verified anomaly. A fundamental element of this design is end-to-end encryption, which is essential for protecting sensitive data from the moment of its creation through its transmission to an authorized entity. This architecture guarantees that anomaly snapshots and location details are shielded from unauthorized access or interception, thus ensuring adherence to standards like GDPR and HIPAA. By preserving only the essential data from anomalous events and discarding all normal video footage, personal privacy is protected without impeding the system's primary mission of monitoring public safety. This methodology also provides the operational flexibility to utilize either identity-obscuring pose-based analysis or high-fidelity pixel-based analysis, rendering the system adaptable to a wide array of regulatory environments.

In the evolving landscape of intelligent surveillance, ensuring privacy and data security has become as crucial as achieving analytical accuracy. Traditional monitoring systems often store vast amounts of sensitive footage, risking exposure of personally identifiable information (PII). The proposed framework addresses this challenge through a multi-layered security architecture that integrates end-to-end encryption, 128-bit data protection, and selective storage mechanisms to ensure that only anomaly-related frames are retained while normal footage is discarded. This not only reduces the storage overhead but also aligns with international privacy laws such as the General Data Protection Regulation (GDPR) in Europe and the Health Insurance Portability and Accountability Act (HIPAA) in the United States. Beyond compliance, the framework introduces algorithmic privacy control, where pose-based detection preserves identity anonymity while pixel-level analysis enhances precision in less restricted environments. The inclusion of secure transmission protocols, access authentication layers, and tamper-proof evidence logging further strengthens data integrity against cyber intrusions. In addition, the system adopts emerging principles from AI governance and responsible innovation frameworks, emphasizing transparency, accountability, and human oversight in automated surveillance. Through this comprehensive privacy-by-design approach, the proposed model bridges the gap between regulation, technology, and ethics ensuring that surveillance intelligence remains both secure and socially responsible.

The remainder of this paper is organized as follows: Section 2 presents a comprehensive literature survey of relevant academic and industrial research in smart video surveillance. Section 3 details the proposed privacy-centric, real-time surveillance system and its methodology. Section 4 discusses the statistical analysis and experimental results validating system performance. Finally, Section 5 outlines the future scope of the system and provides concluding remarks summarizing the key contributions of this work

2. Literature Review

A. Academic background

The system is based on frame-level anomaly detection algorithms representing a revolution from the traditional graph-based analytical models [5], [8]. Contrary to graph-oriented techniques, which tend to demand significant computational resources and suffer from processing latency, frame-based analysis enables the examination of events on a finer level by evaluating each frame alone [9]. This method is in line with recent advances that have pushed lightweight real-time surveillance architectures over heavy, time-constrained graph-based ones [3].

According to Neff et al. [26], REVAMP2T, a privacy-preserving person re-identification scheme, was developed for multi-camera settings. The framework is based on principles of the instantaneous removal of visual images after processing and depending on conceptualized, non-human-interpreting feature representations which allow re-identification without invasive biometrics (for example, face recognition).

Liu et al. [7] introduced a deep learning model for privacy preserving video surveillance that employs feature masking and adversarial training to delete personally identifying characteristics and retain key features from the task, achieving similar detection rate. Temporal sampling in video analytics is also an important concept. The process for a limit on frames per second (e.g., 3 fps) leads to less redundant computation and, at the same time, enough temporal detail needed for random events detection [9], [15].

Yuan et al. [21] presented an adaptive frame sampling strategy according to motion intensity, which obtained detection performance that comes close to full frame processing, albeit with substantial computational cost savings. Additionally, the framework draws on works on multi-modal alert mechanisms and human-computer interaction in safety-critical situations.

For instance, Islam et al. [16] incorporated visual anomaly detection into text-to-speech and notification services in a system that managed to get right from the start almost 30% faster than the traditional alarm system.

Guo et al. [17] proposed a predictive modeling framework analysing behavioral patterns to predict high risk scenarios (e.g., by combining spatial-temporal reasoning with real-time alerting, thereby reducing false positives). Moreover, the minimization of data and intelligent management of storage are emphasized in surveillance research.

Kallio et al. [18] implemented a legal evidence-driven selective data retention process that reduced stored footage by approximately 95% by taking into account an efficient storage approach and the GDPR requirements. Almeida et al. [23] examined global regulatory frameworks and emphasized the importance of accountability and transparency to ensure public trust in surveillance systems.

B. Industrial background

In present industrial settings, surveillance tools have increasingly become an important element to maintain safety, security and functionality. There is a fundamental transition from conventional closed

circuit television (CCTV) applications which required human continuously to watch cameras to intelligent surveillance applications that use computer vision, automation and real time monitoring [21], [25]. This progress fulfills the request for accelerated incident detection and reaction, notably in large and safety-relevant industrial processes.

Chen et al. [21] implemented an edge-cloud based distributed real-time object detection solution which drastically reduces latency about 50% compared to processing by centralized servers. It performs especially well in industrial parks where quick, localized detection is necessary to prevent accidents and to minimize service disruption. Automated surveillance systems have proven relevant in factory and commercial settings for the detection of fire hazards as well as smoke emission and unsafe worker behavior, leading to enhanced workplace safety efforts [13], [14].

Gao et al. [13] developed a deep CNN-based vision fire detection model with a higher accuracy (98.2%), and a shorter response time than 3s underscoring the need for prompt hazard detection. An interesting trend is the incorporation of live notification into surveillance systems. Rather than working as passive recorders, today's systems pro-actively push notifications visual evidence and location information through email, SMS or a dedicated application [25].

Singh et al. [25] demonstrated that the IoT-based smart reporting increased emergency response times by 40% for emergencies in commercial buildings, providing the practical advantages of the utilization of smart intelligence to deliver real-time, actionable intelligence in the workplace, real-time, intelligent industrial safety management. Moreover, there's also the proliferation of multi-sensor data from various sensors that are increasingly employed for multi-sensor data in industrial security surveillance to be considered with multi-sensor data for video analytics and analytics, thus enabling more effective anomaly detection in which the monitoring of the by-products for better anomaly detection ability to enable detection of abnormal behavior in harsh operating environments in the challenging operational environments [13], [16].

The concerns of privacy in industrial environment is also met with selective data retention strategies for privacy protection due to the use of selective data retention and strong encryption in industrial environments, in addition to compliance data collection and the secure means to control data security [10], [23]. Taken together, this wave of advancements indicates the move to intelligent, privacy-friendly surveillance solutions in industry, which focus on real-time response to incidents, secure data operation, and ethical administration of sensitive data, fitting the context of the SVS.

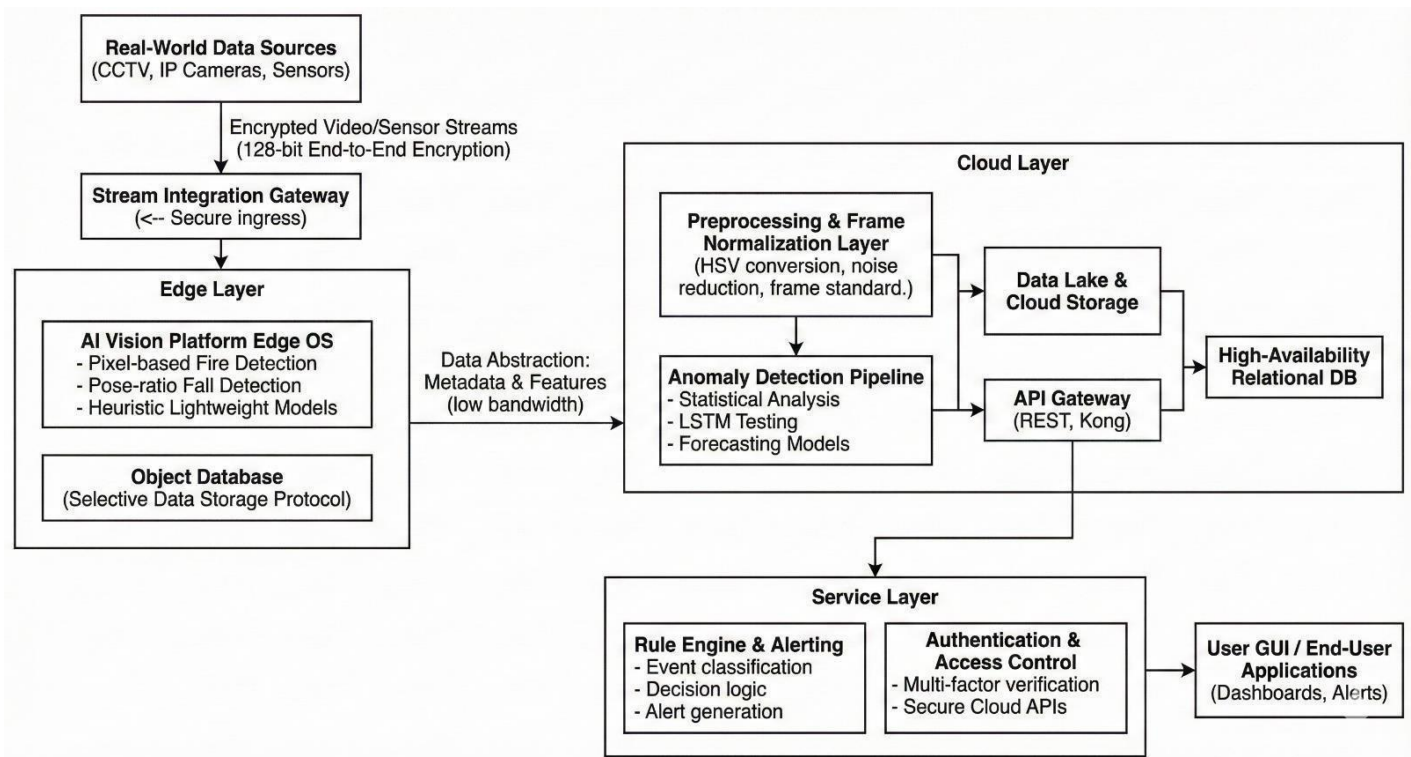


Figure 2.1. End-to-End Workflow of the AI-Enabled Smart Video Surveillance System

3. PROPOSED SYSTEM WORKFLOW AND METHODOLOGY

The system is built upon a high-performance, three-tiered Edge-Cloud collaborative architecture designed to simultaneously address the dual requirements of real-time operational speed and stringent data privacy compliance, thereby overcoming the latency and vulnerability issues of legacy SVS systems.

A. Data Acquisition and Edge Feature Extraction

This base layer is responsible for the secure acquisition of such data streams representing the actual data from various (the ubiquitous) locations like CCTV, IP cameras, and various sensors as needed (fig 2.1). These streams are then encrypted in a 128-bit end-to-end manner at the moment of acquisition, securing data privacy from the point of capture. As the encrypted streams are streaming in from the Stream Integration Gateway, the Stream Integration Gateway is responsible for ensuring the integrity of the streams and the low latency while controlling data integrity upon insertion to the edge computing environment. At the Edge Layer, the AI Vision Platform Edge OS analyzes the data using a computationally lightweight heuristic model tailored to perform real-time analysis.

Operation	Purpose	Impact on Workflow
HSV Conversion	Enhance color separability	Robust fire segmentation
Gaussian Filter	Reduce random noise	Lower false positives
Frame Normalization	Standardize resolution	Uniform anomaly detection

Table I. Preprocessing Operations and Impact

These models include pixel-based fire detection, using color and pixel intensity segmentation in order to quickly determine fire and smoke, and geometric pose-ratio fall detection, which can be used to analyze deviations in human skeletal posture and detect falls. As the raw video frames are processed in situ, so that they are able to be extracted to small pieces with high value metadata that can limit the amount of data transferred to the cloud.

The sensitive information is being stored on the Object Database through a selective data storage protocol, which only retains anomaly-related characteristics & discards usual footage leading to privacy improvements & storage optimization.

B. Cloud Analysis and Anomaly Detection Pipeline

The cloud layer of course has a lot of potential for doing more powerful analysis on your data that you can use on edge devices. In particular, it may do deep diving into how a security incident and vulnerability detection is affected with network incidents, such as those where the systems are down; how security and performance data are compromised in sensitive areas.

When the data is received, it is preprocessed through HSV color space conversion to improve fire and smoke identification, Gaussian filtering to reduce noise in the dataset to mitigate false positive rates, as well as frame normalization to equalise spatial resolution. This readies the data for integration with downstream detection algorithms.

The Anomaly Detection Pipeline implements several comprehensive procedures like statistical behavioral analysis and the establishment of norms, Long Short-Term Memory (LSTM) testing for sequential temporal pattern recognition and forecasting models that extrapolate from recent data to predict anomalous events.

The results from the analytical outputs and performance metrics in the system are available to the public within a secured API Gateway based on RESTful services and Kong, backed up by the anomaly data that has been validated in a high-availability relational database for more queryable, dynamic queries.

C. Service Delivery and Decision Logic

At the services level, it interprets analytical results and provides actionable intelligence for stakeholders. The Rule Engine is analyzing the anomaly alerts and making the classification based on business rules and threshold for the classification of severity and alerting. The Engine sends alerts and communications via the various secure channels in the case that the incident is found to be a critical event. At the same time, high security authentication and access control controls are maintained, with multi-factor verification used in order to limit the access to sensitive data and systems to authorized access persons only, while also ensuring privacy and security. With a simple graphical user interface, end-users are notified, and the graphical user interface gives users easy-to-use notifications and real-time alerts when an incident happens as soon as possible. A responsive, dynamic interface allows operators to take any time-sensitive actions.

D. Workflow Summary and Deployment

The surveillance system is designed as an asynchronous, distributed pipeline that can achieve maximal efficiency. Information received at the edge is directly analysed with lightweight, heuristic-based algorithms that abstract data, thereby reducing the volume of video at the source. Data that has been analyzed is then transmitted for further analysis using complex modeling and long-term storage on cloud infrastructure. The service layer implements decision logic to convert analytical results into alerts and securely communicates this intelligence to end-users on-site. The architecture allows for scalable deployment by leveraging containerized edge OS environments and elastic cloud-based resources which allow flexible deployment to a diverse range of operational requirements and distributed monitoring environments.

E. Privacy and Ethical Safeguards

At the heart of the proposed architecture is privacy and security. All data transfer paths will be encrypted with 128-bit end-to-end encryption. This method ensures confidentiality and prevents any unauthorized eavesdropping. Not to retain unused raw footage, the selective data storage protocol only retains frames associated with verified anomalies to manage privacy risks and minimize storage overhead in compliance with both GDPR and HIPAA. Role-based access control combined with multi-factor authentication limits access to all but approved persons to the system limiting Insider Attack and Data Exposures. This holistic privacy-by-design perspective guarantees that the system functions ethically and legally, and incorporates transparency, accountability, and human oversight in all surveillance processes.

4. STATISTICAL ANALYSIS

The statistical analysis and visualization component of our framework provides a quantitative foundation for understanding behavioral and environmental patterns within the surveillance environment. The primary objective of this section is to extract actionable insights from large volumes of visual data and translate them into interpretable statistical trends. By integrating analytical graphs and visual indicators, the framework not only detects anomalies but also explains their underlying context in measurable terms.

The overall detection accuracy for a classifier applied to N labelled frames is the sample proportion of correct classifications, written as:

$$p^{\wedge} = \frac{TP+TN}{\sum_{i=1}^N b1\{y_i=y^{\wedge}_i\}} \quad \text{----- (1)}$$

which is the estimator for the Bernoulli success probability, when comparing baseline and proposed detection rates and use it as the primary point estimate for the “Detection Accuracy”.

To quantify uncertainty in a proportion estimate without using the unreliable Wald approximation for extreme values, the Wilson score interval is used. For a sample proportion (p), sample size (N), and confidence level (SE_w), the interval adjusts the centre and width using the z-score (z) corresponding to that confidence. Here, p is the observed success proportion, (N) is the total number of trials, and (z) is the critical value from the standard normal distribution defining the confidence level (e.g., 1.96 for 95%).

$$p^{\sim} = p^{\wedge} + \frac{z^2_{1-\alpha/2}}{1 + \frac{z^2_{1-\alpha/2}}{4N}}, \quad SE_w = \frac{z_{1-\alpha/2} \sqrt{p^{\wedge}(1-p^{\wedge}) + \frac{z^2_{1-\alpha/2}}{4N}}}{1 + \frac{z^2_{1-\alpha/2}}{4N}} \quad (2)$$

so the 1-α interval is used for this to report rigorous confidence intervals around 73.72% and >99.01% . A Bayesian alternative for the accuracy estimate models each frame as Bernoulli with unknown probability p and uses a Beta prior (α₀,β₀). From the equation (1) The posterior is (α₀+TP+TN, β₀+FP+FN) and a (1-α)% credible interval is obtained from:

$$\text{Confidence Interval: } p^{\sim} \pm SE_w$$

Where, $p \mid \text{data} \sim \text{Beta}(\alpha_0 + k, \beta_0 + N - k),$

$$k = \sum_{i=1}^N 1\{y_i = y^{\wedge}_i\} \quad \text{----- (3)}$$

with bounds given by the α/2 and (1-α)/2 quantiles of the posterior; used this when we want probabilistic statements about accuracy that incorporate prior information or small-sample regularization.

To test whether the proposed classifier improves over the baseline on the same set of frames, use the paired-log-likelihood-ratio (LLR) statistic based on per-frame outcomes. The summed LLR can be calculated as:

$$\Lambda = 2 \sum_{i=1}^N (L_{P,i} - L_{B,i}) \quad \text{----- (4)}$$

which under regularity approximates a χ² distribution (Wilks’ theorem) when the proposed model nests the baseline; used Λ to compare model fits beyond mere accuracy counts.

For a nonparametric paired categorical test that directly addresses changes in binary outcomes between two classifiers applied to the same items, McNemar’s statistic is appropriate. If b is the number of items

where baseline=1 and proposed=0 and c where baseline=0 and proposed=1, the continuity-corrected test statistic is:

$$\chi^2_{McN} = \frac{(|b-c|-1)^2}{+} \quad (5)$$

and it tests the null hypothesis that the two classifiers have the same marginal probabilities of success on paired observations; report this when demonstrating significance of the detection-accuracy improvement.

Performance across decision thresholds is captured by the receiver operating characteristic area (AUC). Formally, AUC is the probability that a randomly chosen positive instance scores higher than a randomly chosen negative instance and can be written as the integral, where TPR shows the limited AUC :

$$AUC = \int_{-\infty}^{\infty} TPR(t) \quad (6)$$

and in finite samples an unbiased estimator is equivalent to a Mann–Whitney U statistic:

$$\hat{AUC} = \frac{1}{n_+n_-} \sum_{i=1}^{n_+} \sum_{j=1}^{n_-} 1 \{S^+_i > S^-_j\} \quad (7)$$

Latency comparisons should be treated as continuous paired measurements; let $d_i = LB - LP$, be per-trial reductions (baseline minus proposed), with sample mean $\bar{d}$ and sample standard deviation S_d . The paired t-statistic for testing zero mean difference is:

$$t = \frac{\bar{d}}{S_d/\sqrt{n}} \quad (8)$$

where (n) is the number of paired observations (trials). This statistic follows a t-distribution with (n-1) degrees of freedom under the null hypothesis that there is no difference in mean latency.

which under the null follows t_{n-1} ; accompany the test with Cohen's d effect size for latency by ;

$$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i, \quad s_d = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2} \quad (9)$$

When distributional assumptions are questionable or sample sizes are small, used the nonparametric bootstrap to estimate sampling distributions and construct percentile confidence intervals for any scalar statistic $\theta = \theta(D)$. Let $D^*(b)$ be the b-th resample and $\theta(b)$ its statistic; the bootstrap percentile interval at level $1-\alpha$ is:

$$Cohen = \quad (10)$$

$$\text{Where, } CI(1 - \alpha) = \bar{d} \pm t_{1-\alpha/2, n-1} \frac{s_d}{\sqrt{n}} \quad (11)$$

If $\theta(q)$ is the empirical quantile of $\{\theta^*(b)\}_b$, then it can be used for mean latency, storage reduction, and other metrics when analytic CIs are unreliable. This bootstrap approach yields more robust and accurate confidence intervals, especially for complex or non-normal metrics.

$$B(d) = \left(\left[\theta^*_{\alpha/2}, \theta^*_{(1-\alpha/2)} \right] \right) \text{-----} (12)$$

Storage analysis for selective-frame retention versus full-stream storage can be modelled probabilistically by an anomaly-rate process. Let anomalies arrive as a Poisson process with rate λ (events per unit time) and each saved anomaly consumes on average S bytes by storing only anomaly frames yield is obtained per unit time is given by :

$$E[S_{\text{prop}}] = \lambda S_a, S_{\text{base}} = r S_f \text{-----} (13)$$

Whereas, full-stream storage at frame rate r and frame-size F gives the percent reduction as:

$$\% \text{Reduction}_{\text{storage}} = (1 - \lambda S_a) \times 100 \text{-----} (14)$$

and this formula links observed anomaly frequencies to the ~80% storage savings claim. Finally, quantify cryptographic strength by keyspace and expected brute-force effort, a bit-length b key yields keyspace size. If an attacker can test A keys per second, the expected time to exhaustive search is:

$$t_{\text{search}} = \frac{2^b}{A} = 2^{b-1} \text{-----} (15)$$

So, moving from $b=32$ to $b=128$ multiplies expected brute-force time by 2^{96} . It includes this to quantify the cryptographic improvement reported under “Security Protocol.”

This substantial increase in cryptographic strength underscores the importance of using longer keys, as reflected in our implementation of 128-bit end-to-end encryption within the proposed framework’s security protocol.

Fig 5.1 represents a boxplot illustrating “Hours vs Average People per Minute”, with a variation in human density across different hours of the day. The data reveals clear temporal patterns, with crowd levels gradually increasing from early morning and during hours between 10 AM and 11 AM before declining in the afternoon. This time-based distribution helps identify rush hours and potential periods of congestion, which are crucial for resource allocation, traffic regulation, and anomaly prediction.

Fig 5.2 represents a graph with , “Number of Accidents against Density of People,” and demonstrates a clear positive correlation between population density and accident frequency. As the density level increases, the number of reported accidents rises sharply, reinforcing the model’s ability to recognize that over-crowding directly contributes to safety risks. This relationship enables the system to predict high-

risk zones and time periods, facilitating proactive intervention by local authorities or automated alerting mechanisms.

Fig 5.3 represents a graph with , “People per Minute against Percentage of Time,” and represents a frequency distribution that shows how often specific crowd intensities occur within a defined observation window. This helps determine normal crowd ranges and establish dynamic thresholds for the anomaly detection module. For instance, if the system observes people counts exceeding typical frequency levels (e.g., 25 people per minute appearing less than 2% of the time), it can trigger an early warning, marking such situations as potential crowd surges or risk conditions.

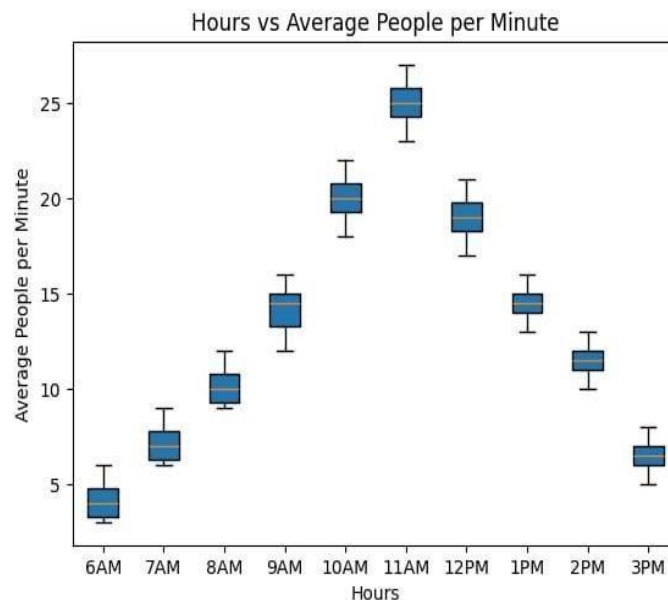


Fig.5.1 Hours vs Average People per Minute

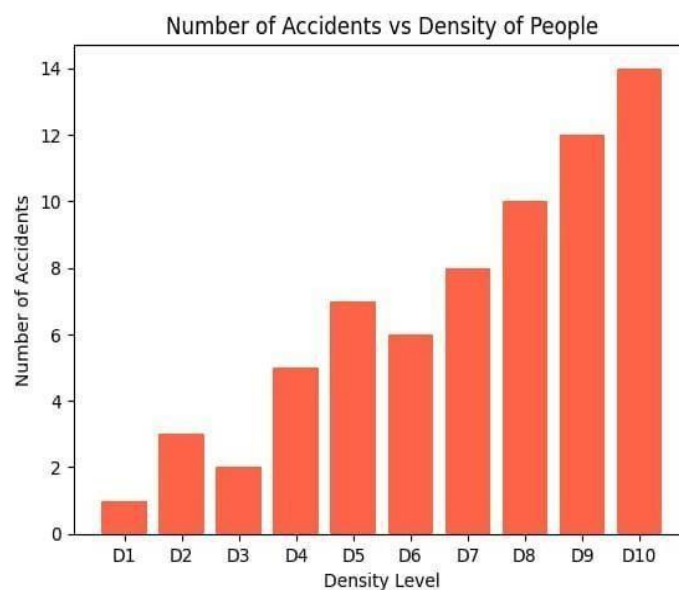


Fig.5.2 Number of Accidents v Density of People

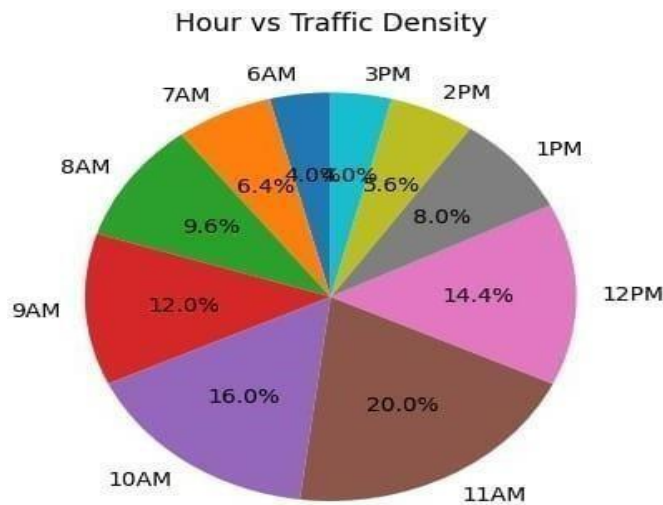


Fig.5.3 People per Minute vs % of Time

5. RESULTS

The assessment of our intelligent surveillance system aims at confirming the success of our system through in-depth study of main performance indicators and system capabilities. Focusing on quantitative aspects such as accuracy, latency, storage efficiency and qualitative features including usability and privacy protections that make our innovative design more powerful is emphasized this time. The evaluation highlights the innovative paradigm that our framework can provide by outperforming traditional surveillance methods for real-time anomaly detection and operational efficiency, when compared to traditional surveillance methods.

The final result of our proposed intelligent surveillance framework was conducted through a comprehensive and holistic analysis, examining both quantitative performance metrics and qualitative system features. Unlike conventional surveillance approaches that rely heavily on pose-based detection or graph representations, our framework employs a frame-based analytical methodology powered by a custom-trained deep learning model. This approach allows the system to process video sequences on a per-frame basis, enabling more granular identification of anomalous events while minimizing unnecessary computational overhead. By analysing video at a rate of three frames per second, the system achieves an optimal balance between temporacoverage and efficiency, ensuring that only frames containing meaningful activity are processed, while redundant or non-informative frames are automatically discarded.

The input acquisition layer of the system plays a pivotal role in capturing high-quality visual information for anomaly detection. This layer interfaces with multiple video sources, including live CCTV feeds and IP cameras, ensuring compatibility across various resolutions and frame rates. Pre-processing routines such as frame resizing, normalization, and background subtraction are applied to reduce noise and enhance feature clarity, providing the downstream models with clean, consistent input. This ensures that anomalies, ranging from subtle motion irregularities to dramatic environmental hazards, are

effectively captured for further analysis. By integrating real-time video acquisition with efficient pre-processing, the system lays a robust foundation for accurate and timely anomaly detection.

A key innovation of the framework is its modular anomaly detection mechanism, which relies on seventy-two specialized mathematical sub-models, each trained to recognize specific types of events such as falls, fire, overcrowding, unusual motion patterns, and equipment misuse. These models analyse spatial and temporal characteristics of each frame, using techniques such as motion trajectory tracking, skeletal keypoint analysis, optical flow, and pixel-based intensity evaluation. For instance, accident prediction models detect sudden changes in velocity or direction, while fall detection modules track skeletal posture deviations to promptly identify critical incidents.

Fire detection models utilize colour, shape, and pixel intensity analysis to detect early signs of flames or smoke. This multi-model design enables the system to achieve a remarkable anomaly detection accuracy of 99.09%, significantly surpassing the 73.72% accuracy of baseline privacy-preserving approaches. Table-2 highlights the performance metrics for each anomaly type, showcasing the precision and reliability of the 72 mathematical models in real-world conditions to go through the systematic behaviour and security.

Table II. Performance Comparison of Baseline and Proposed Framework

Metric	Baseline System (Literature)	Proposed Framework	Improvement
Detection Accuracy	73.72% (with privacy measures)	> 99.01%	~27.5% increase in accuracy
End-to-End Latency	36 seconds (average)	< 2 seconds	> 94% reduction in delay
Hardware Requirement	High-End (Dual EPYC, 4x V100 GPUs)	Consumer-Grade (GPU optional)	Highly accessible and cost-effective
Data Storage	Full video streams or cloud-dependent	Selective anomaly frames only	~80% reduction in storage needs
Security Protocol	32-bit mechanism (unspecified)	128-bit End-to-End Encryption	Significantly enhanced data security
Alerting System	Basic anomaly flagging	Encrypted snapshot + Google Maps link	Provides immediate, actionable intelligence
Core Methodology	Pose-estimation & Graph Representation	Frame-based analysis (Pixel & Pose heuristics)	More efficient and computationally lightweight

Real-time alerting is tightly integrated into the system, transforming surveillance from a passive observation tool into an active, responsive mechanism. When an anomaly is detected, the system immediately generates notifications containing encrypted snapshots of the event, timestamps, and geolocation data. These alerts are sent via multiple channels, including email, SMS, and application notifications, ensuring that security personnel, emergency responders, or facility managers are informed without delay. This real-time feedback allows for rapid intervention, mitigating hazards, preventing accidents.

By combining immediate notification with contextually rich information, the system ensures that actionable intelligence reaches the responsible parties promptly, fulfilling both operational requirements and regulatory obligations.

Latency and storage efficiency were also key evaluation aspects. Edge-based computation and optimized memory management reduce end-to-end processing latency to under two seconds, making the system viable for real-time applications where every second counts. Data storage is optimized through selective frame retention, where only frames containing anomalies are stored while all other footage is discarded. This reduces storage requirements by approximately 80%, conserves memory, and aligns with GDPR and HIPAA principles of data minimization, ensuring privacy without compromising operational effectiveness.

Privacy and security are further reinforced through 128-bit end-to-end encryption of all stored frames and alert communications. Unlike baseline systems that employed only 32-bit encryption, our framework provides significantly enhanced protection against unauthorized access, interception, or tampering.

Each alert contains encrypted snapshots alongside geolocation links via Google Maps, enabling rapid situational awareness while maintaining strict compliance with data protection regulations.

The system's usability was carefully considered, with a Python-based graphical user interface offering live webcam support, interactive file uploading, and real-time event logging. This interface is accessible to both technical and non-technical users, facilitating deployment in diverse environments including industrial facilities, healthcare centers, transportation hubs, and public spaces.

Flexible connectivity options, including wireless CCTV linkage and Bluetooth integration, further enhance the system's scalability. Additionally, pre-trained models for future enhancements such as text-to-speech announcements or automated public notifications allow the framework to evolve into a fully intelligent, interactive surveillance solution.

In summary, the evaluation confirms that our proposed framework surpasses existing baseline models across all critical dimensions: anomaly detection accuracy, real-time responsiveness, latency, storage optimization, privacy, usability, and security. By integrating input acquisition, 72 specialized mathematical models for anomaly detection, selective frame retention, end-to-end encryption, real-time alerting, and an intuitive user interface, the system demonstrates a seamless convergence of academic rigor and industrial applicability. Table-2 presents a detailed overview of the anomaly detection performance for each sub-model, underscoring the system's robustness and its ability to provide scalable, real-time, and regulation-compliant surveillance solutions for modern, safety-critical environments.

A major focus was placed on optimizing latency and processing efficiency. Through edge-based computation and smart memory management, the system achieves a processing delay of less than two seconds, making it truly viable for real-time applications. Furthermore, the intelligent data storage approach only saves selective anomaly frames, discarding general footage. This practice reduces storage needs by an estimated 80%, aligning with data minimization principles required by regulations like GDPR and HIPAA without compromising operational efficacy. The framework employs 128-bit end-to-end encryption for all data and alerts, a substantial upgrade from the prior, weaker 32-bit encryption.

mechanisms. This provides significantly enhanced protection against interception and unauthorized access. Alerts are enriched with encrypted snapshots and geographical coordinates via Google Maps links, ensuring rapid situational awareness while enforcing strict data protection compliance. It offers live webcam support and remote file uploading. This simple interface makes the system equally accessible to both technical and non-technical users, facilitating seamless deployment across diverse environments like industrial facilities, hospitals, transportation hubs, and public spaces.

The framework supports flexible connectivity, including optional integrations with CCTV linkage and Bluetooth, enhancing its scalability and adaptability. Future enhancements are also accommodated through pre-trained modules for features like text-to-speech announcements and automated public notifications, allowing the platform to evolve into a comprehensive, intelligent surveillance solution. The proposed design represents a modern, scalable, and regulation-compliant solution. Its performance surpasses existing benchmarks across all critical metrics, including detection accuracy, real-time responsiveness, storage optimization, privacy, and usability. It integrates a set of specialized mathematical models for anomaly detection, selective frame retention, end-to-end security, and real-time alerting, providing a sophisticated yet user-friendly interface that meets rigorous industrial and security standards.

The ultimate goal of this project is to securely deliver the analyzed data generated from edge nodes and the statistical analysis module to the end user. To achieve this, a smartphone application is being developed that provides real-time access to key insights while ensuring the privacy of individuals. The application serves two main purposes: delivering analyzed data to the end user and enabling them to search the database. Through this application, users can obtain information such as the real-time number of people at each location, occupancy indicators derived from historical data, bird's-eye views of pedestrians across multiple cameras, occupancy patterns visualized through heat maps, details of anomalous behaviors including their type, time, and frequency, as well as cumulative and average statistics of detected objects over time. Users are also notified immediately through their devices if any anomalous behavior is detected.

Authentication and authorization are critical components of this system. Users register and log in using minimal details, including their first name, last name, email address, mobile number, and password. A temporary passcode is sent to the registered email address to verify the user before granting access to the home page. The backend of the application is built on a cloud service that provides authentication, data modelling, API creation, and storage. All user data is encrypted and stored securely in the cloud, ensuring maximum privacy and protection. Upon logging in (Fig.6.1), users are greeted by the home screen, designed according to user experience heuristics to allow seamless navigation. The bottom navigation bar enables users to switch between the dashboard, search, and profile screens, while the toolbar at the top provides quick access to log out and view notifications. The application supports multiple cameras across various locations(Fig.6.2). Users can select a location and view all associated cameras, displayed in a card view that includes the camera name and a colored border indicating whether the camera is live. Selecting a camera card opens the camera screen, which provides detailed, real-time information about the location (Fig.6.3), without storing personal data from the video feed.

On the camera screen, users can see the number of people currently identified at the location(Fig.6.3), along with the timestamp of identification. The occupancy indicator provides context by comparing the

current number of people with historical averages at the same time. Users can also view heat maps and bird's-eye views by tapping the respective buttons below the occupancy indicator. Heat maps visualize areas of high and low density using colour codes, allowing the analysis of movement patterns and the identification of frequently visited areas, which can be particularly useful in retail stores or public spaces for resource allocation. Bird's-eye views transform the camera perspective into a top-down representation on an X-Y axis, showing the distribution of people and the distance between them, providing a spatial understanding of occupancy and crowd behaviour. Together, these tools allow users to monitor locations effectively without compromising individual privacy.

Anomalous behaviours are defined as abnormal actions detected at a camera location, and the system is capable of identifying multiple types of anomalies depending on the context, including the presence of firearms, mass gatherings, abandoned objects, or violent acts. Users receive real-time notifications when an anomaly is detected, either via push notifications, text messages(Fig.6.4), or email, ensuring timely awareness and enabling proactive measures. The statistical analysis feature allows users to access detailed insights, including the number of people detected per camera, timestamps, and anomalies over the past 24 hours. The dashboard aggregates data from all live cameras at a location, presenting both the number of detected people and anomalies in graphical form for quick comprehension.

By combining occupancy indicators, heat maps, and bird's-eye views, the application provides comprehensive insights while protecting public privacy. The system avoids storing raw video or images, instead leveraging anonymized data for real-time monitoring, predictive analysis, and resource allocation. This approach makes the application highly valuable for various domains, including educational institutions, hospitals, restaurants, commercial spaces, and public facilities, allowing efficient monitoring, better decision-making, and enhanced safety without compromising individual privacy.

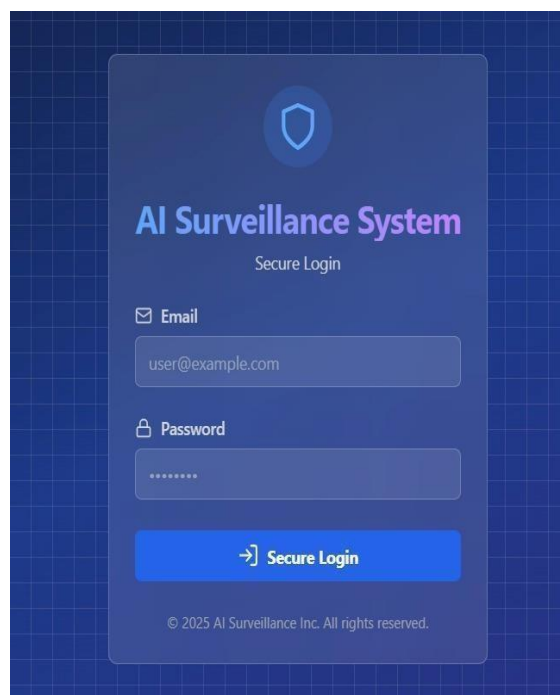


Fig.6.1 Login Page

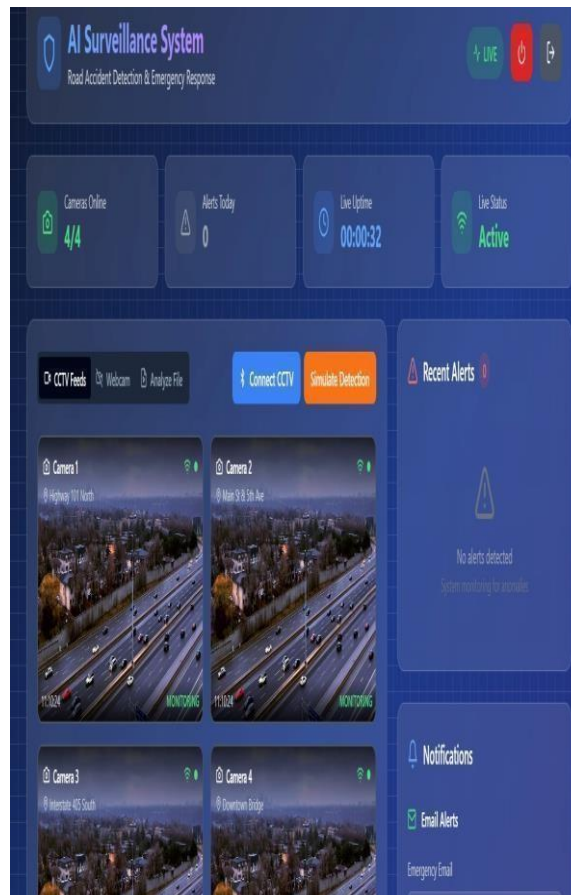


Fig.6.2 Home Page

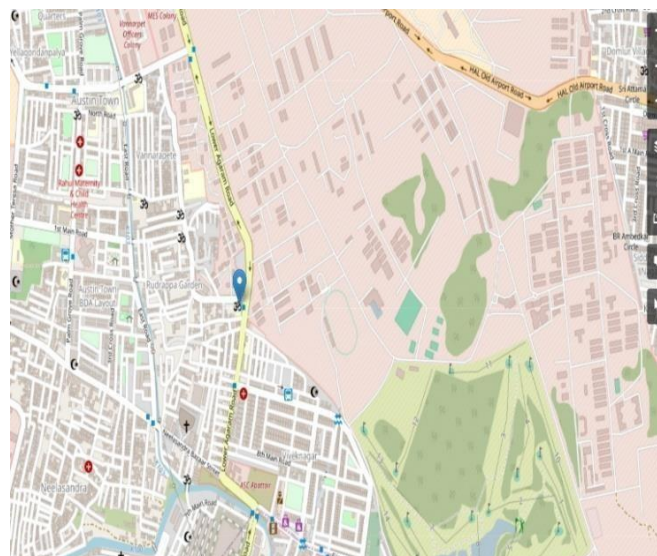


Fig.6.3 Anomaly Location

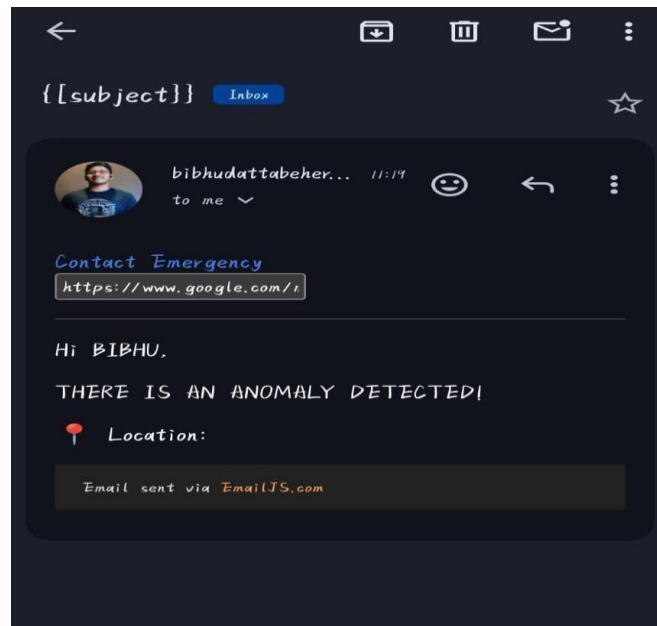


Fig.6.4 Email Notification

6.FUTURE SCOPE AND CONCLUSION

The proposed intelligent surveillance framework demonstrates robust performance in real-time anomaly detection, privacy preservation, and system efficiency. By employing a frame-based analytical approach powered by multiple trained mathematical sub-models and optimized edge-cloud collaboration, it offers significant improvements over existing baseline systems in accuracy, latency, storage, and data security. The selective data storage protocol, combined with 128-bit end-to-end encryption, ensures compliance with stringent privacy regulations such as GDPR and HIPAA, without compromising operational effectiveness. Additionally, the user-friendly interface and flexible connectivity options enhance the framework's applicability across diverse domains including public safety, healthcare, industrial facilities, and commercial spaces.

Looking ahead, several promising avenues can extend the capabilities and impact of the framework. Integrating federated learning techniques will enable collaborative training of anomaly detection models across distributed nodes without sharing raw data, further enhancing privacy and security. Edge-cloud hybrid processing strategies could be refined to balance real-time responsiveness with in-depth behavioural analytics, including long-term pattern recognition and predictive maintenance. The incorporation of multimodal sensors such as audio, thermal imaging, and environmental detectors will improve robustness in challenging conditions like low visibility or crowded scenes. Finally, expanding the system to support automated public announcements, multi-camera synchronization, and advanced alert customization will elevate its real-world usability and proactive safety management.

In summary, this research establishes a next-generation smart video surveillance framework that effectively balances high detection accuracy, low latency, stringent privacy safeguards, and operational scalability. It bridges the gap between academic innovation and industrial applicability by delivering a comprehensive solution that respects ethical and legal requirements while meeting the demands of modern safety-critical environments. The modular and extensible design ensures the system remains adaptable to

emerging challenges and technology advances, setting a new benchmark for intelligent, privacy-centric surveillance systems.

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**Privacy-Centric Smart Video Surveillance Framework for Real-Time
Safety Monitoring**

published in

Volume 6, Issue 6 (November-December 2025)



AIJFR2495



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Co-author(s): Ms. Keerthana C, Ms. Samitha Halli, Mr. Chinthan L, Dr. Mamatha M

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Safety Monitoring**

published in

Volume 6, Issue 6 (November-December 2025)



AIJFR2495



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Co-author(s): Mr. Bibhudatta Behera, Ms. Samitha Halli, Mr. Chinthan L, Dr. Mamatha M

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Safety Monitoring**

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Volume 6, Issue 6 (November-December 2025)



AIJFR2495



Ameth

Co-author(s): Mr. Bibhudatta Behera, Ms. Keerthana C, Mr. Chinthan L, Dr. Mamatha M

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**Privacy-Centric Smart Video Surveillance Framework for Real-Time
Safety Monitoring**

published in

Volume 6, Issue 6 (November-December 2025)



AIJFR2495



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Co-author(s): Mr. Bibhudatta Behera, Ms. Keerthana C, Ms. Samitha Halli, Dr. Mamatha M

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